

EGH 404 Assessment 2 – Data Analysis

You should draw across the unit content for Data Analysis—data wrangling, data visualisation, statistics, correlation and regression – to answer this assessment. Please provide your answers directly on this template. Do not delete the template wording as this assists markers in knowing which question you are answering. You have the option of completing this assessment either using MATLAB, Python or Excel.

Remember to provide evidence to justify your responses. Evidence includes but is not limited to any forms of visualisation, source code (for MATLAB or Python users), screenshots of Excel formulas, and screenshots of Excel “Data Analysis” tools. Please include your supporting evidence at the end of the report as an appendix (copy and paste the code into the document, or if using live scripts, merge the document PDF with the exported live script PDF). Please note that your code will not be run except under exceptional circumstances, but may be used to evaluate the process that you used to answer the question. Figures should be well formatted and include clear titles, axis labels, legends (where applicable), and appropriate units.

SECTION 1 – PROCESSING AND VISUALISATION OF REAL-WORLD DATA

Brisbane City Council (BCC) is considering upgrades to a number of bikeway networks. To do so, they need to understand how different bikeways are being used by both cyclists and pedestrians, so they can use this information to plan suitable bikeway upgrades. They are using data they have gathered from sensors placed along the bikeway paths, which automatically record the daily number of cyclists and pedestrians.

BCC has requested that you conduct an initial assessment regarding cyclist *and* pedestrian usage of two specific bikeway paths - Bicentennial Bikeway in Brisbane and North Brisbane Bikeway in Windsor. BCC is investigating historical bikeway usage of these two bikeway paths from 2014-2018 (5 years), and if there has been any relationship between different weather factors (rainfall, solar exposure and maximum temperature) and the use of these bikeway paths by both cyclists and pedestrians.

We have provided you with the BCC Bikeway Data for 2014-2018 (between 01 January 2014 and 31 December 2018) for daily cyclist and pedestrian usage of two locations: Bicentennial Bikeway in Brisbane and North Brisbane Bikeway in Windsor. Furthermore, Bureau of Meteorology (BOM) weather data captured at Brisbane City’s weather station for the same period above for three weather factors: daily rainfall, solar exposure and maximum temperatures is also provided.

Please refer to the Assessment page in Canvas for a copy of this dataset and scripts to load the data (only for Python and MATLAB users)

Download the brisbane_bikeway_counters.csv and brisbane_weather.csv datasets.

S1.T1: Bikeway Data Inspection

- a) Create a **single** box plot visualisation (separate box plots for each bikeway in a SINGLE visualisation) to help you identify outliers across daily bikeway usage data. Your box plot should include box data for:
 - Cyclist usage on Bicentennial Bikeway, Brisbane

- Pedestrian usage on Bicentennial Bikeway, Brisbane
- Cyclist usage on North Brisbane Bikeway, Windsor
- Pedestrian usage on North Brisbane Bikeway, Windsor

(1 pts)

Hint: Module on - Introduction to Visualising Data & Statistics along with the subsequent MATLAB, Python, and Excel modules (depending on the tool that you are using), could be beneficial as you develop your response.

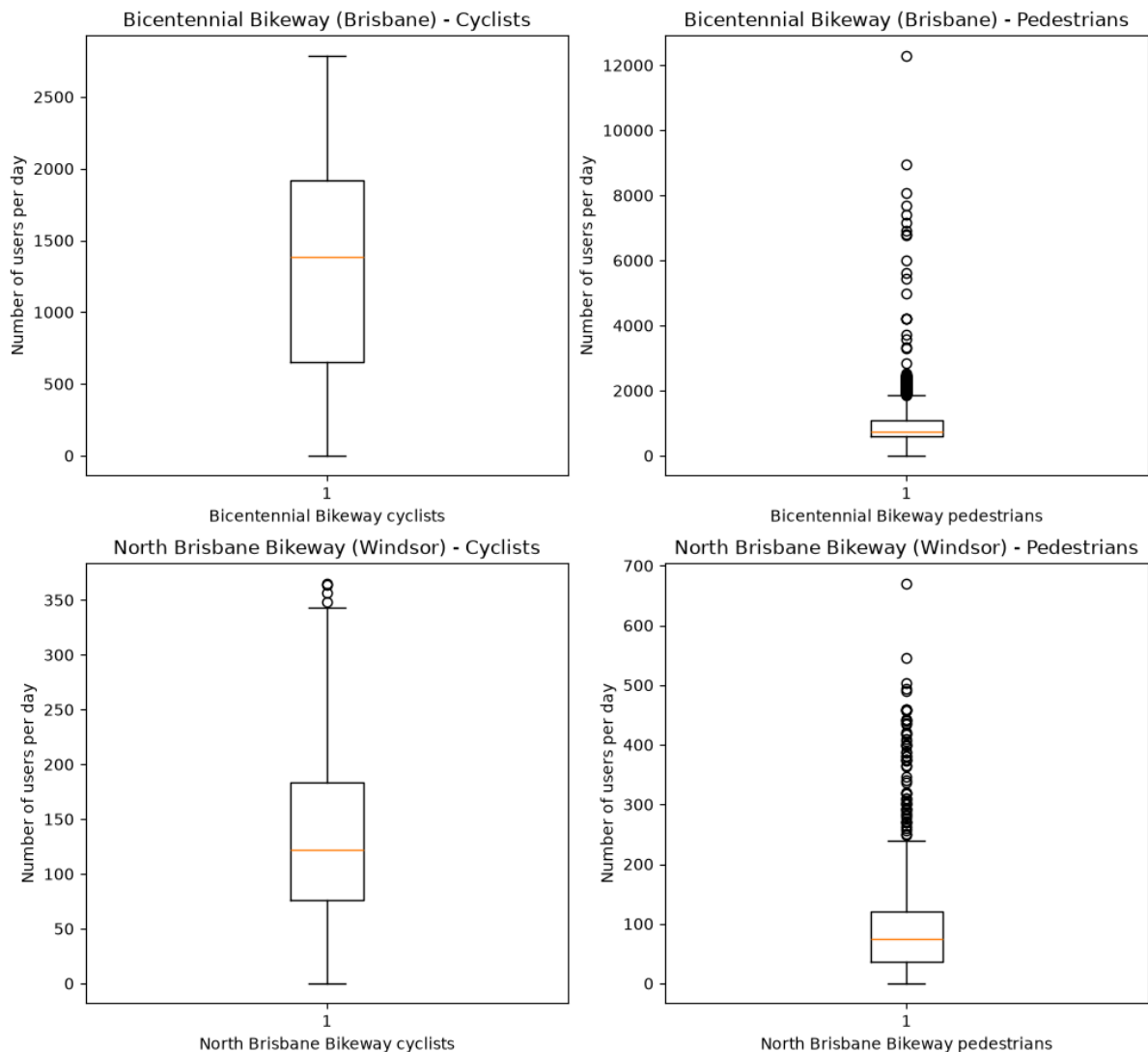


Figure 1: Box plots of daily cyclist and pedestrian counts on the Bicentennial Bikeway and North Brisbane Bikeway, 1 January 2014 to 31 December 2018. Circles represent values beyond 1.5 times the interquartile range.

- b) Create a **single** line graph visualisation to help identify any errors, missing or other problematic data across daily bikeway usage data. Your line graph visualisation should include 4 separate lines to represent usage over time for:
- Cyclist usage on Bicentennial Bikeway, Brisbane
 - Pedestrian usage on Bicentennial Bikeway, Brisbane
 - Cyclist usage on North Brisbane Bikeway, Windsor
 - Pedestrian usage on North Brisbane Bikeway, Windsor

(1 pts)

Hint: Module on - Introduction to Visualising Data & Statistics along with the subsequent MATLAB, Python, and Excel modules (depending on the tool that you are using), could be beneficial as you develop your response.

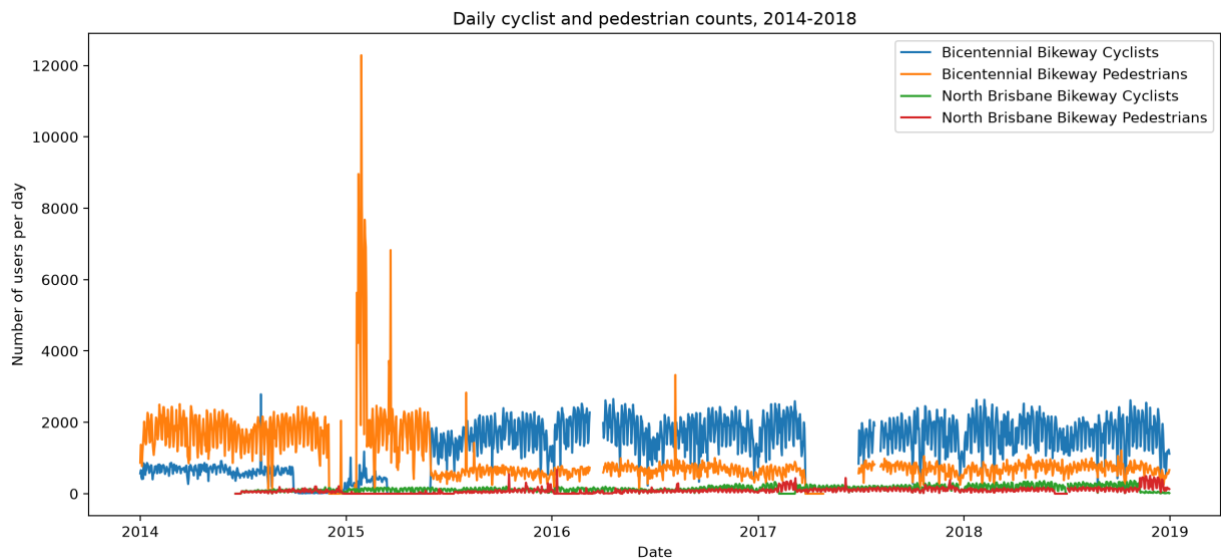


Figure 2: Daily cyclist and pedestrian counts on the Bicentennial Bikeway and North Brisbane Bikeway, 1 January 2014 to 31 December 2018. Gaps indicate missing values and flat zero periods indicate counters recording no users.

- c) Based on your answers in (a) and (b) above, have you identified any problematic data (i.e. outliers, missing data, errors, etc.) in your bikeways data? If yes, please identify the problems you have found, the dates (or date ranges if the identified problem persists for a consecutive period) of the problematic data points, and how you intend to handle the problematic data in your final (cleaned) dataset, with justification for the method of handling. Provide your answers in the table below (you can add more rows if needed), and for each bikeway and problem type. Provide **up to 3 examples for each problem type** (e.g., outliers, missing data, errors) for **each bikeway**. i.e. If more than 3 instances exist → report only 3; If fewer than 3 exist → report all that you can find.

(4 pts)

Hint: Module on - Introduction to Visualising Data & Statistics could be beneficial as you develop your response.

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All date ranges are inclusive. Remove or leave missing means that only the affected values are treated as missing; entire date rows are retained and the raw CSV remains unchanged.

Name of the column with the problematic data, type of the problem (outliers, missing data, errors) and the value.	Specify the date or date range (for continuing missing values) with problematic data	How do you intend to handle the problematic data identified?	Justification
Bicentennial pedestrians and cyclists: missing (NaN), 23 days	9–31 Mar 2016	Leave missing; do not impute.	Both channels are absent for the same continuous block (Box 9). The gap is too long for linear interpolation, and a single mean would erase weekday/weekend and weather variation.
Bicentennial pedestrians and cyclists: 32 zero days followed by 62 missing days	26 Mar–27 Jun 2017	Treat as one 94-day outage and leave both channels missing.	Both channels fall to zero together for 32 days and then stop reporting for 62 days (Boxes 9–10). A bikeway-wide zero for a month is implausible and the whole interval is too long to reconstruct reliably.
Bicentennial pedestrians and cyclists: missing, 10 days	27 Jul–5 Aug 2017	Leave missing.	This short gap is visible in Box 9, but interpolation over 10 days would invent unsupported daily counts. Excluding it has little effect on the five-year record.
North Brisbane pedestrians and cyclists: 169 missing days followed by 10 zero days	1 Jan–28 Jun 2014	Treat the entire period as not operating; begin analysis on 29 Jun 2014.	Both channels are blank from the start of the file through 18 Jun, then both are zero on 19–28 Jun (Boxes 9–10). Combining these avoids counting the same start-up failure twice.
North Brisbane pedestrians: 22 zero days then missing; cyclists: 34 on 30 Jun, two zero days, then missing	Pedestrians 11 Jun–3 Jul 2018; cyclists 30 Jun–3 Jul 2018	Leave pedestrian values missing for 11 Jun–3 Jul and cyclist values missing for 30 Jun–3 Jul.	Pedestrians are zero while cyclists remain normal through 29 Jun. Cyclists then fall to 34 on 30 Jun, compared with 83–122 on other June Saturdays, before both channels record zero and then no value (Box 9).
Bicentennial cyclists: spike (inside IQR bound), 2,786	3 Aug 2014	Keep.	About 4.8 times adjacent days while pedestrians fall to 928 (Box 15), consistent with a cycling event rather than a shared fault. It lies inside the IQR bound but is clear on the line graph.

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Problematic data table (continued)

Name of the column with the problematic data, type of the problem (outliers, missing data, errors) and the value.	Specify the date or date range (for continuing missing values) with problematic data	How do you intend to handle the problematic data identified?	Justification
Bicentennial pedestrians: outlier, 6,827	21 Mar 2015	Keep but flag.	Nearby pedestrian days are also high and this is outside the January multiple-of-128 fault (Box 15). The cyclist channel is zero, so the split between user types remains uncertain.
Bicentennial pedestrians: outlier, 3,328	7 Aug 2016	Keep.	The only 2016 value above the 1,867 bound and cyclists fall to 720 (Boxes 13 and 15). The date matches UQ St Lucia Open Day, supporting a genuine pedestrian event (The University of Queensland, 2016).
North Brisbane pedestrians: outlier, 546	17 Oct 2015	Remove/set missing.	Adjacent pedestrian counts are 28–85 and cyclists do not rise (Box 15). The isolated pedestrian-only spike is more consistent with a recording error.
North Brisbane pedestrians: outlier, 670	10 Jan 2016	Remove/set missing.	Pedestrians are zero on 5–9 and 12 Jan and only 5–6 on 11 and 13 Jan, while cyclists remain normal (Box 15), showing an intermittent pedestrian-channel fault.
North Brisbane cyclists: outlier, 365	23 Oct 2018	Keep.	Only slightly above the 346 bound; nearby weekdays are 304, 285 and 330 (Box 15). It matches the higher 2018 usage level.
Bicentennial pedestrians: counter fault (zeros, exact multiples of 128, then 5,452)	2 Dec 2014–6 Feb 2015	Remove all pedestrian values in this period.	Most days through 19 Jan are zero; every value from 20 Jan–5 Feb is an exact multiple of 128, and 6 Feb is still far above the following days while the other counters do not rise (Box 11).

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Problematic data table (continued)

Name of the column with the problematic data, type of the problem (outliers, missing data, errors) and the value.	Specify the date or date range (for continuing missing values) with problematic data	How do you intend to handle the problematic data identified?	Justification
Bicentennial cyclists: undercount, median 23.5/day	30 Sep–28 Dec 2014	Remove/set missing.	Counts collapse from 586 to 145 and then 56, with a period median of 23.5 and maximum of 145, while pedestrians remain active (Box 11). The 29 Dec–14 Mar partial recovery is retained but flagged as uncertain.
Bicentennial cyclists: 77 consecutive zeros	15 Mar–30 May 2015	Remove/set missing.	Pedestrians remain active while cyclists are zero for 77 days, followed by 422 cyclists on 31 May and then 1,709 on 1 Jun (Boxes 10–11).
North Brisbane pedestrians: intermittent low/zero counter fault	29 Dec 2014–11 Jul 2015	Remove/set missing.	There are 161 zero days in this 195-day period and the median is zero, while cyclists remain normal; before/after pedestrian medians are about 61–63/day (Box 12).
North Brisbane cyclists drop while pedestrians jump	6 Feb–8 Mar 2017	Remove both North Brisbane channels.	Cyclists are zero from 7 Feb–7 Mar while pedestrians jump to a median of 271, indicating users were probably reclassified into the pedestrian channel (Boxes 12–13).
North Brisbane cyclists drop while pedestrians jump	9 Nov–31 Dec 2018	Remove both North Brisbane channels.	Cyclists fall from 320 to 76 and then remain low (median 32, maximum 76), while pedestrians jump from 144 to 442 (median 283), repeating the earlier misclassification pattern (Box 12).

The 1.5 x IQR upper bounds are 1,867 Bicentennial pedestrians, 3,814.5 Bicentennial cyclists, 244.5 North Brisbane pedestrians and 346 North Brisbane cyclists. All lower bounds are negative.

The IQR rule also flags 58 North Brisbane pedestrian days above 244.5, but 18 of them fall in 6 Feb–8 Mar 2017 and 28 in 9 Nov–31 Dec 2018 (Box 13), the two periods where cyclists appear to have been counted as pedestrians. They are handled with those periods under Errors rather than as individual outliers.

The rule flags 220 Bicentennial pedestrian days: 147 in 2014, 72 in 2015 and only one in 2016. These are not removed as a group because they mainly describe the older operating level. Mean daily use changes from about 1,711 pedestrians and 654 cyclists before 30 September 2014 to about 593 pedestrians and 1,582 cyclists from June–December 2015, while combined use is within about 8%. This indicates a likely counter classification change, which must be considered when interpreting later correlations.

S1.T2: Weather Data Inspection

- a) Create **separate boxplots** for each weather factor to help identify outliers for:
 - daily rainfall
 - daily maximum temperatures
 - daily solar exposure

(1 pts)

Hint: Module on - Introduction to Visualising Data & Statistics along with the subsequent MATLAB, Python, and Excel modules (depending on the tool that you are using), could be beneficial as you develop your response.

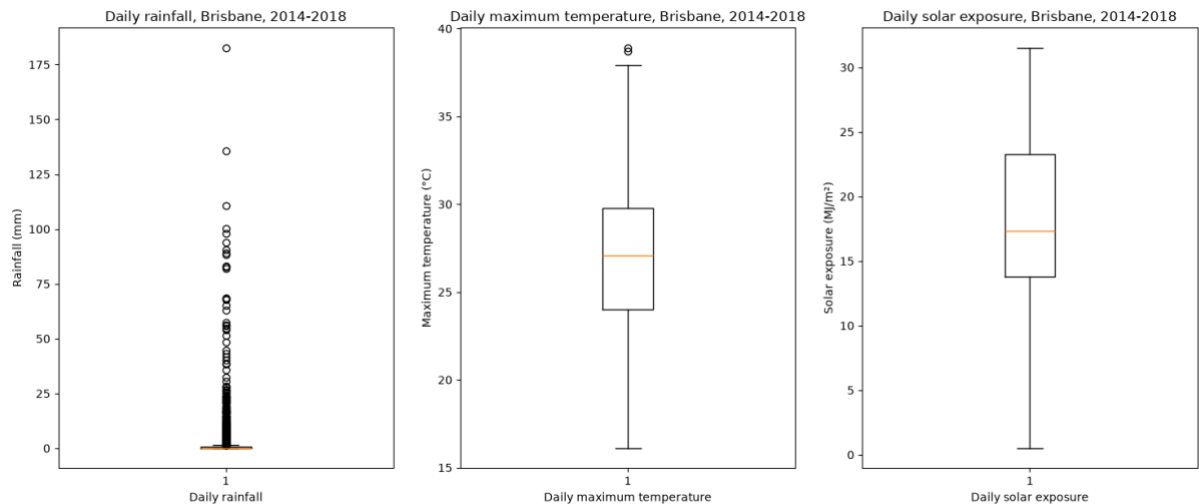


Figure 3: Box plots of daily rainfall, maximum temperature and solar exposure in Brisbane, 1 January 2014 to 31 December 2018. Circles represent values beyond 1.5 times the interquartile range.

b) Create **separate line graphs** to help identify any errors, missing or other problematic data across time for each of the following weather factors

- daily rainfall
- daily maximum temperatures
- daily solar exposure

(1 pts)

Hint: Module on - Introduction to Visualising Data & Statistics along with the subsequent MATLAB, Python, and Excel modules (depending on the tool that you are using), could be beneficial as you develop your response.

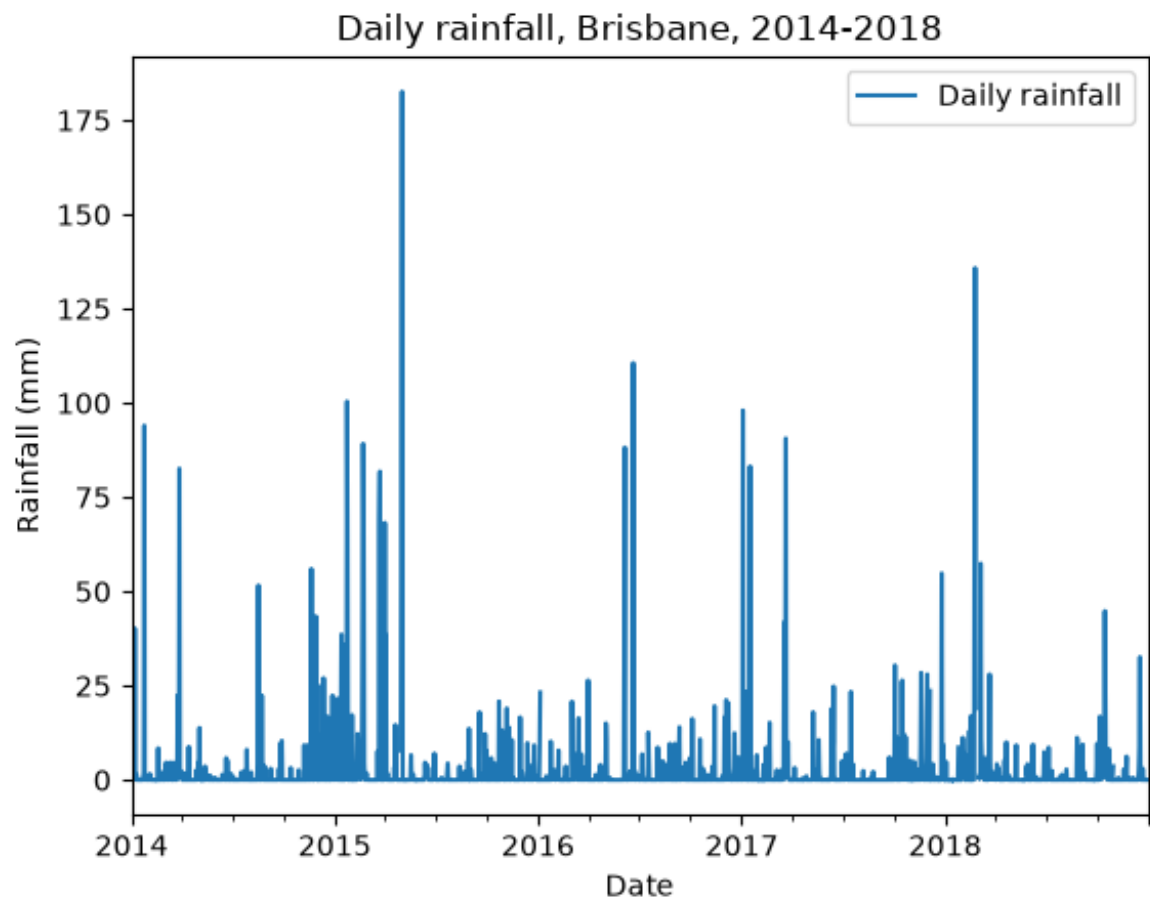


Figure 4: Daily rainfall in Brisbane, 1 January 2014 to 31 December 2018.

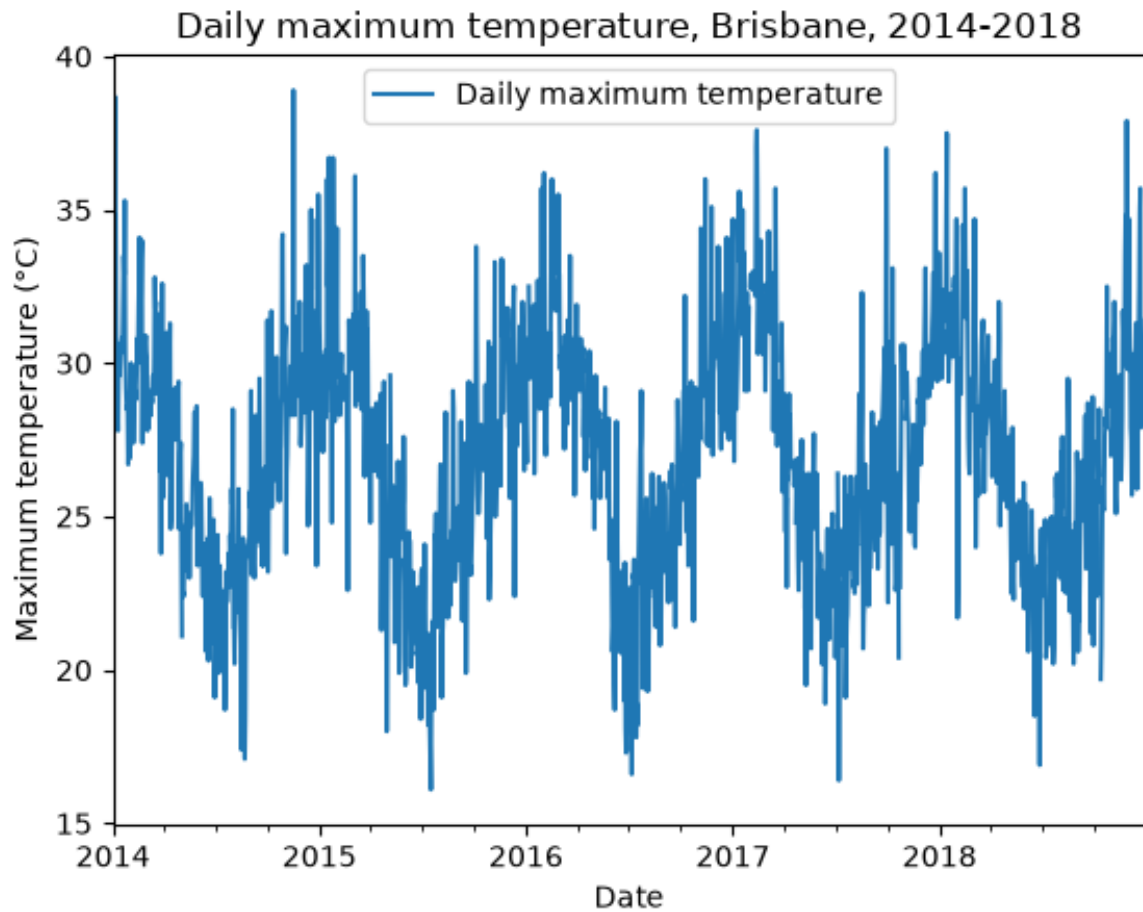


Figure 5: Daily maximum temperature in Brisbane, 1 January 2014 to 31 December 2018.

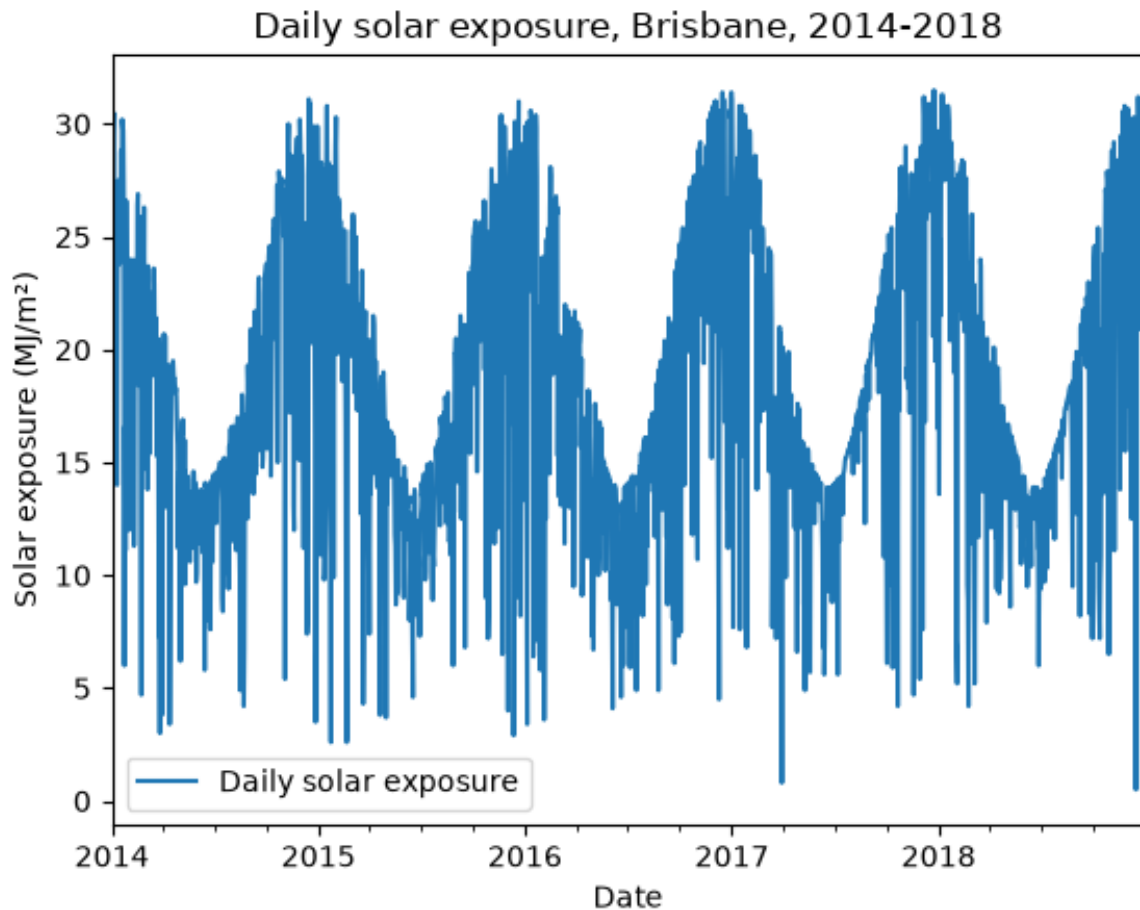


Figure 6: Daily solar exposure in Brisbane, 1 January 2014 to 31 December 2018. Small single-day gaps may not be visible at this scale.

- d) Based on your answers in (a) and (b) above, have you identified any problematic data (i.e. outliers, missing data, errors, etc.) in your weather data? If yes, please identify the problems you have found, the dates (or date ranges if the identified problem persists for a consecutive period) of the problematic data points, and how you intend to handle the problematic data in your final (cleaned) dataset, with justification for the method of handling. Provide your answers in the table below (you can add more rows if needed), and for each bikeway and problem type. Provide **up to 3 examples for each problem type** (e.g., outliers, missing data, errors) for **each bikeway**. i.e. If more than 3 instances exist → report only 3; If fewer than 3 exist → report all that you can find.

(4 pts)

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Rainfall is missing on 39 of 1,826 days (2.1%), maximum temperature on 24 days (1.3%), and solar exposure on one day. Leave missing means the affected value is excluded from calculations; interpolate means estimate between adjacent observations.

Name of the column with the problematic data, type of the problem (outliers, missing data, errors) and the value.	Specify the date or date range (for continuing missing values) with problematic data	How do you intend to handle the problematic data identified?	Justification
Daily rainfall: missing, 2 days	30–31 Mar 2017	Leave missing; do not impute.	Ex-Tropical Cyclone Debbie affected Brisbane and BOM reported a Brisbane City gauge outage (Bureau of Meteorology, 2017). Solar exposure fell to 0.8 MJ/m ² , so interpolation between dry readings would misrepresent severe weather (Box 16).
Daily rainfall: missing, 3 days	19–21 Feb 2018	Leave missing; do not impute.	The longest rainfall gap lies within an unsettled fortnight containing 17.0, 62.8 and 135.8 mm days; rainfall does not vary smoothly enough for interpolation (Box 16).
Daily rainfall: missing, 1 day	6 Jan 2016	Leave missing.	Maximum temperature is also missing on 5 Jan. That day was very dark (3.4 MJ/m ²) after 8.8 mm and 23.2 mm of rain, so rain probably fell but its amount cannot be estimated (Box 16).
Daily maximum temperature: missing, 1 day	4 May 2014	Interpolate between 21.1 °C and 22.5 °C.	Temperature changes gradually and solar exposure on 4 and 5 May is almost identical, so adjacent days provide a reasonable estimate (Box 16).
Daily maximum temperature: missing, 2 days	25–26 Feb 2018	Interpolate between 29.1 °C and 28.5 °C.	The surrounding temperatures differ by only 0.6 °C and both missing days have similar high solar exposure (Box 16).
Daily maximum temperature: missing, 2 days	29–30 Oct 2018	Leave missing; do not interpolate.	Solar exposure falls to 6.5 MJ/m ² and rain is recorded within the gap, so a straight line may overestimate the cloudy day (Box 16).
Daily solar exposure: missing, 1 day	26 Nov 2017	Interpolate between 24.2 and 26.4 MJ/m ² .	No rain is recorded and maximum temperature stays steady, making an intermediate clear-day value reasonable (Box 16).

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Problematic data table (continued)

Name of the column with the problematic data, type of the problem (outliers, missing data, errors) and the value.	Specify the date or date range (for continuing missing values) with problematic data	How do you intend to handle the problematic data identified?	Justification
Daily rainfall: outlier, 182.6 mm	2 May 2015	Keep.	BOM lists 56.2 mm on 1 May and 183 mm on 2 May, matching this data (Bureau of Meteorology, 2015). Solar exposure on 1 May is only 3.7 MJ/m ² (Box 19), and BOM also reports flooding.
Daily rainfall: outlier, 135.8 mm	24 Feb 2018	Keep.	BOM reports severe thunderstorms and heavy rainfall in Brisbane during the last week of February (Bureau of Meteorology, 2018a). Low solar exposure on 23 February supports the observation.
Daily rainfall: outlier, 110.6 mm	20 Jun 2016	Keep.	BOM reports 110.6 mm to 9 am on 20 June, exactly matching the dataset (Bureau of Meteorology, 2016). Solar exposure on the previous day was also the lowest of that week.
Daily maximum temperature: outlier, 38.9 °C	16 Nov 2014	Keep.	The day was dry and sunny, and BOM reported very hot conditions across Greater Brisbane on 15–16 November (Bureau of Meteorology, 2014). The value is meteorologically plausible.
Daily maximum temperature: outlier, 38.7 °C	4 Jan 2014	Keep.	The observation occurred between other hot days, with no rainfall and high solar exposure (Box 19). The surrounding values support a genuine summer heat event.
Daily solar exposure: lowest value, 0.5 MJ/m ² (inside IQR bounds)	16 Dec 2018	Keep.	Rainfall was 9.0 mm and maximum temperature fell from 31.0 to 25.9 °C. BOM links the period to ex-Tropical Cyclone Owen, supporting a genuine very cloudy day (Bureau of Meteorology, 2018b).
Daily solar exposure: second-lowest value, 0.8 MJ/m ² (inside IQR bounds)	30 Mar 2017	Keep.	The value occurred as ex-Tropical Cyclone Debbie affected Brisbane (Bureau of Meteorology, 2017); maximum temperature also dropped. The other variables support a genuine low-solar day.

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Problematic data table (continued)

Name of the column with the problematic data, type of the problem (outliers, missing data, errors) and the value.	Specify the date or date range (for continuing missing values) with problematic data	How do you intend to handle the problematic data identified?	Justification
Daily rainfall - no recording errors found	-	No action.	No negative readings were found. The largest values agree with BOM event records and surrounding solar and temperature measurements.
Daily maximum temperature - no recording errors found	-	No action.	No negative readings were found and all recorded values lie between 16.1 and 38.9 °C, a plausible Brisbane range (Boxes 7 and 21).
Daily solar exposure - no recording errors found	-	No action.	No negative readings were found and values lie between 0.5 and 31.5 MJ/m ² . The two lowest days agree with rain, lower temperatures and documented weather events (Boxes 7, 19 and 21).

The 1.5 x IQR limits are rainfall above 1.5 mm, maximum temperature below 15.3 or above 38.5 °C, and solar exposure below -0.45 or above 37.55 MJ/m². Rainfall is strongly zero-inflated: 1,161 of 1,787 recorded days have no rain, so the IQR rule flags 338 ordinary wet days. The largest falls were therefore checked against the other weather variables and BOM records rather than removed automatically (Boxes 7, 18 and 19).

Section 1 references

Bureau of Meteorology. (2014). Brisbane in spring 2014. <https://www.bom.gov.au/climate/current/season/qld/archive/201411.brisbane.shtml>

Bureau of Meteorology. (2015). Brisbane in May 2015. <https://www.bom.gov.au/climate/current/month/qld/archive/201505.brisbane.shtml>

Bureau of Meteorology. (2016). Brisbane in June 2016. <https://www.bom.gov.au/climate/current/month/qld/archive/201606.brisbane.shtml>

Bureau of Meteorology. (2017). Brisbane in March 2017. <https://www.bom.gov.au/climate/current/month/qld/archive/201703.brisbane.shtml>

Bureau of Meteorology. (2018a). Brisbane in February 2018.
<https://www.bom.gov.au/climate/current/month/qld/archive/201802.brisbane.shtml>

Bureau of Meteorology. (2018b). Brisbane in December 2018.
<https://www.bom.gov.au/climate/current/month/qld/archive/201812.brisbane.shtml>

The University of Queensland. (2016). 2017 Business and Economics undergraduate guide.
https://bel.uq.edu.au/files/9723/2017_B%26E_UG_Guide_web.pdf

SECTION 2 – CORRELATION ANALYSIS USING REAL-WORLD DATA

Using your cleaned data, please answer the following questions.

BCC wants to further investigate the relationship between daily cyclist and pedestrian usage on the Bicentennial Bikeway in Brisbane and the North Brisbane Bikeway in Windsor for 2014-2018. To answer Sections 2 and 3, please use the cleaned datasets (brisbane_bikeway_counters_cleaned.csv, brisbane_weather_cleaned.csv) provided in Canvas Assessment 2 – Details page.

Please answer the following questions:

S2.T1: Correlation Coefficient

Please compute the correlation coefficients for each of the relationships listed in the table below.

Hint: Module on - Correlation Analysis along with the subsequent MATLAB, Python, and Excel modules (depending on the tool that you are using), could be beneficial as you develop your response.

(2 pts)

Please compute the correlation coefficient between:	Correlation coefficient
(a) Cyclists and Pedestrians on the Bicentennial Bikeway in Brisbane	-0.35539
(b) Cyclists and Pedestrians on the North Brisbane Bikeway in Windsor	0.27612
(c) Cyclists on Bicentennial Bikeway in Brisbane and Cyclists on the North Brisbane Bikeway in Windsor	0.48184
(d) Pedestrians on Bicentennial Bikeway in Brisbane and Cyclists on the North Brisbane Bikeway in Windsor	-0.02775

Pearson correlation calculated from all 1,461 paired rows in the provided cleaned bikeway CSV (S2 Boxes 2 and 5). Values are rounded to five decimal places. The weather file is not needed for the relationships specified in Section 2.

S2.T2: Correlation Visualisation

Please create a scatter plot to visualise the relationship between daily cyclist usage of North Brisbane Bikeway in Windsor (x-axis) and daily pedestrian usage (y-axis) for the Bicentennial Bikeway in Brisbane.

(2 pts)

Hint: Module on - Correlation Analysis along with the subsequent MATLAB, Python, and Excel modules (depending on the tool that you are using), could be beneficial as you develop your response.

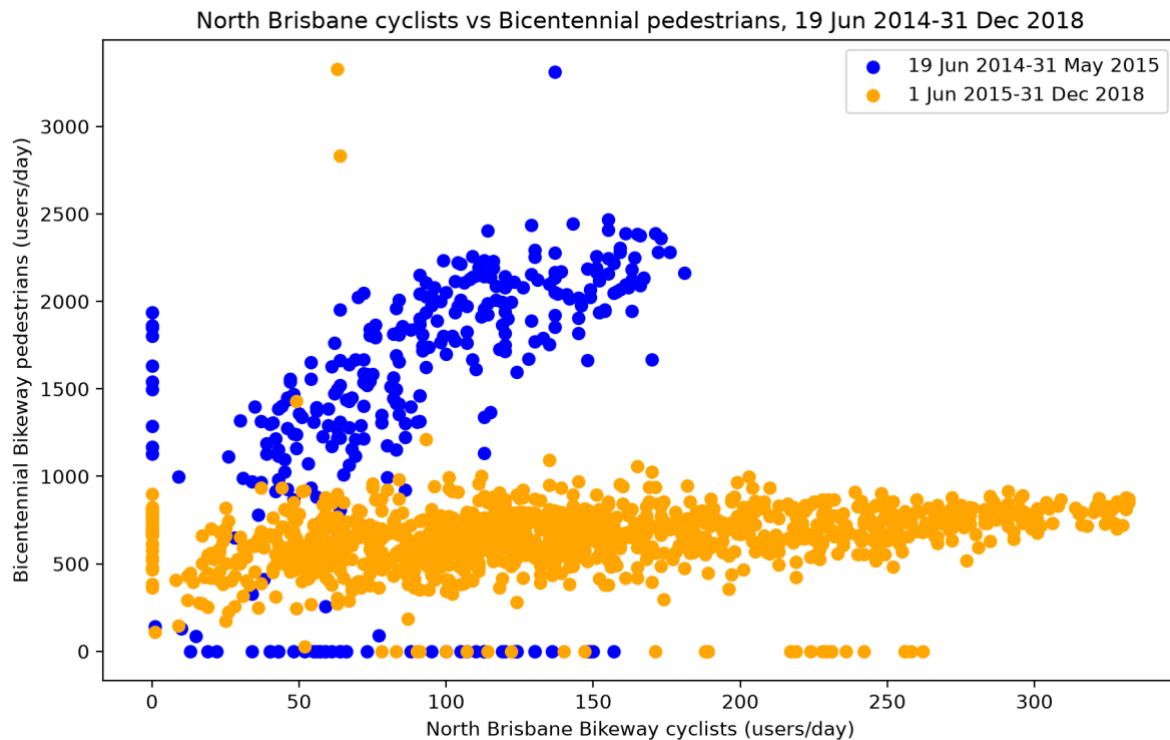


Figure 7: Daily North Brisbane Bikeway cyclists (x-axis) against Bicentennial Bikeway pedestrians (y-axis) in the supplied cleaned file, 19 June 2014 to 31 December 2018 (1,461 days). Blue points are before the mid-2015 Bicentennial counter change (19 Jun 2014–31 May 2015); orange points are from 1 Jun 2015. Generated by S2 Box 6.

S2.T3: Correlation Analysis Interpretation (400 words)

Using the correlation coefficient calculated in 2.1(a) and the visualisation produced in 2.2 above, what is your interpretation of the relationship between cyclist usage of North Brisbane Bikeway in Windsor and pedestrian usage on Bicentennial Bikeway in Brisbane between 2014-2018? Please use your visualisation, correlation coefficient outputs and data patterns over time to support your interpretation of bikeway usage.

(4 pts)

Hint: Module on - Correlation Analysis could be beneficial as you develop your response.

The pair named here is row (d) in 2.1, which is also the pair plotted in Figure 7.

Across all 1,461 days in the supplied cleaned file (19 June 2014 to 31 December 2018), North Brisbane cyclists and Bicentennial pedestrians show almost no linear relationship ($r = -0.028$, S2 Box 5). The negative sign is too small to describe as an inverse relationship.

Figure 7 shows that this near-zero value hides two separate groups. Before June 2015, most Bicentennial pedestrian counts sit between about 1,000 and 2,500 a day and rise with North Brisbane cycling ($r = 0.469$). From June 2015 they sit almost entirely below 1,000, with a weaker positive relationship ($r = 0.216$) (S2 Box 7). The correlation is also positive within every calendar year, from 0.109 in 2017 to 0.382 in 2015 (S2 Box 8).

The pooled coefficient is weak because the two series moved in opposite directions between these periods. Average Bicentennial pedestrians fell from 1,440 to 648 a day, while North Brisbane cyclists rose from 91 to 144 (S2 Box 7). Combining the two groups cancels the positive day-to-day association, which is the subgroup problem covered in Week 8. The pedestrian drop matches the mid-2015 change at the Bicentennial counter identified in Section 1, when recorded pedestrians fell and recorded cyclists rose by a similar amount, so it probably reflects how users were classified rather than fewer people walking. The same shift explains the negative coefficient in row (a) ($r = -0.355$).

Counter faults kept in the supplied file weaken the coefficient further. The line of points at zero pedestrians and the line at zero cyclists in Figure 7 are the 72 and 40 zero days identified as recording problems in Section 1 (S2 Box 9).

The positive relationship within each period most likely reflects influences that affect both paths on the same days, such as weekday commuting, holidays and weather, rather than cycling at Windsor changing walking on the Bicentennial Bikeway. Correlation does not show causation.

For BCC, the pooled coefficient should not be read as meaning the two paths are unrelated. Once the counter change is accounted for, daily use at the two sites rises and falls together to a weak-to-moderate degree. Neither count predicts the other well enough to replace direct counting, so each upgrade should be planned from that path's own corrected counts.

SECTION 3 – MULTIPLE LINEAR REGRESSION ANALYSIS

Using your cleaned data, please answer the following questions.

To answer Sections 2 and 3, please use the cleaned datasets (brisbane_bikeway_counters_cleaned.csv, brisbane_weather_cleaned.csv) provided in Canvas Assessment 2 – Details page.

BCC wants to further investigate how daily weather factors (i.e. daily rainfall, solar exposure, maximum temperature) influence usage of the Bicentennial Bikeway, Brisbane – and whether there is any difference in how weather influences cyclist versus pedestrian usage of this bikeway. To answer the questions in this section, please split the dataset as follows: use data from January 1st, 2014, to December 31st, 2017 (inclusive) as the training set, and use all remaining data as the testing set.

S3.T1: Multiple Regression Models & Supporting Visualisations

Please conduct Multiple Regression analyses to investigate if one or more daily weather factors (i.e. rainfall, solar exposure, maximum temperature) influence **pedestrian usage on the Bicentennial Bikeway, Brisbane**. Do not create separate regression models for each independent variable. Instead, build a single regression model that includes all the identified independent variables. You may need to repeat the regression modelling process until all variables have statistically significant p-values.

- Provide model summary table (including P-values, R^2 , adjusted R^2 , RMSE on train and test sets)
- Generate scatter plots illustrating the observed data points for each independent variable against the dependent variable and overlay the fitted regression line to demonstrate the model's representation of the relationship.
- Discuss Any model simplification (i.e. removal of independent variables from the model) with appropriate justification. Discuss the impact of simplifying the model on performance if any.

(3 pts)

Hint: Module on - Regression Analysis along with the subsequent MATLAB, Python, and Excel modules (depending on the tool that you are using), could be beneficial as you develop your response.

Pedestrian regression model

The cleaned bikeway and weather tables were joined by their Date index. The 1,145 observations from 19 June 2014 to 31 December 2017 form the training set; the 316 observations in 2018 form the unchanged testing set (S3 Boxes 2–3). Coefficients are daily users per unit of the listed factor. P-values test whether each coefficient differs from zero in the training model.

Measure	Initial three-factor model	Simplified model
Predictors	Rainfall, solar exposure, maximum temperature	Solar exposure and maximum temperature
Intercept	1278.58 (p <0.001)	1273.46 (p <0.001)
Daily rainfall	-2.67 (p 0.074)	Removed
Solar exposure	14.77 (p <0.001)	15.68 (p <0.001)
Maximum temperature	-25.60 (p <0.001)	-26.30 (p <0.001)
R ²	0.026	0.023
Adjusted R ²	0.023	0.022
F-statistic	10.15 (p <0.001)	13.61 (p <0.001)
Training RMSE	557.25	558.03
Testing RMSE	215.69	218.27

Simplified equation: predicted pedestrians = 1273.46 + 15.68(solar exposure) – 26.30(maximum temperature).

Rainfall was removed because p = 0.074 exceeded 0.05. This reduced R² by 0.003 and increased training and testing RMSE by 0.78 and 2.58 users respectively. The small changes support the simpler model (S3 Boxes 4–7).

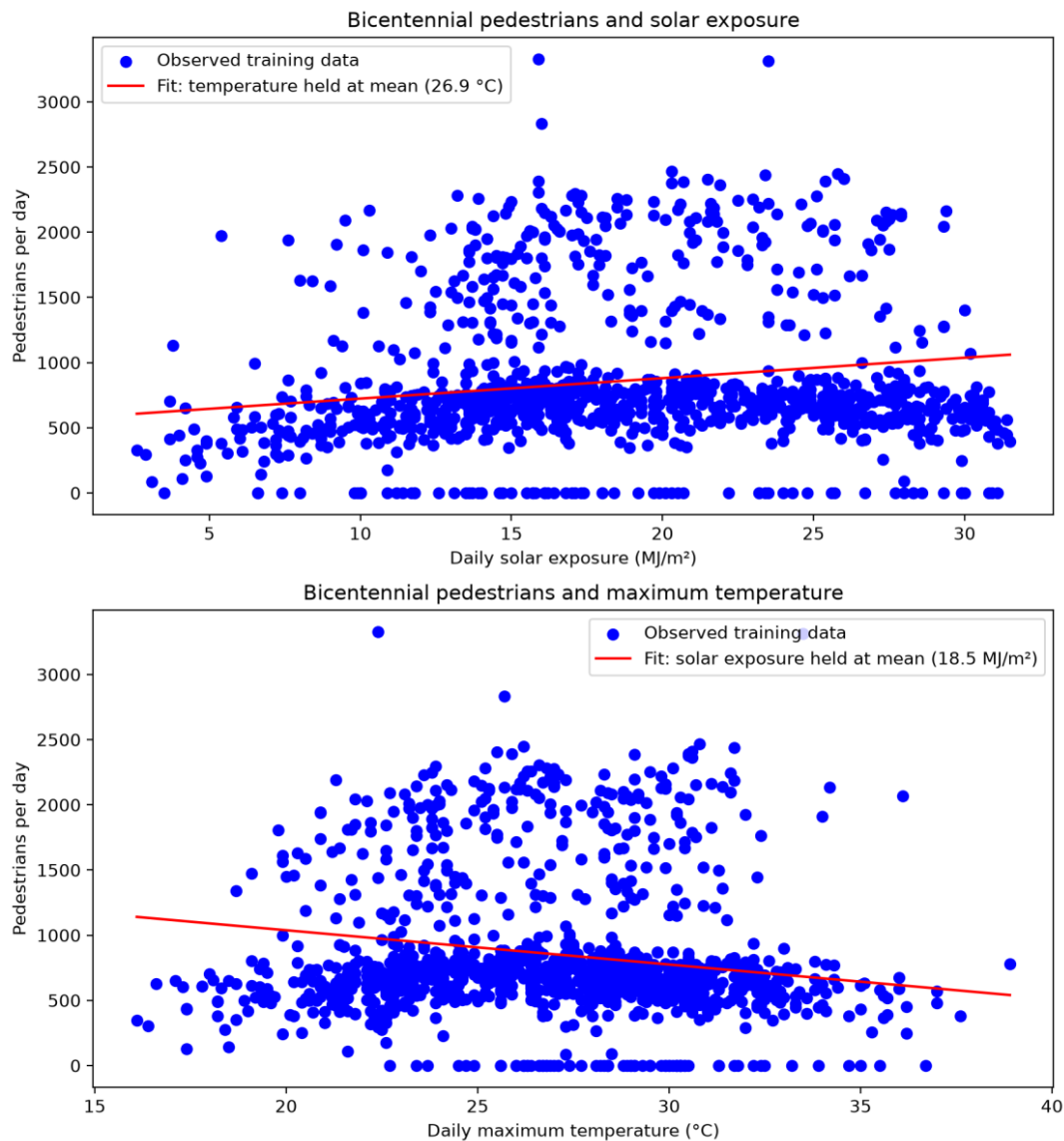


Figure 8: Observed pedestrian counts and fitted directions from the simplified multiple regression. Maximum temperature is held at its training mean of 26.9 °C in the solar-exposure plot; solar exposure is held at its training mean of 18.5 MJ/m² in the temperature plot. Both lines come from the same model (S3 Box 8).

Please conduct Multiple Regression analyses to investigate if one or more daily weather factors (i.e. rainfall, solar exposure, maximum temperature) influence **cyclist usage on the Bicentennial Bikeway, Brisbane**. Provide a model summary table (including P-values, R^2 , adjusted R^2 , RMSE on the test-set). You may need to repeat the regression modelling process until all variables have statistically significant p-values.

- Provide model summary table (including P-values, R^2 , adjusted R^2 , RMSE on train and test sets)
- Generate scatter plots illustrating the observed data points for each independent variable against the dependent variable and overlay the fitted regression line to demonstrate the model's representation of the relationship.
- Discuss Any model simplification (i.e. removal of independent variables from the model) with appropriate justification. Discuss the impact of simplifying the model on performance if any.

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(3 pts)

Hint: Module on - Regression Analysis along with the subsequent MATLAB, Python, and Excel modules (depending on the tool that you are using), could be beneficial as you develop your response.

Cyclist regression model

The cyclist models use the same training and testing dates as the pedestrian models, allowing their performance to be compared on the same observations.

Measure	Initial three-factor model	Simplified model
Predictors	Rainfall, solar exposure, maximum temperature	Rainfall and solar exposure
Intercept	911.38 (p <0.001)	919.58 (p <0.001)
Daily rainfall	-7.87 (p <0.001)	-7.87 (p <0.001)
Solar exposure	18.81 (p <0.001)	18.98 (p <0.001)
Maximum temperature	0.42 (p 0.955)	Removed
R ²	0.043	0.043
Adjusted R ²	0.040	0.041
F-statistic	16.94 (p <0.001)	25.43 (p <0.001)
Training RMSE	760.03	760.03
Testing RMSE	586.73	586.74

Simplified equation: predicted cyclists = 919.58 – 7.87(daily rainfall) + 18.98(solar exposure).

Maximum temperature was removed because p = 0.955. R² remains 0.043 and both RMSE values change by less than 0.01 user, so the variable adds no useful predictive information here (S3 Boxes 9–12).

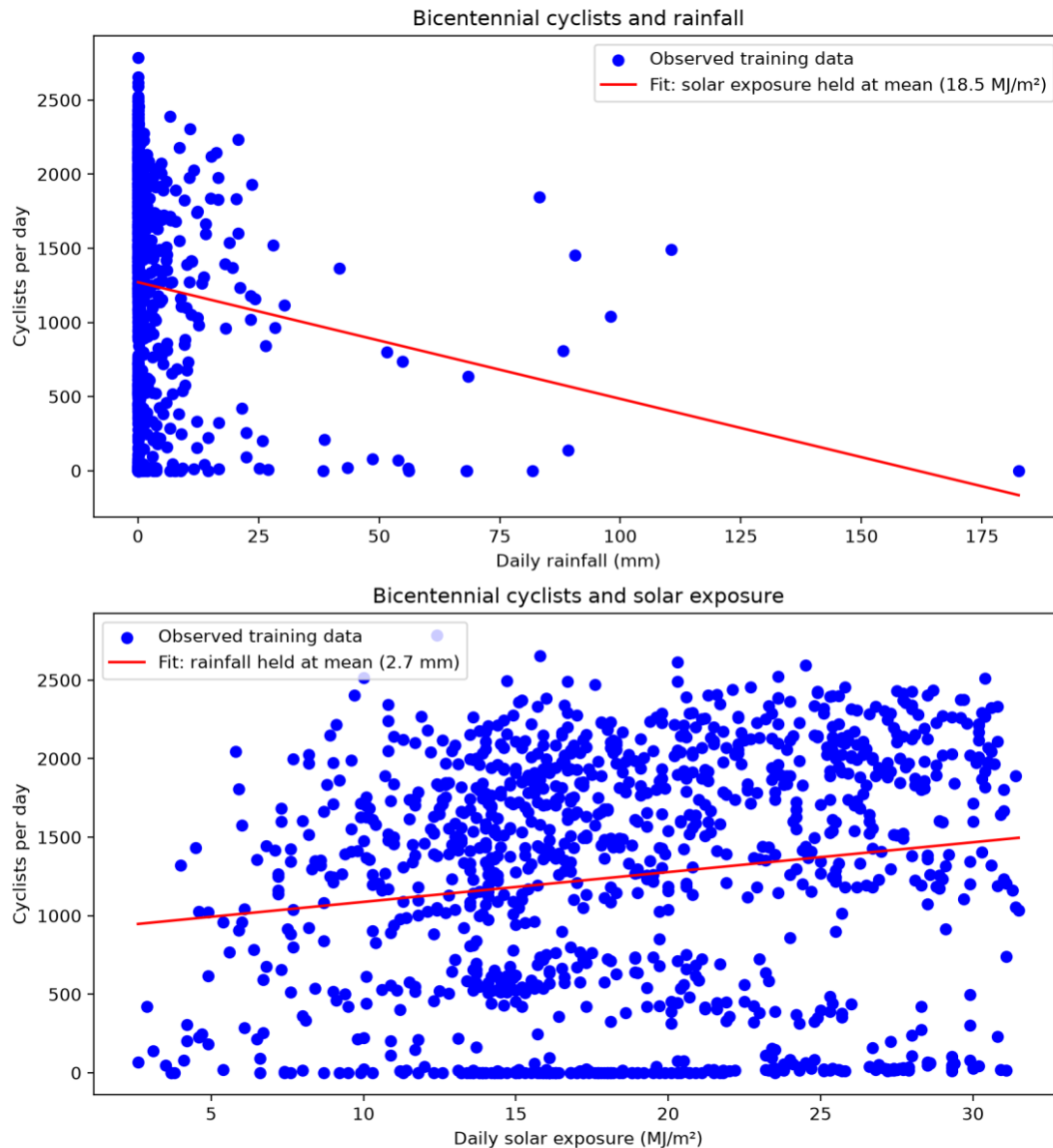


Figure 9: Observed cyclist counts and fitted directions from the simplified multiple regression. Solar exposure is held at its training mean of 18.5 MJ/m² in the rainfall plot; rainfall is held at its training mean of 2.7 mm in the solar-exposure plot. Both lines come from the same model (S3 Box 13).

S3.T2: Multiple Regression Interpretation (400 words)

What is your interpretation of the results produced from the two Multiple Regression Models above, and why? Specifically, provide:

- Predicted vs actual values plot on the test-set and discuss the predictive power of the regression models in S3.T1 (a) and (b).
- Discuss the suitability and the applicability of the model using statistical measures (e.g. regression coefficients, R^2 , adjusted R^2 , RMSE, and p-values).
- Discuss the reasons for any identified performance gap between training and testing, supported by appropriate plots that you have generated above.

(5 pts)

Hint: Module on - Regression Analysis could be beneficial as you develop your response.

Interpretation of the regression results

The pedestrian model retains solar exposure and maximum temperature after rainfall was removed ($p = 0.074 > 0.05$). Solar exposure has a positive coefficient of 15.68 pedestrians per MJ/m² and maximum temperature a negative coefficient of 26.30 pedestrians per °C, holding the other variable constant; both retained p-values are below 0.001. However, R^2 is 0.023 and adjusted R^2 is 0.022, so weather explains only 2.3% of pedestrian variation.

The cyclist model retains rainfall and solar exposure. Its coefficients indicate 7.87 fewer cyclists per millimetre of rainfall and 18.98 more cyclists per MJ/m² of solar exposure, holding the other factor constant; both p-values are below 0.001. R^2 is 0.043 and adjusted R^2 is 0.041, so only 4.3% of cyclist variation is explained. Figures 8 and 9 show the fitted directions, but observations remain widely dispersed. Statistical significance therefore does not imply accurate daily prediction or causation.

Simplification caused little performance loss. Removing rainfall from the pedestrian model reduced R^2 from 0.026 to 0.023 and increased training and testing RMSE from 557.25 to 558.03 and from 215.69 to 218.27 users. Removing maximum temperature from the cyclist model left R^2 at 0.043 and changed both RMSE values by less than 0.01 user. Both final models are significant overall ($F = 13.61$ and 25.43 , $p < 0.001$), so weather does better than predicting the mean, but only slightly.

Figures 10 and 11 show smoother predictions that miss daily peaks and troughs. In 2018, the pedestrian model predicts an average 852.6 users against 694.1 actual, while the cyclist model predicts 1,243.2 against 1,642.5 actual. Figure 12 therefore shows points spread from the perfect-prediction line. Testing RMSE is lower than training RMSE, but this is not stronger generalisation: pedestrian counts vary much more in training (SD 565) than testing (SD 153), while both models retain large mean biases.

The training data also combine the mid-2015 counter-classification change and recording faults identified in Section 1; 2018 is more stable. Weather-only models omit commuting patterns, weekdays, holidays and events. They are suitable for describing limited associations, not forecasting upgrade demand. BCC should use corrected counter data and test additional predictors before relying on regression estimates.

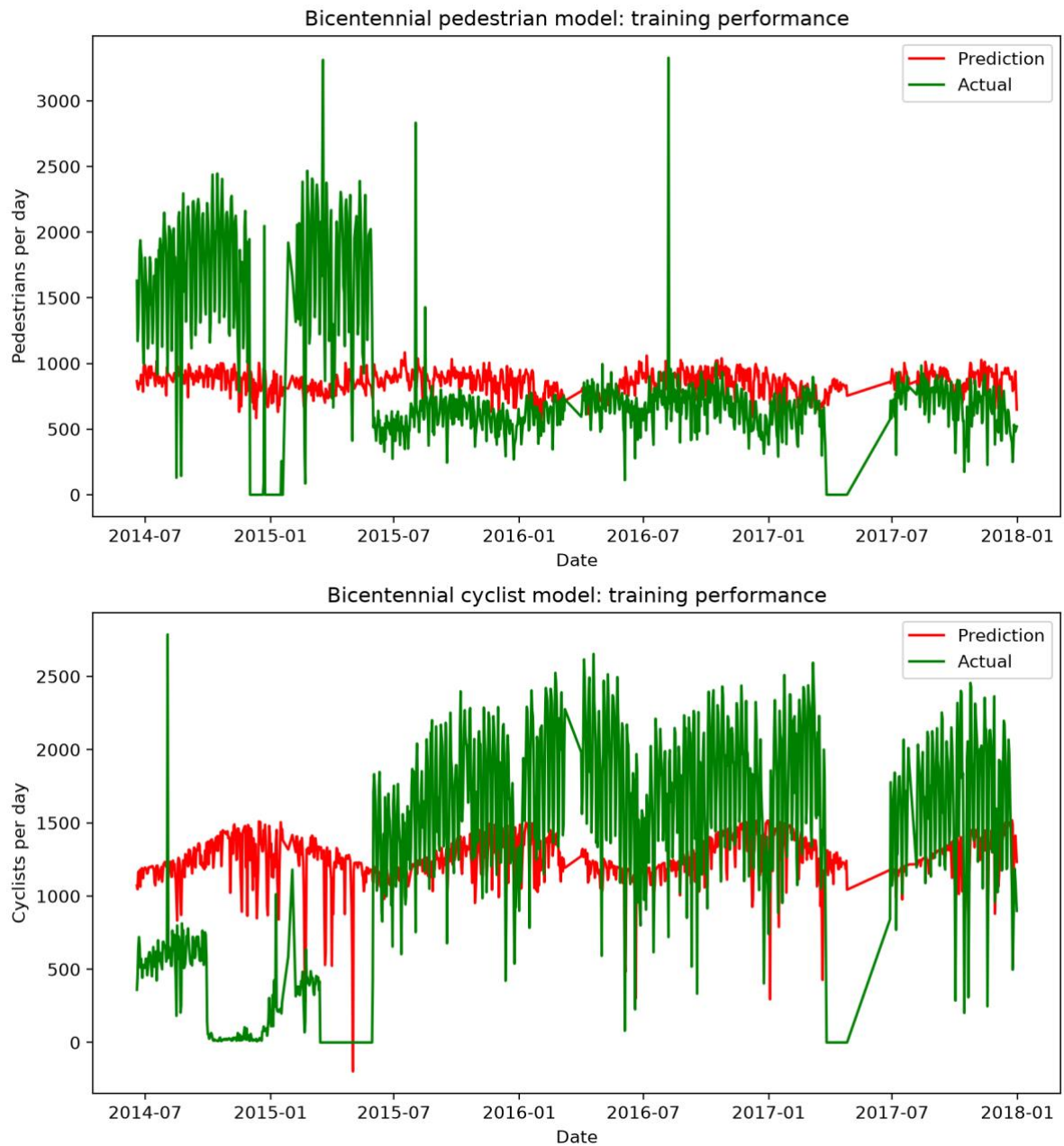


Figure 10: Actual and predicted pedestrian and cyclist counts in the training period, 19 June 2014–31 December 2017 (S3 Box 14).

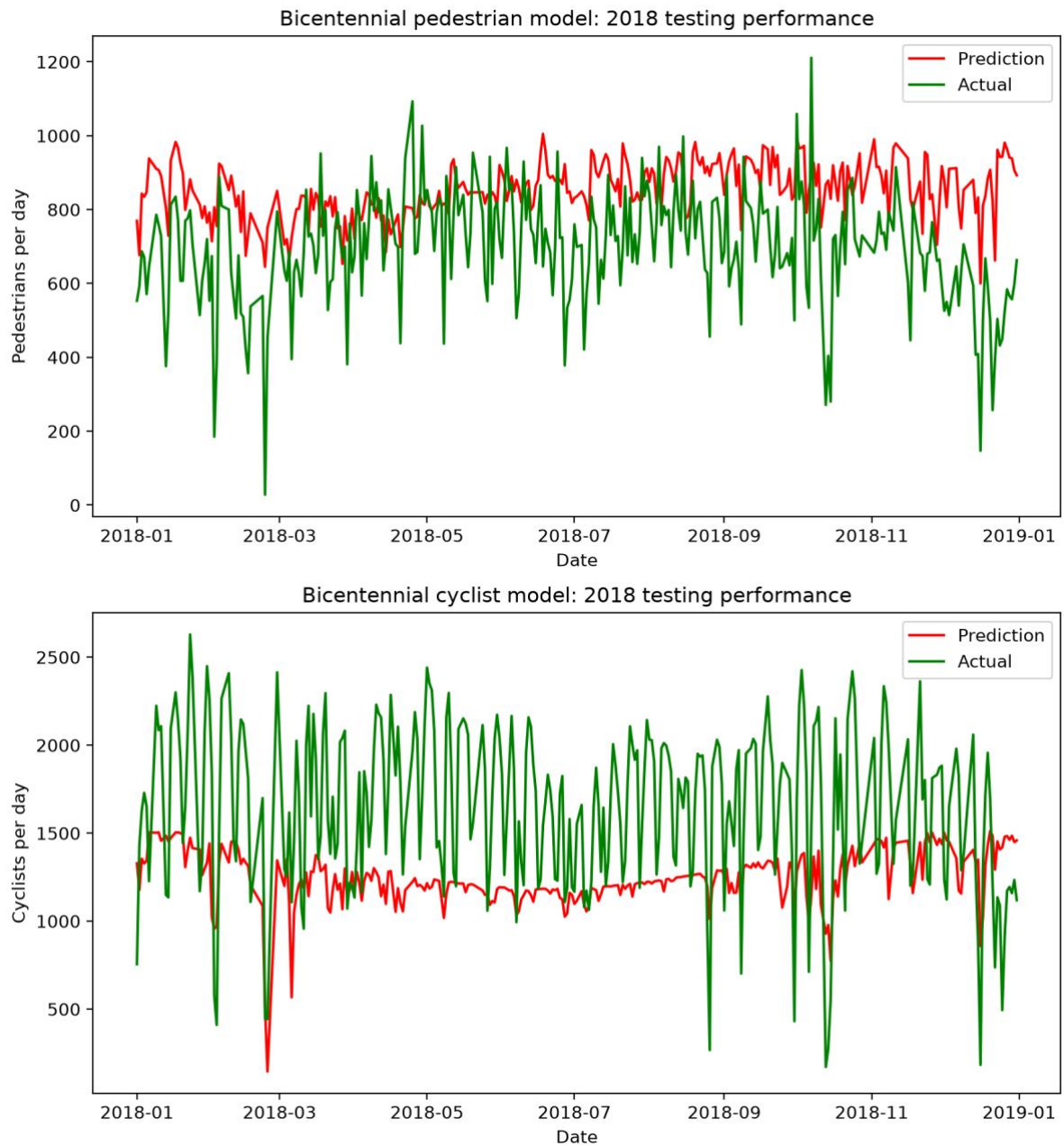


Figure 11: Actual and predicted pedestrian and cyclist counts in the 2018 testing period (S3 Box 15).

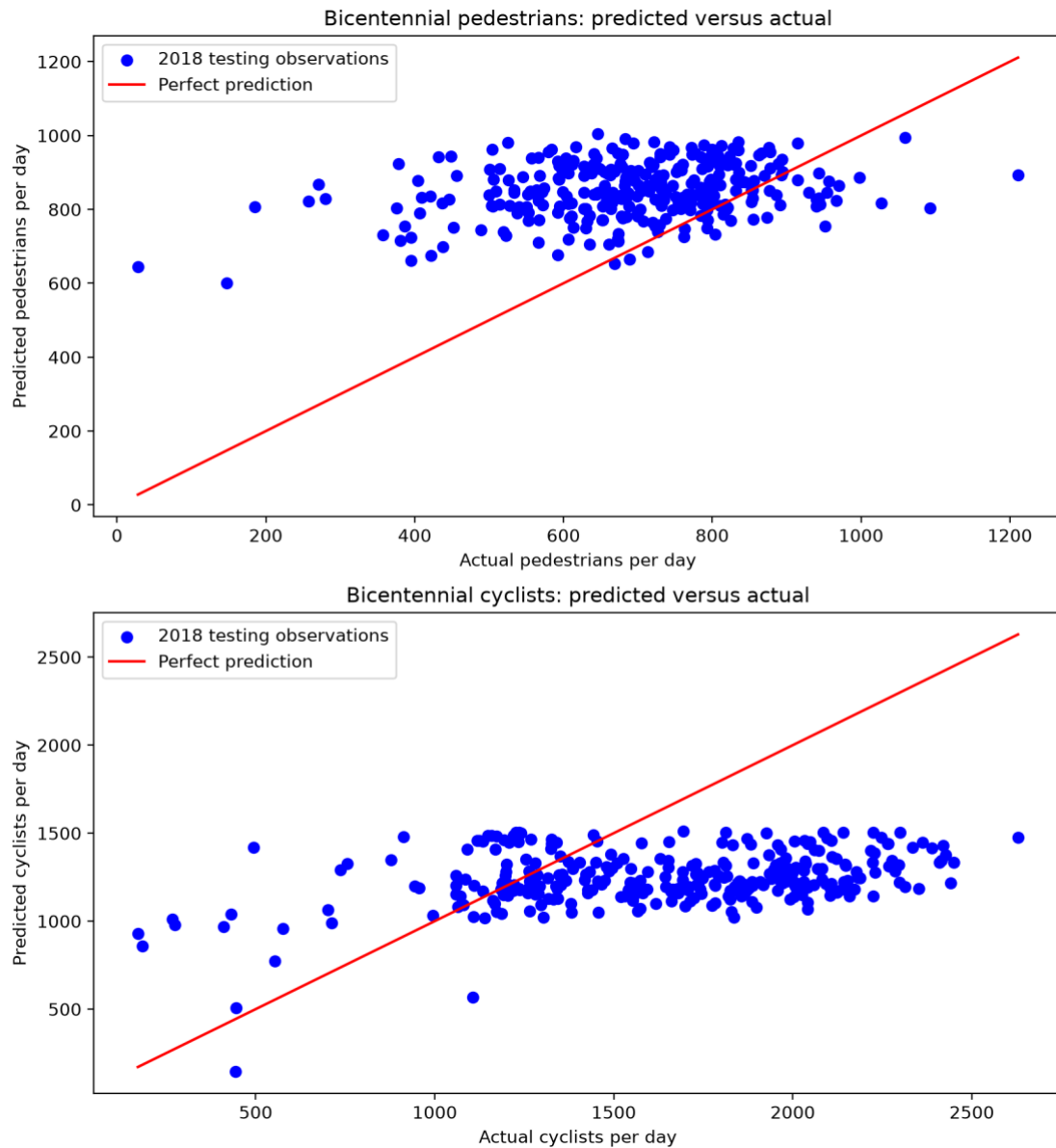


Figure 12: Predicted versus actual 2018 values. The red diagonal represents perfect prediction (S3 Box 16).

SECTION 4 – FINDING REAL-WORLD DATA OR SYNTHESISING DATA FOR PRELIMINARY RESEARCH ANALYSIS

In this section, one pathway must be selected. This response completes S4.TA: locating and justifying a publicly available real-world tabular dataset broadly related to the Assessment 1 research topic.

S4.TA Locating Real-World Dataset

Dataset title: Databases for synergistic multi-performance optimization of recycled aggregate concrete using explainable machine learning and NSGA-III

Selected file: compressive_strength_database.xlsx

Repository: Mendeley Data – <https://data.mendeley.com/datasets/3xvfyycp8d/1>

DOI: 10.17632/3xvfyycp8d.1

Licence: Creative Commons Attribution 4.0 International (CC BY 4.0)

Original dataset dimensions: 1,736 rows and 15 columns

Analysis subset for Section 5: 1,649 rows after retaining outlier_flag = 0, as directed by the dataset authors

Justification (235 words)

This publicly available dataset is useful as preliminary proxy data for my research on CO₂-cured recycled aggregate concrete (RAC). My Assessment 1 question asks whether controlled CO₂ curing alone achieves similar improvements to CO₂ curing combined with alkaline slurry or biochar. Although this dataset does not record CO₂-curing conditions or co-treatment agents, it contains 1,736 RAC mixture records and 11 quantitative mixture and aggregate predictors, with 28-day compressive strength as the target variable. The most relevant variables are RCA replacement ratio, RCA water absorption, crushing value and apparent density. These are the RCA properties that carbonation improves; for example, CO₂ curing reduced RCA water absorption from 5.23% to 3.85–4.46% (Benahmad et al., 2026). Cement, water, fly ash, ground granulated blast-furnace slag, coarse aggregate, sand and superplasticizer contents are also important mix-design factors that could confound treatment comparisons and should therefore be controlled in a future experiment. For Section 5, the 87 records flagged by the dataset authors as potential outliers will be excluded, leaving 1,649 observations. Correlations and scatter plots can then identify whether RCA properties and replacement level are associated with compressive strength and help prioritise variables for the proposed factorial study. However, the literature-derived records were harmonised from 116 publications and may retain differences in materials and test methods. Therefore, the analysis can guide experimental design but cannot determine whether CO₂ curing or a co-treatment agent causes strength improvement (Zhang et al., 2026).

Dataset variables

The table reports the ranges in the 1,649-record analysis subset (outlier_flag = 0). The non-analytical columns record_id and source_reference provide traceability; outlier_flag records the authors' potential-outlier classification.

Variable	Min	Max	Units
RCA apparent density (ρ -RCA)	2,100	2,880	kg/m ³
RCA crushing value (CV-RCA)	6	32	%
RCA water absorption (W-RCA)	0.4	10.1	%
Cement content	142	663	kg/m ³
Fly ash content (FA)	0	376	kg/m ³
Ground granulated blast-furnace slag (GGBS)	0	129	kg/m ³
Mixing-water content	107	304.9	kg/m ³
Sand content	330	952	kg/m ³
Total coarse aggregate content (CA)	824.1	1,427	kg/m ³
Recycled coarse aggregate replacement ratio (RAR)	0	100	%
Superplasticizer content (SP)	0	25	kg/m ³
28-day compressive strength (CS)	14.2	74.3	MPa

Connection to the proposed research

Preliminary analysis will examine how RCA replacement and aggregate-quality indicators relate to 28-day compressive strength. These relationships cannot test the direct effect of CO₂ curing or biochar/alkaline-slurry co-treatment, because those variables are absent. They can identify important baseline effects and confounders that should be held constant in the later factorial experiment comparing untreated RCA, controlled CO₂-only RCA and controlled CO₂ plus a co-treatment agent.

References

- Zhang, M., Liu, J., Chen, Y., Wang, Y., Kang, B., & Lu, D. (2026). Databases for synergistic multi-performance optimization of recycled aggregate concrete using explainable machine learning and NSGA-III (Version 1) [Data set]. Mendeley Data. <https://doi.org/10.17632/3xvfyycp8d.1>
- Benahmad, A., Klathae, T., Tangchirapat, W., Rattanachu, P., & Jaturapitakkul, C. (2026). Utilization of treated recycled aggregate by using carbonation incorporated with highly alkaline slurry technique for producing high strength concrete. *Results in Engineering*, 32, Article 111942. <https://doi.org/10.1016/j.rineng.2026.111942>

SECTION 5 – CORRELATION ANALYSIS USING PRELIMINARY DATA

Using the dataset you used in Section 4 (i.e. real-world data you located or the synthetic dataset you generated) answer the following questions.

S5.T1: Data Modelling Using Preliminary Data

- In your dataset, identify two or more variables that appear to be related (i.e., feature(s) and a target). Provide scatter plots and a correlation matrix.
- Describe the nature of their relationship (e.g., linear, inverse, nonlinear, or no clear pattern). Support your answer using outputs generated using relevant data analysis concepts (e.g., correlation coefficient, scatter plots, etc.).
- Compare the relationships you observe in your data with those reported in the relevant literature for your chosen topic.

(5 pts)

Analysis and results

The downloaded workbook was saved as compressive_strength_database.csv in CSV UTF-8 format so that it could be loaded using the Week 7 `pd.read_csv` method. Following the dataset authors' modelling instructions, the 87 records with `outlier_flag = 1` were excluded, leaving 1,649 observations. The target variable was 28-day compressive strength (CS). A water-to-binder ratio was calculated as water divided by the sum of cement, fly ash and ground granulated blast-furnace slag.

The RCA-property analysis was restricted to the 1,422 mixtures with $\text{RAR} > 0\%$. This removes 227 mixtures that contain no recycled coarse aggregate, because the recorded RCA water absorption, crushing value and density cannot directly affect a mixture in which RCA was not used. Pearson correlation coefficients and scatter plots were then used to examine the relationships between the selected features and compressive strength (S5 Boxes 3–8).

Correlation matrix

Variable	ρ -RCA	CV-RCA	W-RCA	RAR	w/b	CS
ρ -RCA	1.000	-0.235	-0.201	0.105	0.034	0.107
CV-RCA	-0.235	1.000	0.298	0.019	0.100	-0.238
W-RCA	-0.201	0.298	1.000	0.027	0.141	-0.350
RAR	0.105	0.019	0.027	1.000	0.066	-0.082
w/b	0.034	0.100	0.141	0.066	1.000	-0.569
CS	0.107	-0.238	-0.350	-0.082	-0.569	1.000

Table 6: Pearson correlation matrix for the 1,422 unflagged mixtures containing RCA. ρ -RCA = RCA apparent density; CV-RCA = crushing value; W-RCA = water absorption; RAR = RCA replacement ratio; w/b = water-to-binder ratio; CS = 28-day compressive strength. Generated by S5 Box 4.

Scatter plots

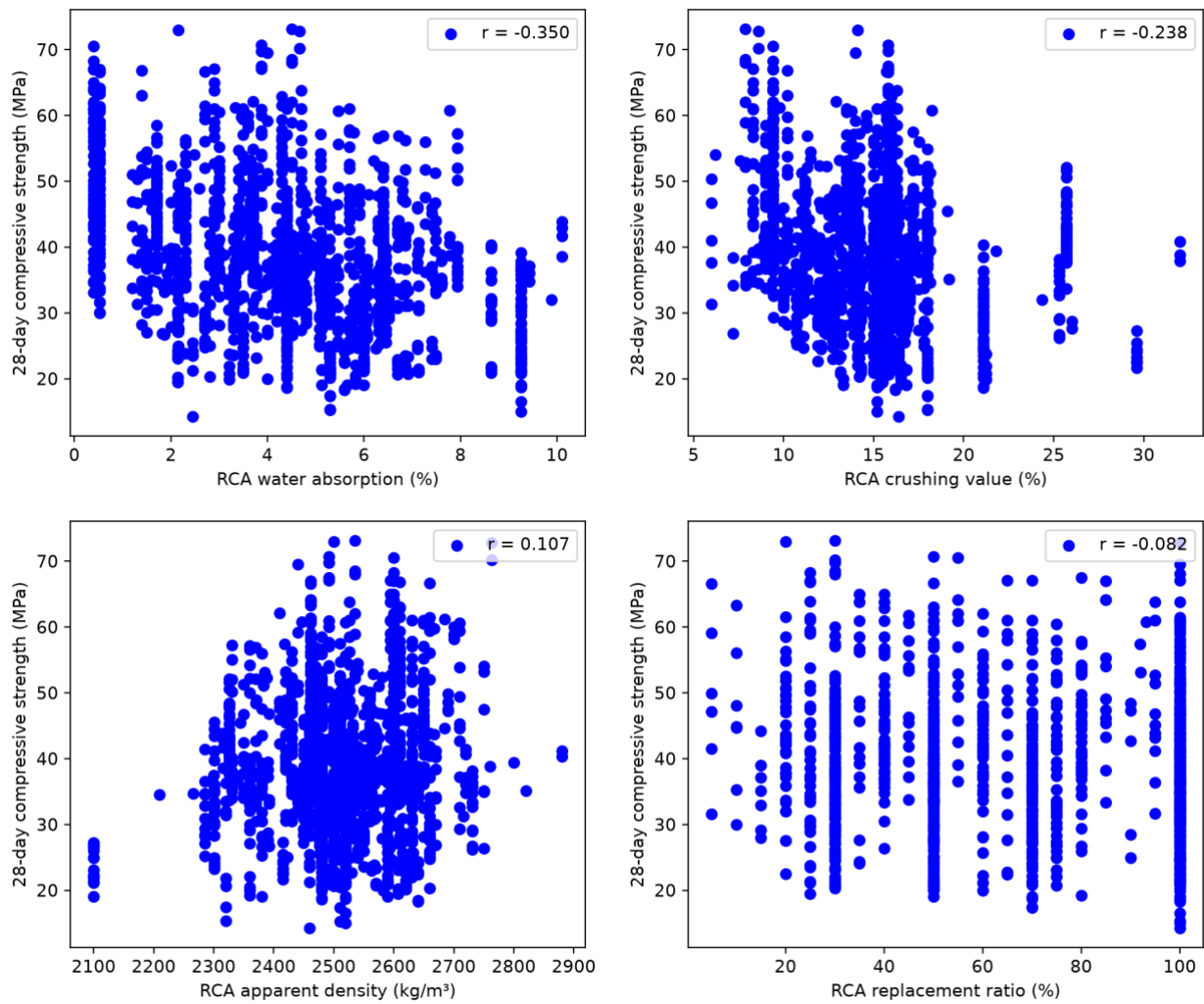


Figure 13: RCA water absorption, crushing value, apparent density and replacement ratio against 28-day compressive strength for the 1,422 unflagged mixtures containing RCA. The legends show Pearson r . Generated by S5 Box 7.

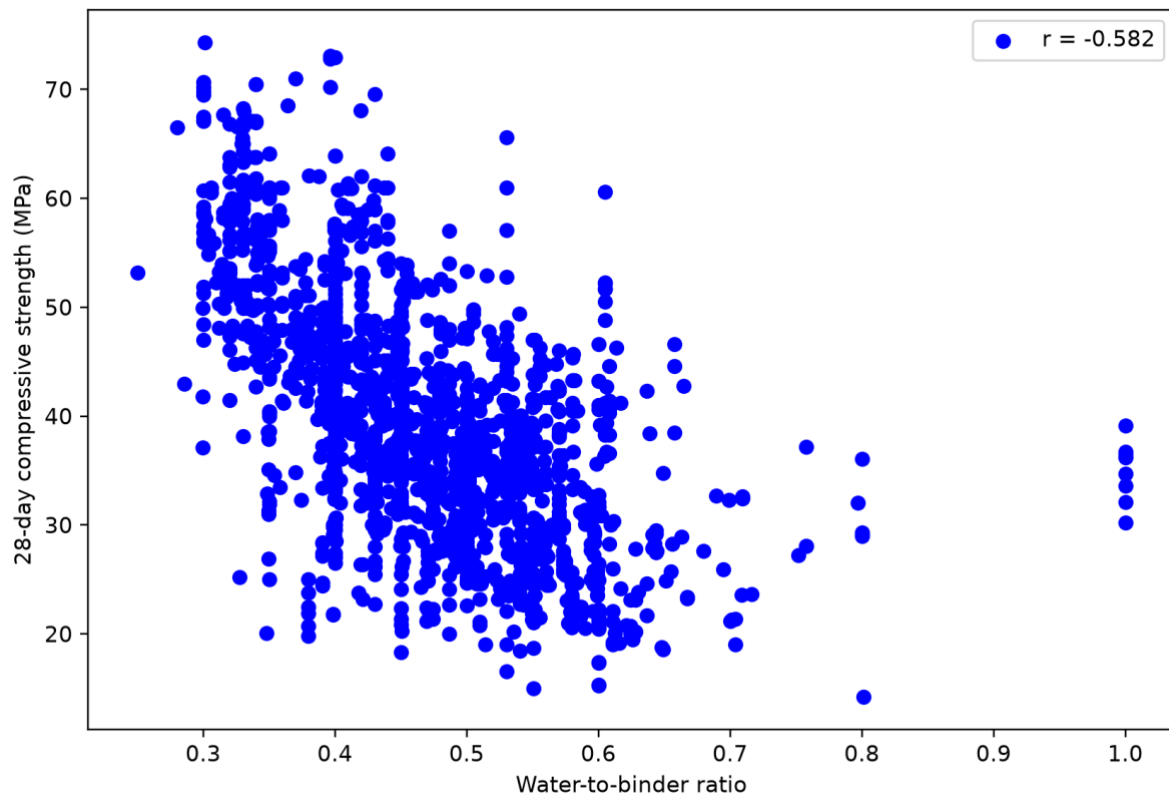


Figure 14: Water-to-binder ratio against 28-day compressive strength for all 1,649 unflagged mixtures. The legend shows Pearson r . Generated by S5 Box 8.

Interpretation and comparison with literature

Water-to-binder ratio had the strongest relationship with compressive strength. It showed a moderate-to-strong inverse correlation across all unflagged mixtures ($r = -0.582$) and within the RCA-containing subset ($r = -0.569$). Figure 14 shows an approximately linear downward pattern, although substantial scatter remains. This means mixes containing more water relative to their binder generally had lower strength, but the ratio alone does not determine the result.

Among the RCA properties, water absorption had the clearest inverse relationship with strength ($r = -0.350$), followed by crushing value ($r = -0.238$). Apparent density had a very weak positive relationship ($r = 0.107$), while RCA replacement ratio had a very weak inverse relationship ($r = -0.082$). The four scatter plots show broad vertical bands rather than narrow lines because many records share the same reported aggregate property or replacement value. No clear nonlinear curve is visible. The weak individual RCA correlations should therefore not be read as evidence that RCA quality is unimportant. Mix design and differences between source studies also affect strength.

The full unflagged dataset supports that interpretation. Superplasticizer and cement content were positively correlated with strength ($r = 0.456$ and $r = 0.358$), while mixing water was negatively correlated ($r = -0.331$) (S5 Box 6). These relationships are as large as, or larger than, the RCA-property correlations. Comparisons of carbonation treatments must therefore control water-to-binder ratio, cement and admixture content rather than attributing every strength difference to the treatment.

The inverse water-absorption relationship agrees with the proposed mechanism in the literature. Carbonation can fill pores in the attached mortar and densify the interfacial region. Benahmad et al.

(2026) reduced RCA water absorption from 5.23% to 3.85–4.46%. The same study reported a 6.0% strength increase from controlled CO₂ curing alone and an 11.0–13.3% increase when alkaline slurry was added. Other reported strength changes range from about 3% for an uncontrolled dry-ice treatment (Shehadah et al., 2026) to about 16–18% for controlled carbonation (Jamil et al., 2025). This variation is consistent with the database scatter and indicates that curing conditions, aggregate source and mix proportions influence the result.

The preliminary analysis narrows the proposed study. RCA water absorption and crushing value should be measured before and after treatment, while water-to-binder ratio, cement content, superplasticizer, RCA replacement, aggregate source and curing conditions should be held constant. The experiment can then compare untreated RCA, controlled CO₂-only RCA and controlled CO₂ plus alkaline slurry or biochar. Because the database contains no carbonation or co-treatment variables and combines records from 116 publications, these correlations guide variable selection but cannot establish that a treatment causes higher strength.

Additional references for Section 5

Jamil, S., Idrees, M., Akbar, A., & Barbhuiya, S. (2025). Optimizing concrete properties with carbonated recycled concrete aggregate: A sustainable approach. *Archives of Civil and Mechanical Engineering*, 25, Article 266. <https://doi.org/10.1007/s43452-025-01319-y>

Shehadah, M., Mukhtar, F., Ibrahim, M., & Hanif, A. (2026). Evaluating the influence of treatment techniques of recycled aggregate on properties of concrete towards sustainable development. *Case Studies in Construction Materials*, 25, Article e06278. <https://doi.org/10.1016/j.cscm.2026.e06278>

APPENDIX

Python source code and outputs used for Section 1 are reproduced below. Box numbers match the evidence cited in the answers. The six figures are included both in the report and beside the corresponding plotting boxes.

S1.T1 Python notebook evidence

Box 1 — Code cell

```
# Box 1 - Import the libraries
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
```

Box 2 — Code cell

```
# Box 2 - Load the raw bikeway data and check its structure
df = pd.read_csv('brisbane_bikeway_counters (1).csv', header=0)
df.info()

df['Date'] = pd.to_datetime(df['Date'], format='%Y-%m-%d')

print("Shape : ", df.shape)
print("Rows : ", df.shape[0])
print("Columns : ", df.shape[1])
```

Output

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1826 entries, 0 to 1825
Data columns (total 5 columns):
#   Column                                Non-Null Count  Dtype
---  -
0   Date                                1826 non-null   str
1   Bicentennial Bikeway Pedestrians    1731 non-null   float64
2   Bicentennial Bikeway Cyclists       1731 non-null   float64
3   North Brisbane Bikeway Pedestrians  1656 non-null   float64
4   North Brisbane Bikeway Cyclists     1656 non-null   float64
dtypes: float64(4), str(1)
memory usage: 71.5+ KB
Shape : (1826, 5)
Rows : 1826
Columns : 5
```

Box 3 — Code cell

```
# Box 3 - First five rows
df.head()
```

Output

	Date	Bicentennial Bikeway Pedestrians	Bicentennial Bikeway Cyclists	North Brisbane Bikeway Pedestrians	North Brisbane Bikeway Cyclists
0	2014-01-01	866.0	569.0	NaN	NaN
1	2014-01-02	1372.0	658.0	NaN	NaN
2	2014-01-03	1163.0	463.0	NaN	NaN
3	2014-01-04	770.0	410.0	NaN	NaN
4	2014-01-05	1153.0	654.0	NaN	NaN

Box 4 — Code cell

```
# Box 4 - Last five rows
df.tail()
```

Output

	Date	Bicentennial Bikeway Pedestrians	Bicentennial Bikeway Cyclists	North Brisbane Bikeway Pedestrians	North Brisbane Bikeway Cyclists
1821	2018-12-27	584.0	1172.0	176.0	23.0
1822	2018-12-28	566.0	1194.0	164.0	23.0
1823	2018-12-29	557.0	1159.0	131.0	20.0
1824	2018-12-30	598.0	1234.0	146.0	32.0
1825	2018-12-31	663.0	1119.0	131.0	17.0

Box 5 — Code cell

```
# Box 5 - Reload with Date as the index and get summary statistics (quartiles for the IQR rule)
df = pd.read_csv('brisbane_bikeway_counters (1).csv', header=0, index_col='Date')
df.index = pd.to_datetime(df.index)

df.describe()
```

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Output

	Bicentennial Bikeway Pedestrians	Bicentennial Bikeway Cyclists	North Brisbane Bikeway Pedestrians	North Brisbane Bikeway Cyclists
count	1731.000000	1731.000000	1656.000000	1656.000000
mean	969.316002	1277.454073	85.371981	85.371981
std	821.610771	740.759388	76.545438	76.545438
min	0.000000	0.000000	0.000000	0.000000
25%	592.000000	654.500000	37.000000	37.000000
50%	735.000000	1382.000000	75.000000	75.000000
75%	1102.000000	1918.500000	120.000000	120.000000
max	12288.000000	2786.000000	670.000000	670.000000

Box 6 — Code cell

```
# Box 6 - Part (a): Box plots of daily bikeway usage
fig, axs = plt.subplots(2, 2, figsize=(12, 11))

axs[0, 0].boxplot([df['Bicentennial Bikeway Cyclists'][np.isfinite(df['Bicentennial Bikeway Cyclists'])]])
axs[0, 0].set_title('Bicentennial Bikeway (Brisbane) - Cyclists')
axs[0, 0].set_xlabel('Bicentennial Bikeway cyclists')
axs[0, 0].set_ylabel('Number of users per day')

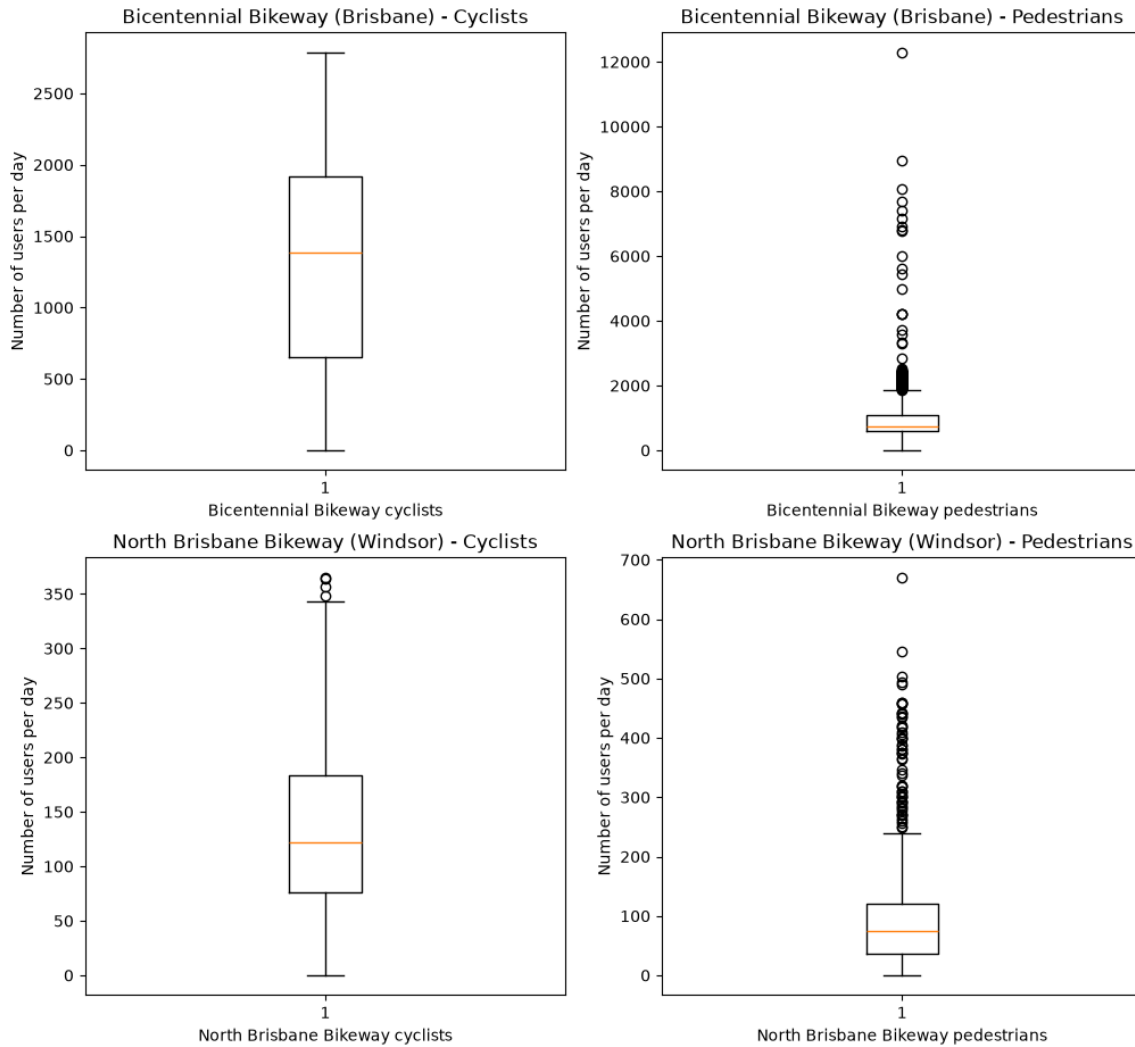
axs[0, 1].boxplot([df['Bicentennial Bikeway Pedestrians'][np.isfinite(df['Bicentennial Bikeway Pedestrians'])]])
axs[0, 1].set_title('Bicentennial Bikeway (Brisbane) - Pedestrians')
axs[0, 1].set_xlabel('Bicentennial Bikeway pedestrians')
axs[0, 1].set_ylabel('Number of users per day')

axs[1, 0].boxplot([df['North Brisbane Bikeway Cyclists'][np.isfinite(df['North Brisbane Bikeway Cyclists'])]])
axs[1, 0].set_title('North Brisbane Bikeway (Windsor) - Cyclists')
axs[1, 0].set_xlabel('North Brisbane Bikeway cyclists')
axs[1, 0].set_ylabel('Number of users per day')

axs[1, 1].boxplot([df['North Brisbane Bikeway Pedestrians'][np.isfinite(df['North Brisbane Bikeway Pedestrians'])]])
axs[1, 1].set_title('North Brisbane Bikeway (Windsor) - Pedestrians')
axs[1, 1].set_xlabel('North Brisbane Bikeway pedestrians')
axs[1, 1].set_ylabel('Number of users per day')

plt.show()
```

Output



Box 7 — Code cell

```
# Box 7 - Part (b): Line graph of daily bikeway usage
plt.figure(figsize=(14, 6))
plt.plot(df['Bicentennial Bikeway Cyclists'], label='Bicentennial Bikeway Cyclists')
plt.plot(df['Bicentennial Bikeway Pedestrians'], label='Bicentennial Bikeway Pedestrians')
plt.plot(df['North Brisbane Bikeway Cyclists'], label='North Brisbane Bikeway Cyclists')
plt.plot(df['North Brisbane Bikeway Pedestrians'], label='North Brisbane Bikeway Pedestrians')
plt.title('Daily cyclist and pedestrian counts, 2014-2018')
plt.xlabel('Date')
plt.ylabel('Number of users per day')
plt.legend()
plt.show()
```

Bicentennial Bikeway Pedestrians - missing days			
Bicentennial Bikeway Pedestrians	Bicentennial Bikeway Cyclists	North Brisbane Bikeway Pedestrians	North Brisbane Bikeway Cyclists
Date			
2016-03-09	NaN	NaN	2.0
107.0			
2016-03-10	NaN	NaN	15.0
145.0			
2016-03-11	NaN	NaN	0.0
129.0			
2016-03-12	NaN	NaN	0.0
60.0			
2016-03-13	NaN	NaN	0.0
46.0			
...	...	...	...
...			
2017-08-01	NaN	NaN	182.0
217.0			
2017-08-02	NaN	NaN	176.0
201.0			
2017-08-03	NaN	NaN	127.0
188.0			
2017-08-04	NaN	NaN	124.0
158.0			
2017-08-05	NaN	NaN	68.0
107.0			
[95 rows x 4 columns]			
(95, 4)			
Bicentennial Bikeway Cyclists - missing days			
Bicentennial Bikeway Pedestrians	Bicentennial Bikeway Cyclists	North Brisbane Bikeway Pedestrians	North Brisbane Bikeway Cyclists
Date			
2016-03-09	NaN	NaN	2.0
107.0			
2016-03-10	NaN	NaN	15.0
145.0			
2016-03-11	NaN	NaN	0.0
129.0			
2016-03-12	NaN	NaN	0.0
60.0			
2016-03-13	NaN	NaN	0.0
46.0			

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```

...
...
2017-08-01      NaN      NaN      182.0
217.0
2017-08-02      NaN      NaN      176.0
201.0
2017-08-03      NaN      NaN      127.0
188.0
2017-08-04      NaN      NaN      124.0
158.0
2017-08-05      NaN      NaN      68.0
107.0

[95 rows x 4 columns]
(95, 4)
North Brisbane Bikeway Pedestrians - missing days
      Bicentennial Bikeway Pedestrians  Bicentennial Bikeway Cyclists  North Brisbane Bikeway Pedestrians  North
Brisbane Bikeway Cyclists
Date
2014-01-01      866.0      569.0      NaN
NaN
2014-01-02      1372.0      658.0      NaN
NaN
2014-01-03      1163.0      463.0      NaN
NaN
2014-01-04      770.0      410.0      NaN
NaN
2014-01-05      1153.0      654.0      NaN
NaN
...
...
2014-06-15      1332.0      631.0      NaN
NaN
2014-06-16      1917.0      598.0      NaN
NaN
2014-06-17      2017.0      559.0      NaN
NaN
2014-06-18      2028.0      594.0      NaN
NaN
2018-07-03      785.0      1899.0      NaN
NaN

[170 rows x 4 columns]
(170, 4)
North Brisbane Bikeway Cyclists - missing days
      Bicentennial Bikeway Pedestrians  Bicentennial Bikeway Cyclists  North Brisbane Bikeway Pedestrians  North
Brisbane Bikeway Cyclists
Date
2014-01-01      866.0      569.0      NaN
NaN
2014-01-02      1372.0      658.0      NaN
NaN
2014-01-03      1163.0      463.0      NaN
NaN
2014-01-04      770.0      410.0      NaN
NaN
2014-01-05      1153.0      654.0      NaN
NaN
...
...
2014-06-15      1332.0      631.0      NaN
NaN
2014-06-16      1917.0      598.0      NaN
NaN
2014-06-17      2017.0      559.0      NaN
NaN
2014-06-18      2028.0      594.0      NaN
NaN
2018-07-03      785.0      1899.0      NaN
NaN

[170 rows x 4 columns]
(170, 4)

```

Box 9 — Code cell

```

# Box 9 - Part (c): Start and end of each missing period
print('Bicentennial: missing 9-31 Mar 2016')
print(df.iloc[797:822, :])

print('Bicentennial: zeros from 26 Mar 2017, then missing from 27 Apr 2017')
print(df.iloc[1178:1214, :])

print('Bicentennial: missing period ends 27 Jun 2017')
print(df.iloc[1270:1276, :])

print('Bicentennial: missing 27 Jul-5 Aug 2017')
print(df.iloc[1302:1314, :])

print('North Brisbane: missing until 18 Jun 2014, then zeros 19-28 Jun 2014')
print(df.iloc[165:182, :])

print('North Brisbane: pedestrians zero 11 Jun-2 Jul 2018; cyclists only 34 on 30 Jun, zero 1-2 Jul; both missing 3 Jul')
print(df.iloc[1620:1646, :])

```

EGH404 – Assessment 2 Template

Output

Bicentennial: missing 9-31 Mar 2016

	Bicentennial Bikeway Pedestrians	Bicentennial Bikeway Cyclists	North Brisbane Bikeway Pedestrians	North Brisbane Bikeway Cyclists
Date				
2016-03-08	726.0	2276.0	35.0	
134.0				
2016-03-09	NaN	NaN	2.0	
107.0				
2016-03-10	NaN	NaN	15.0	
145.0				
2016-03-11	NaN	NaN	0.0	
129.0				
2016-03-12	NaN	NaN	0.0	
60.0				
2016-03-13	NaN	NaN	0.0	
46.0				
2016-03-14	NaN	NaN	5.0	
157.0				
2016-03-15	NaN	NaN	59.0	
181.0				
2016-03-16	NaN	NaN	17.0	
150.0				
2016-03-17	NaN	NaN	12.0	
151.0				
2016-03-18	NaN	NaN	22.0	
117.0				
2016-03-19	NaN	NaN	31.0	
67.0				
2016-03-20	NaN	NaN	34.0	
52.0				
2016-03-21	NaN	NaN	63.0	
122.0				
2016-03-22	NaN	NaN	59.0	
151.0				
2016-03-23	NaN	NaN	85.0	
144.0				
2016-03-24	NaN	NaN	63.0	
117.0				
2016-03-25	NaN	NaN	27.0	
72.0				
2016-03-26	NaN	NaN	21.0	
51.0				
2016-03-27	NaN	NaN	20.0	
18.0				
2016-03-28	NaN	NaN	47.0	
66.0				
2016-03-29	NaN	NaN	84.0	
151.0				
2016-03-30	NaN	NaN	55.0	
140.0				
2016-03-31	NaN	NaN	82.0	
181.0				
2016-04-01	594.0	1982.0	15.0	
133.0				

Bicentennial: zeros from 26 Mar 2017, then missing from 27 Apr 2017

	Bicentennial Bikeway Pedestrians	Bicentennial Bikeway Cyclists	North Brisbane Bikeway Pedestrians	North Brisbane Bikeway Cyclists
Date				
2017-03-24	488.0	1827.0	88.0	
173.0				
2017-03-25	382.0	1101.0	61.0	
87.0				
2017-03-26	0.0	0.0	75.0	
114.0				
2017-03-27	0.0	0.0	112.0	
256.0				
2017-03-28	0.0	0.0	131.0	
262.0				
2017-03-29	0.0	0.0	100.0	
217.0				
2017-03-30	0.0	0.0	21.0	
34.0				
2017-03-31	0.0	0.0	75.0	
96.0				
2017-04-01	0.0	0.0	74.0	
107.0				
2017-04-02	0.0	0.0	83.0	
122.0				
2017-04-03	0.0	0.0	133.0	
256.0				
2017-04-04	0.0	0.0	138.0	
258.0				
2017-04-05	0.0	0.0	127.0	
236.0				
2017-04-06	0.0	0.0	106.0	
188.0				
2017-04-07	0.0	0.0	107.0	
228.0				
2017-04-08	0.0	0.0	61.0	
91.0				
2017-04-09	0.0	0.0	76.0	
114.0				
2017-04-10	0.0	0.0	160.0	
231.0				
2017-04-11	0.0	0.0	154.0	

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242.0			
2017-04-12	0.0	0.0	136.0
229.0			
2017-04-13	0.0	0.0	119.0
171.0			
2017-04-14	0.0	0.0	141.0
100.0			
2017-04-15	0.0	0.0	78.0
90.0			
2017-04-16	0.0	0.0	64.0
83.0			
2017-04-17	0.0	0.0	77.0
78.0			
2017-04-18	0.0	0.0	142.0
189.0			
2017-04-19	0.0	0.0	140.0
231.0			
2017-04-20	0.0	0.0	126.0
219.0			
2017-04-21	0.0	0.0	124.0
180.0			
2017-04-22	0.0	0.0	52.0
64.0			
2017-04-23	0.0	0.0	70.0
140.0			
2017-04-24	0.0	0.0	101.0
189.0			
2017-04-25	0.0	0.0	104.0
147.0			
2017-04-26	0.0	0.0	135.0
224.0			
2017-04-27	NaN	NaN	126.0
213.0			
2017-04-28	NaN	NaN	114.0
188.0			
Bicentennial: missing period ends 27 Jun 2017			
Bicentennial Bikeway Pedestrians	Bicentennial Bikeway Cyclists	North Brisbane Bikeway Pedestrians	North
Brisbane Bikeway Cyclists			
Date			
2017-06-24	NaN	NaN	65.0
97.0			
2017-06-25	NaN	NaN	89.0
75.0			
2017-06-26	NaN	NaN	137.0
195.0			
2017-06-27	NaN	NaN	159.0
187.0			
2017-06-28	572.0	840.0	123.0
198.0			
2017-06-29	717.0	1777.0	112.0
170.0			
Bicentennial: missing 27 Jul-5 Aug 2017			
Bicentennial Bikeway Pedestrians	Bicentennial Bikeway Cyclists	North Brisbane Bikeway Pedestrians	North
Brisbane Bikeway Cyclists			
Date			
2017-07-26	822.0	1911.0	149.0
202.0			
2017-07-27	NaN	NaN	121.0
219.0			
2017-07-28	NaN	NaN	142.0
162.0			
2017-07-29	NaN	NaN	75.0
92.0			
2017-07-30	NaN	NaN	73.0
108.0			
2017-07-31	NaN	NaN	172.0
196.0			
2017-08-01	NaN	NaN	182.0
217.0			
2017-08-02	NaN	NaN	176.0
201.0			
2017-08-03	NaN	NaN	127.0
188.0			
2017-08-04	NaN	NaN	124.0
158.0			
2017-08-05	NaN	NaN	68.0
107.0			
2017-08-06	763.0	1195.0	70.0
113.0			
North Brisbane: missing until 18 Jun 2014, then zeros 19-28 Jun 2014			
Bicentennial Bikeway Pedestrians	Bicentennial Bikeway Cyclists	North Brisbane Bikeway Pedestrians	North
Brisbane Bikeway Cyclists			
Date			
2014-06-15	1332.0	631.0	NaN
NaN			
2014-06-16	1917.0	598.0	NaN
NaN			
2014-06-17	2017.0	559.0	NaN
NaN			
2014-06-18	2028.0	594.0	NaN
NaN			
2014-06-19	1630.0	360.0	0.0
0.0			
2014-06-20	1170.0	460.0	0.0
0.0			
2014-06-21	1289.0	645.0	0.0
0.0			

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2014-06-22 0.0	1542.0	720.0	0.0
2014-06-23 0.0	1862.0	656.0	0.0
2014-06-24 0.0	1938.0	512.0	0.0
2014-06-25 0.0	1849.0	573.0	0.0
2014-06-26 0.0	1801.0	521.0	0.0
2014-06-27 0.0	1498.0	442.0	0.0
2014-06-28 0.0	1127.0	530.0	0.0
2014-06-29 9.0	997.0	527.0	37.0
2014-06-30 62.0	1473.0	509.0	73.0
2014-07-01 74.0	1806.0	514.0	54.0
North Brisbane: pedestrians zero 11 Jun-2 Jul 2018; cyclists only 34 on 30 Jun, zero 1-2 Jul; both missing 3 Jul			
Bicentennial Bikeway Pedestrians Bicentennial Bikeway Cyclists North Brisbane Bikeway Pedestrians North			
Brisbane Bikeway Cyclists			
Date			
2018-06-09 122.0	754.0	1332.0	82.0
2018-06-10 101.0	930.0	1204.0	63.0
2018-06-11 283.0	772.0	1958.0	0.0
2018-06-12 286.0	856.0	2158.0	0.0
2018-06-13 275.0	747.0	2105.0	0.0
2018-06-14 258.0	732.0	1886.0	0.0
2018-06-15 188.0	655.0	1741.0	0.0
2018-06-16 83.0	775.0	1185.0	0.0
2018-06-17 76.0	866.0	1236.0	0.0
2018-06-18 221.0	646.0	1545.0	0.0
2018-06-19 251.0	748.0	1679.0	0.0
2018-06-20 230.0	710.0	1832.0	0.0
2018-06-21 232.0	684.0	1744.0	0.0
2018-06-22 194.0	568.0	1593.0	0.0
2018-06-23 102.0	696.0	1238.0	0.0
2018-06-24 111.0	957.0	1230.0	0.0
2018-06-25 245.0	723.0	1724.0	0.0
2018-06-26 226.0	725.0	1825.0	0.0
2018-06-27 154.0	378.0	1108.0	0.0
2018-06-28 151.0	534.0	1187.0	0.0
2018-06-29 188.0	555.0	1580.0	0.0
2018-06-30 34.0	611.0	1191.0	0.0
2018-07-01 0.0	760.0	1167.0	0.0
2018-07-02 0.0	699.0	1553.0	0.0
2018-07-03 NaN	785.0	1899.0	NaN
2018-07-04 272.0	704.0	1660.0	149.0

Box 10 — Code cell

```
# Box 10 - Part (c): Zero counts in each column
print('Bicentennial Bikeway Pedestrians - zero days')
print(df[df['Bicentennial Bikeway Pedestrians'] == 0])
print(df[df['Bicentennial Bikeway Pedestrians'] == 0].shape)

print('Bicentennial Bikeway Cyclists - zero days')
print(df[df['Bicentennial Bikeway Cyclists'] == 0])
print(df[df['Bicentennial Bikeway Cyclists'] == 0].shape)

print('North Brisbane Bikeway Pedestrians - zero days')
print(df[df['North Brisbane Bikeway Pedestrians'] == 0])
print(df[df['North Brisbane Bikeway Pedestrians'] == 0].shape)

print('North Brisbane Bikeway Cyclists - zero days')
print(df[df['North Brisbane Bikeway Cyclists'] == 0])
print(df[df['North Brisbane Bikeway Cyclists'] == 0].shape)
```

EGH404 – Assessment 2 Template

Output

```

Bicentennial Bikeway Pedestrians - zero days
      Bicentennial Bikeway Pedestrians  Bicentennial Bikeway Cyclists  North Brisbane Bikeway Pedestrians  North
Brisbane Bikeway Cyclists
Date
2014-12-02          0.0          12.0          55.0
149.0
2014-12-03          0.0          59.0          59.0
157.0
2014-12-04          0.0          32.0          51.0
150.0
2014-12-05          0.0          22.0          35.0
88.0
2014-12-06          0.0          12.0          16.0
22.0
...
...
2017-04-22          0.0          0.0          52.0
64.0
2017-04-23          0.0          0.0          70.0
140.0
2017-04-24          0.0          0.0          101.0
189.0
2017-04-25          0.0          0.0          104.0
147.0
2017-04-26          0.0          0.0          135.0
224.0

[78 rows x 4 columns]
(78, 4)
Bicentennial Bikeway Cyclists - zero days
      Bicentennial Bikeway Pedestrians  Bicentennial Bikeway Cyclists  North Brisbane Bikeway Pedestrians  North
Brisbane Bikeway Cyclists
Date
2015-03-15          1447.0          0.0          0.0
82.0
2015-03-16          2080.0          0.0          0.0
162.0
2015-03-17          1666.0          0.0          0.0
170.0
2015-03-18          3722.0          0.0          0.0
142.0
2015-03-19          3312.0          0.0          0.0
137.0
...
...
2017-04-22          0.0          0.0          52.0
64.0
2017-04-23          0.0          0.0          70.0
140.0
2017-04-24          0.0          0.0          101.0
189.0
2017-04-25          0.0          0.0          104.0
147.0
2017-04-26          0.0          0.0          135.0
224.0

[109 rows x 4 columns]
(109, 4)
North Brisbane Bikeway Pedestrians - zero days
      Bicentennial Bikeway Pedestrians  Bicentennial Bikeway Cyclists  North Brisbane Bikeway Pedestrians  North
Brisbane Bikeway Cyclists
Date
2014-06-19          1630.0          360.0          0.0
0.0
2014-06-20          1170.0          460.0          0.0
0.0
2014-06-21          1289.0          645.0          0.0
0.0
2014-06-22          1542.0          720.0          0.0
0.0
2014-06-23          1862.0          656.0          0.0
0.0
...
...
2018-06-28          534.0          1187.0          0.0
151.0
2018-06-29          555.0          1580.0          0.0
188.0
2018-06-30          611.0          1191.0          0.0
34.0
2018-07-01          760.0          1167.0          0.0
0.0
2018-07-02          699.0          1553.0          0.0
0.0

[229 rows x 4 columns]
(229, 4)
North Brisbane Bikeway Cyclists - zero days
      Bicentennial Bikeway Pedestrians  Bicentennial Bikeway Cyclists  North Brisbane Bikeway Pedestrians  North
Brisbane Bikeway Cyclists
Date
2014-06-19          1630.0          360.0          0.0
0.0
2014-06-20          1170.0          460.0          0.0
0.0

```

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2014-06-21 0.0	1289.0	645.0	0.0
2014-06-22 0.0	1542.0	720.0	0.0
2014-06-23 0.0	1862.0	656.0	0.0
2014-06-24 0.0	1938.0	512.0	0.0
2014-06-25 0.0	1849.0	573.0	0.0
2014-06-26 0.0	1801.0	521.0	0.0
2014-06-27 0.0	1498.0	442.0	0.0
2014-06-28 0.0	1127.0	530.0	0.0
2017-02-07 0.0	745.0	2505.0	287.0
2017-02-08 0.0	722.0	2228.0	239.0
2017-02-09 0.0	615.0	1702.0	260.0
2017-02-10 0.0	524.0	2110.0	210.0
2017-02-11 0.0	478.0	1263.0	59.0
2017-02-12 0.0	382.0	1075.0	67.0
2017-02-13 0.0	366.0	1792.0	189.0
2017-02-14 0.0	736.0	2177.0	317.0
2017-02-15 0.0	677.0	2340.0	319.0
2017-02-16 0.0	685.0	2229.0	311.0
2017-02-17 0.0	534.0	2019.0	278.0
2017-02-18 0.0	578.0	1513.0	117.0
2017-02-19 0.0	466.0	1283.0	99.0
2017-02-20 0.0	825.0	2118.0	341.0
2017-02-21 0.0	794.0	2427.0	336.0
2017-02-22 0.0	774.0	2408.0	320.0
2017-02-23 0.0	748.0	2289.0	271.0
2017-02-24 0.0	613.0	2096.0	256.0
2017-02-25 0.0	695.0	1497.0	122.0
2017-02-26 0.0	819.0	1631.0	145.0
2017-02-27 0.0	811.0	2113.0	347.0
2017-02-28 0.0	801.0	2439.0	366.0
2017-03-01 0.0	802.0	2334.0	301.0
2017-03-02 0.0	676.0	2290.0	281.0
2017-03-03 0.0	667.0	2164.0	219.0
2017-03-04 0.0	665.0	1472.0	113.0
2017-03-05 0.0	769.0	1334.0	108.0
2017-03-06 0.0	785.0	2287.0	306.0
2017-03-07 0.0	897.0	2593.0	303.0
2018-07-01 0.0	760.0	1167.0	0.0
2018-07-02 0.0	699.0	1553.0	0.0
(41, 4)			

Box 11 — Code cell

```
# Box 11 - Part (c): Bicentennial Bikeway fault periods
print('Bicentennial pedestrians: zeros start 2 Dec 2014')
print(df.iloc[332:338, :])

print('Bicentennial pedestrians: odd values on 22-23 Dec 2014')
print(df.iloc[353:359, :])

print('Bicentennial pedestrians: zeros end 19 Jan 2015, multiples of 128 from 20 Jan to 5 Feb, 5452 on 6 Feb,
normal from 7 Feb 2015')
print(df.iloc[378:405, :])

print('Bicentennial cyclists: undercount starts 30 Sep 2014')
print(df.iloc[268:277, :])
```

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```
print('Bicentennial cyclists: near-zero counts end 28 Dec 2014, then stay below normal')
print(df.iloc[357:366, :])
```

```
bc_fault = df.iloc[272:362, :]
print('Bicentennial cyclists, 30 Sep-28 Dec 2014')
print(bc_fault[['Bicentennial Bikeway Cyclists']].describe())
```

```
bc_partial = df.iloc[362:438, :]
print('Bicentennial cyclists, 29 Dec 2014-14 Mar 2015 (partial recovery)')
print(bc_partial[['Bicentennial Bikeway Cyclists']].describe())
```

```
print('Bicentennial cyclists: zeros start 15 Mar 2015')
print(df.iloc[435:441, :])
```

```
print('Bicentennial cyclists: zeros end 30 May 2015')
print(df.iloc[512:518, :])
```

Output

Bicentennial pedestrians: zeros start 2 Dec 2014

	Bicentennial Bikeway Pedestrians	Bicentennial Bikeway Cyclists	North Brisbane Bikeway Pedestrians	North Brisbane Bikeway Cyclists
2014-11-29	1067.0	45.0	102.0	67.0
2014-11-30	1279.0	49.0	84.0	67.0
2014-12-01	1946.0	11.0	67.0	154.0
2014-12-02	0.0	12.0	55.0	149.0
2014-12-03	0.0	59.0	59.0	157.0
2014-12-04	0.0	32.0	51.0	150.0

Bicentennial pedestrians: odd values on 22-23 Dec 2014

	Bicentennial Bikeway Pedestrians	Bicentennial Bikeway Cyclists	North Brisbane Bikeway Pedestrians	North Brisbane Bikeway Cyclists
2014-12-20	0.0	75.0	66.0	43.0
2014-12-21	0.0	76.0	39.0	73.0
2014-12-22	90.0	109.0	86.0	77.0
2014-12-23	2047.0	82.0	69.0	72.0
2014-12-24	0.0	79.0	55.0	56.0
2014-12-25	0.0	83.0	36.0	19.0

Bicentennial pedestrians: zeros end 19 Jan 2015, multiples of 128 from 20 Jan to 5 Feb, 5452 on 6 Feb, normal from 7 Feb 2015

	Bicentennial Bikeway Pedestrians	Bicentennial Bikeway Cyclists	North Brisbane Bikeway Pedestrians	North Brisbane Bikeway Cyclists
2015-01-14	0.0	218.0	0.0	113.0
2015-01-15	0.0	304.0	0.0	135.0
2015-01-16	0.0	232.0	0.0	120.0
2015-01-17	256.0	198.0	0.0	59.0
2015-01-18	0.0	275.0	0.0	43.0
2015-01-19	0.0	313.0	0.0	130.0
2015-01-20	5632.0	298.0	0.0	125.0
2015-01-21	4224.0	346.0	0.0	124.0
2015-01-22	4224.0	191.0	0.0	106.0
2015-01-23	8960.0	32.0	0.0	24.0
2015-01-24	7424.0	210.0	0.0	36.0
2015-01-25	4992.0	807.0	0.0	65.0
2015-01-26	3584.0	489.0	0.0	54.0
2015-01-27	1920.0	587.0	0.0	137.0
2015-01-28	12288.0	248.0	0.0	104.0
2015-01-29	8064.0	536.0	0.0	146.0
2015-01-30	6784.0	438.0	0.0	116.0
2015-01-31	6016.0	599.0	0.0	77.0
2015-02-01	4224.0	735.0	0.0	66.0
2015-02-02	1664.0	1180.0	0.0	

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148.0			
2015-02-03	7680.0	497.0	0.0
164.0			
2015-02-04	7168.0	449.0	0.0
162.0			
2015-02-05	6912.0	717.0	0.0
150.0			
2015-02-06	5452.0	522.0	0.0
116.0			
2015-02-07	1316.0	315.0	0.0
91.0			
2015-02-08	1514.0	352.0	0.0
81.0			
2015-02-09	2056.0	376.0	0.0
158.0			
Bicentennial cyclists: undercount starts 30 Sep 2014			
Bicentennial Bikeway Pedestrians Bicentennial Bikeway Cyclists North Brisbane Bikeway Pedestrians North			
Brisbane Bikeway Cyclists			
Date			
2014-09-26	1748.0	578.0	58.0
92.0			
2014-09-27	1468.0	752.0	72.0
48.0			
2014-09-28	1373.0	734.0	142.0
56.0			
2014-09-29	1958.0	586.0	146.0
113.0			
2014-09-30	2221.0	145.0	139.0
104.0			
2014-10-01	2140.0	56.0	85.0
115.0			
2014-10-02	2009.0	59.0	66.0
105.0			
2014-10-03	1718.0	25.0	75.0
92.0			
2014-10-04	1159.0	39.0	42.0
49.0			
Bicentennial cyclists: near-zero counts end 28 Dec 2014, then stay below normal			
Bicentennial Bikeway Pedestrians Bicentennial Bikeway Cyclists North Brisbane Bikeway Pedestrians North			
Brisbane Bikeway Cyclists			
Date			
2014-12-24	0.0	79.0	55.0
56.0			
2014-12-25	0.0	83.0	36.0
19.0			
2014-12-26	0.0	97.0	26.0
48.0			
2014-12-27	0.0	46.0	3.0
13.0			
2014-12-28	0.0	92.0	2.0
34.0			
2014-12-29	0.0	161.0	0.0
52.0			
2014-12-30	0.0	304.0	0.0
66.0			
2014-12-31	0.0	159.0	12.0
57.0			
2015-01-01	0.0	154.0	1.0
43.0			
Bicentennial cyclists, 30 Sep-28 Dec 2014			
Bicentennial Bikeway Cyclists			
count	90.000000		
mean	34.677778		
std	27.634382		
min	7.000000		
25%	15.250000		
50%	23.500000		
75%	46.750000		
max	145.000000		
Bicentennial cyclists, 29 Dec 2014-14 Mar 2015 (partial recovery)			
Bicentennial Bikeway Cyclists			
count	76.000000		
mean	378.157895		
std	191.544463		
min	32.000000		
25%	244.000000		
50%	378.500000		
75%	444.500000		
max	1180.000000		
Bicentennial cyclists: zeros start 15 Mar 2015			
Bicentennial Bikeway Pedestrians Bicentennial Bikeway Cyclists North Brisbane Bikeway Pedestrians North			
Brisbane Bikeway Cyclists			
Date			
2015-03-12	2157.0	444.0	0.0
155.0			
2015-03-13	1889.0	357.0	0.0
129.0			
2015-03-14	1222.0	417.0	0.0
86.0			
2015-03-15	1447.0	0.0	0.0
82.0			
2015-03-16	2080.0	0.0	0.0
162.0			
2015-03-17	1666.0	0.0	0.0
170.0			
Bicentennial cyclists: zeros end 30 May 2015			
Bicentennial Bikeway Pedestrians Bicentennial Bikeway Cyclists North Brisbane Bikeway Pedestrians North			

EGH404 – Assessment 2 Template

```
Brisbane Bikeway Cyclists
Date
2015-05-28          2023.0          0.0          2.0
149.0
2015-05-29          1771.0          0.0          0.0
130.0
2015-05-30          1312.0          0.0          0.0
90.0
2015-05-31          806.0          422.0          0.0
64.0
2015-06-01          513.0          1709.0          0.0
148.0
2015-06-02          587.0          1832.0          0.0
156.0
```

Box 12 — Code cell

```
# Box 12 - Part (c): North Brisbane Bikeway fault periods
print('North Brisbane pedestrians: fault starts 29 Dec 2014')
print(df.iloc[358:377, :])

print('North Brisbane pedestrians: fault ends 11 Jul 2015')
print(df.iloc[550:560, :])

nb_ped_fault = df.iloc[362:557, :]
print('North Brisbane pedestrians, 29 Dec 2014-11 Jul 2015')
print(nb_ped_fault[['North Brisbane Bikeway Pedestrians']].describe())
print(nb_ped_fault[nb_ped_fault['North Brisbane Bikeway Pedestrians'] == 0].shape)
print('North Brisbane cyclists over the same period')
print(nb_ped_fault[['North Brisbane Bikeway Cyclists']].describe())

print('North Brisbane: pedestrians jump on 6 Feb 2017; cyclists are zero from 7 Feb')
print(df.iloc[1129:1136, :])

print('North Brisbane: cyclists zero until 7 Mar 2017; both normal again from 9 Mar')
print(df.iloc[1158:1165, :])

nb_ped_before = df.iloc[179:362, :]
print('North Brisbane pedestrians before the fault, 29 Jun-28 Dec 2014')
print(nb_ped_before[['North Brisbane Bikeway Pedestrians']].describe())

nb_ped_after = df.iloc[557:730, :]
print('North Brisbane pedestrians after the fault, 12 Jul-31 Dec 2015')
print(nb_ped_after[['North Brisbane Bikeway Pedestrians']].describe())

nb_2017 = df.iloc[1132:1163, :]
print('North Brisbane pedestrians, 6 Feb-8 Mar 2017')
print(nb_2017[['North Brisbane Bikeway Pedestrians']].describe())

print('North Brisbane: cyclists drop and pedestrians jump from 9 Nov 2018')
print(df.iloc[1769:1777, :])

nb_before_2018 = df.iloc[1734:1773, :]
print('North Brisbane, 1 Oct-8 Nov 2018')
print(nb_before_2018[['North Brisbane Bikeway Pedestrians', 'North Brisbane Bikeway Cyclists']].describe())

nb_2018 = df.iloc[1773:1826, :]
print('North Brisbane, 9 Nov-31 Dec 2018')
print(nb_2018[['North Brisbane Bikeway Pedestrians', 'North Brisbane Bikeway Cyclists']].describe())
```

Output

```
North Brisbane pedestrians: fault starts 29 Dec 2014
Bicentennial Bikeway Pedestrians  Bicentennial Bikeway Cyclists  North Brisbane Bikeway Pedestrians  North
Brisbane Bikeway Cyclists
Date
2014-12-25          0.0          83.0          36.0
19.0
2014-12-26          0.0          97.0          26.0
48.0
2014-12-27          0.0          46.0          3.0
13.0
2014-12-28          0.0          92.0          2.0
34.0
2014-12-29          0.0          161.0          0.0
52.0
2014-12-30          0.0          304.0          0.0
66.0
2014-12-31          0.0          159.0          12.0
57.0
2015-01-01          0.0          154.0          1.0
43.0
2015-01-02          0.0          110.0          0.0
64.0
2015-01-03          0.0          147.0          0.0
59.0
2015-01-04          0.0          324.0          0.0
55.0
2015-01-05          0.0          111.0          0.0
113.0
2015-01-06          0.0          420.0          0.0
107.0
2015-01-07          0.0          428.0          0.0
136.0
```

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2015-01-08	0.0	246.0	0.0
119.0			
2015-01-09	0.0	1013.0	6.0
123.0			
2015-01-10	0.0	238.0	0.0
64.0			
2015-01-11	0.0	213.0	60.0
61.0			
2015-01-12	0.0	224.0	0.0
124.0			
North Brisbane pedestrians: fault ends 11 Jul 2015			
	Bicentennial Bikeway Pedestrians	Bicentennial Bikeway Cyclists	North Brisbane Bikeway Pedestrians
North Brisbane Bikeway Cyclists			
Date			
2015-07-05	607.0	1143.0	0.0
68.0			
2015-07-06	566.0	1557.0	0.0
128.0			
2015-07-07	634.0	1765.0	0.0
148.0			
2015-07-08	486.0	1518.0	0.0
122.0			
2015-07-09	522.0	1411.0	0.0
123.0			
2015-07-10	378.0	1356.0	0.0
89.0			
2015-07-11	519.0	1090.0	0.0
65.0			
2015-07-12	596.0	603.0	26.0
55.0			
2015-07-13	488.0	1257.0	61.0
143.0			
2015-07-14	523.0	1559.0	44.0
118.0			
North Brisbane pedestrians, 29 Dec 2014-11 Jul 2015			
	North Brisbane Bikeway Pedestrians		
count	195.000000		
mean	4.117949		
std	15.648241		
min	0.000000		
25%	0.000000		
50%	0.000000		
75%	0.000000		
max	133.000000		
(161, 4)			
North Brisbane cyclists over the same period			
	North Brisbane Bikeway Cyclists		
count	195.000000		
mean	110.548718		
std	41.442190		
min	15.000000		
25%	72.000000		
50%	120.000000		
75%	147.000000		
max	177.000000		
North Brisbane: pedestrians jump on 6 Feb 2017; cyclists are zero from 7 Feb			
	Bicentennial Bikeway Pedestrians	Bicentennial Bikeway Cyclists	North Brisbane Bikeway Pedestrians
North Brisbane Bikeway Cyclists			
Date			
2017-02-03	537.0	2006.0	97.0
224.0			
2017-02-04	503.0	1383.0	46.0
122.0			
2017-02-05	644.0	1612.0	61.0
100.0			
2017-02-06	664.0	2191.0	203.0
84.0			
2017-02-07	745.0	2505.0	287.0
0.0			
2017-02-08	722.0	2228.0	239.0
0.0			
2017-02-09	615.0	1702.0	260.0
0.0			
North Brisbane: cyclists zero until 7 Mar 2017; both normal again from 9 Mar			
	Bicentennial Bikeway Pedestrians	Bicentennial Bikeway Cyclists	North Brisbane Bikeway Pedestrians
North Brisbane Bikeway Cyclists			
Date			
2017-03-04	665.0	1472.0	113.0
0.0			
2017-03-05	769.0	1334.0	108.0
0.0			
2017-03-06	785.0	2287.0	306.0
0.0			
2017-03-07	897.0	2593.0	303.0
0.0			
2017-03-08	862.0	2374.0	436.0
71.0			
2017-03-09	795.0	2291.0	112.0
286.0			
2017-03-10	619.0	1986.0	91.0
206.0			
North Brisbane pedestrians before the fault, 29 Jun-28 Dec 2014			
	North Brisbane Bikeway Pedestrians		
count	183.000000		
mean	66.513661		
std	35.640583		
min	2.000000		

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```

25%          45.000000
50%          61.000000
75%          77.500000
max          210.000000
North Brisbane pedestrians after the fault, 12 Jul-31 Dec 2015
  North Brisbane Bikeway Pedestrians
count        173.000000
mean         75.236994
std          60.914508
min          15.000000
25%          43.000000
50%          63.000000
75%          87.000000
max          546.000000
North Brisbane pedestrians, 6 Feb-8 Mar 2017
  North Brisbane Bikeway Pedestrians
count         31.000000
mean         242.774194
std          97.816396
min          59.000000
25%          167.000000
50%          271.000000
75%          314.000000
max          436.000000
North Brisbane: cyclists drop and pedestrians jump from 9 Nov 2018
  Bicentennial Bikeway Pedestrians  Bicentennial Bikeway Cyclists  North Brisbane Bikeway Pedestrians  North
Brisbane Bikeway Cyclists
Date
2018-11-05          734.0          2005.0          137.0
283.0
2018-11-06          738.0          2335.0          116.0
271.0
2018-11-07          691.0          2247.0          111.0
271.0
2018-11-08          794.0          1993.0          144.0
320.0
2018-11-09          655.0          1992.0          442.0
76.0
2018-11-10          743.0          1325.0          202.0
26.0
2018-11-11          915.0          1577.0          186.0
51.0
2018-11-12          875.0          2066.0          420.0
52.0
North Brisbane, 1 Oct-8 Nov 2018
  North Brisbane Bikeway Pedestrians  North Brisbane Bikeway Cyclists
count          39.000000          39.000000
mean          116.307692          229.128205
std           43.334776          101.025578
min           19.000000          17.000000
25%           83.000000          126.000000
50%          128.000000          260.000000
75%          153.000000          321.000000
max           179.000000          365.000000
North Brisbane, 9 Nov-31 Dec 2018
  North Brisbane Bikeway Pedestrians  North Brisbane Bikeway Cyclists
count          53.000000          53.000000
mean          281.094340          34.584906
std           143.330208          16.065292
min           44.000000           8.000000
25%           144.000000          23.000000
50%           283.000000          32.000000
75%           409.000000          45.000000
max           503.000000          76.000000

```

Box 13 — Code cell

```

# Box 13 - Part (c): Outliers using the 1.5 x IQR rule
# Upper bounds from the quartiles in Box 5 (Q3 + 1.5 x IQR):
# BP upper = 1102 + 1.5*(1102-592) = 1867
# BC upper = 1918.5 + 1.5*(1918.5-654.5) = 3814.5
# NP upper = 120 + 1.5*(120-37) = 244.5
# NC upper = 184 + 1.5*(184-76) = 346
# All lower bounds are negative and the lowest counts are 0, so there are no low outliers.

print('Bicentennial pedestrians above 1867')
print(df[df['Bicentennial Bikeway Pedestrians'] > 1867].shape)
year_2014 = df.iloc[0:365, :]
year_2015 = df.iloc[365:730, :]
year_2016 = df.iloc[730:1096, :]
print(year_2014[year_2014['Bicentennial Bikeway Pedestrians'] > 1867].shape)
print(year_2015[year_2015['Bicentennial Bikeway Pedestrians'] > 1867].shape)
print(year_2016[year_2016['Bicentennial Bikeway Pedestrians'] > 1867])

print('Bicentennial pedestrians above 3000')
print(df[df['Bicentennial Bikeway Pedestrians'] > 3000])

print('Bicentennial cyclists above 3814.5')
print(df[df['Bicentennial Bikeway Cyclists'] > 3814.5])

print('North Brisbane pedestrians above 244.5')
print(df[df['North Brisbane Bikeway Pedestrians'] > 244.5].shape)
print(nb_2017[nb_2017['North Brisbane Bikeway Pedestrians'] > 244.5].shape)
print(nb_2018[nb_2018['North Brisbane Bikeway Pedestrians'] > 244.5].shape)

```

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```
print('North Brisbane pedestrians above 500')
print(df[df['North Brisbane Bikeway Pedestrians'] > 500])

print('North Brisbane cyclists above 346')
print(df[df['North Brisbane Bikeway Cyclists'] > 346])
```

Output

```
Bicentennial pedestrians above 1867
(220, 4)
(147, 4)
(72, 4)
      Bicentennial Bikeway Pedestrians  Bicentennial Bikeway Cyclists  North Brisbane Bikeway Pedestrians  North
Brisbane Bikeway Cyclists
Date
2016-08-07                3328.0                720.0                48.0
63.0
Bicentennial pedestrians above 3000
      Bicentennial Bikeway Pedestrians  Bicentennial Bikeway Cyclists  North Brisbane Bikeway Pedestrians  North
Brisbane Bikeway Cyclists
Date
2015-01-20                5632.0                298.0                0.0
125.0
2015-01-21                4224.0                346.0                0.0
124.0
2015-01-22                4224.0                191.0                0.0
106.0
2015-01-23                8960.0                 32.0                0.0
24.0
2015-01-24                7424.0                210.0                0.0
36.0
2015-01-25                4992.0                807.0                0.0
65.0
2015-01-26                3584.0                489.0                0.0
54.0
2015-01-28                12288.0                248.0                0.0
104.0
2015-01-29                8064.0                536.0                0.0
146.0
2015-01-30                6784.0                438.0                0.0
116.0
2015-01-31                6016.0                599.0                0.0
77.0
2015-02-01                4224.0                735.0                0.0
66.0
2015-02-03                7680.0                497.0                0.0
164.0
2015-02-04                7168.0                449.0                0.0
162.0
2015-02-05                6912.0                717.0                0.0
150.0
2015-02-06                5452.0                522.0                0.0
116.0
2015-03-18                3722.0                 0.0                0.0
142.0
2015-03-19                3312.0                 0.0                0.0
137.0
2015-03-21                6827.0                 0.0                0.0
39.0
2016-08-07                3328.0                720.0                48.0
63.0
Bicentennial cyclists above 3814.5
Empty DataFrame
Columns: [Bicentennial Bikeway Pedestrians, Bicentennial Bikeway Cyclists, North Brisbane Bikeway Pedestrians, North
Brisbane Bikeway Cyclists]
Index: []
North Brisbane pedestrians above 244.5
(58, 4)
(18, 4)
(28, 4)
North Brisbane pedestrians above 500
      Bicentennial Bikeway Pedestrians  Bicentennial Bikeway Cyclists  North Brisbane Bikeway Pedestrians  North
Brisbane Bikeway Cyclists
Date
2015-10-17                652.0                1376.0                546.0
92.0
2016-01-10                660.0                1582.0                670.0
105.0
2018-11-27                796.0                2412.0                503.0
34.0
North Brisbane cyclists above 346
      Bicentennial Bikeway Pedestrians  Bicentennial Bikeway Cyclists  North Brisbane Bikeway Pedestrians  North
Brisbane Bikeway Cyclists
Date
2018-05-15                874.0                2444.0                162.0
348.0
2018-05-23                772.0                2031.0                146.0
357.0
2018-10-23                918.0                2619.0                167.0
365.0
2018-10-30                939.0                2448.0                158.0
364.0
```

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Box 14 — Code cell

```
# Box 14 - Part (c): Bicentennial Bikeway level change, 2014 vs 2015
print('Bicentennial Bikeway, 1 Jan-29 Sep 2014')
print(df.iloc[0:272, 0:2].describe())

print('Bicentennial Bikeway, 1 Jun-31 Dec 2015')
print(df.iloc[516:730, 0:2].describe())
```

Output

```
Bicentennial Bikeway, 1 Jan-29 Sep 2014
      Bicentennial Bikeway Pedestrians  Bicentennial Bikeway Cyclists
count                                272.000000                      272.000000
mean                                1710.801471                      653.801471
std                                 429.808690                      177.279724
min                                 129.000000                      182.000000
25%                                1383.750000                      571.750000
50%                                1769.500000                      655.500000
75%                                2071.500000                      733.250000
max                                 2511.000000                      2786.000000

Bicentennial Bikeway, 1 Jun-31 Dec 2015
      Bicentennial Bikeway Pedestrians  Bicentennial Bikeway Cyclists
count                                214.000000                      214.000000
mean                                592.953271                      1581.537383
std                                 200.093283                      399.017780
min                                 244.000000                      421.000000
25%                                506.500000                      1297.750000
50%                                579.000000                      1595.500000
75%                                662.750000                      1891.750000
max                                 2833.000000                      2398.000000
```

Box 15 — Code cell

```
# Box 15 - Part (c): Values around each outlier
print('3 Aug 2014')
print(df.iloc[211:218, :])

print('21 Mar 2015')
print(df.iloc[441:448, :])

print('7 Aug 2016')
print(df.iloc[946:953, :])

print('17 Oct 2015')
print(df.iloc[651:658, :])

print('10 Jan 2016')
print(df.iloc[734:743, :])

print('23 Oct 2018')
print(df.iloc[1753:1760, :])
```

Output

```
3 Aug 2014
      Bicentennial Bikeway Pedestrians  Bicentennial Bikeway Cyclists  North Brisbane Bikeway Pedestrians  North
Brisbane Bikeway Cyclists
Date
2014-07-31                                1962.0                      635.0                                68.0
83.0
2014-08-01                                1804.0                      552.0                                45.0
76.0
2014-08-02                                1277.0                      571.0                                29.0
46.0
2014-08-03                                928.0                      2786.0                                24.0
46.0
2014-08-04                                1460.0                      578.0                                57.0
91.0
2014-08-05                                2044.0                      655.0                                72.0
91.0
2014-08-06                                2163.0                      636.0                                64.0
71.0
21 Mar 2015
      Bicentennial Bikeway Pedestrians  Bicentennial Bikeway Cyclists  North Brisbane Bikeway Pedestrians  North
Brisbane Bikeway Cyclists
Date
2015-03-18                                3722.0                      0.0                                0.0
142.0
2015-03-19                                3312.0                      0.0                                0.0
137.0
2015-03-20                                1764.0                      0.0                                0.0
107.0
2015-03-21                                6827.0                      0.0                                0.0
39.0
2015-03-22                                867.0                      0.0                                0.0
48.0
2015-03-23                                1598.0                      0.0                                0.0
124.0
2015-03-24                                2376.0                      0.0                                0.0
166.0
7 Aug 2016
      Bicentennial Bikeway Pedestrians  Bicentennial Bikeway Cyclists  North Brisbane Bikeway Pedestrians  North
Brisbane Bikeway Cyclists
Date
2016-08-04                                501.0                      1309.0                                59.0
```

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86.0			
2016-08-05	649.0	1698.0	95.0
102.0			
2016-08-06	651.0	1194.0	41.0
48.0			
2016-08-07	3328.0	720.0	48.0
63.0			
2016-08-08	780.0	1910.0	89.0
143.0			
2016-08-09	694.0	1996.0	101.0
120.0			
2016-08-10	958.0	1644.0	270.0
75.0			
17 Oct 2015			
Bicentennial Bikeway Pedestrians Bicentennial Bikeway Cyclists North Brisbane Bikeway Pedestrians North Brisbane Bikeway Cyclists			
Date			
2015-10-14	666.0	2094.0	65.0
148.0			
2015-10-15	701.0	2043.0	85.0
151.0			
2015-10-16	578.0	2025.0	85.0
129.0			
2015-10-17	652.0	1376.0	546.0
92.0			
2015-10-18	667.0	1638.0	28.0
63.0			
2015-10-19	677.0	1859.0	66.0
94.0			
2015-10-20	759.0	2262.0	82.0
111.0			
10 Jan 2016			
Bicentennial Bikeway Pedestrians Bicentennial Bikeway Cyclists North Brisbane Bikeway Pedestrians North Brisbane Bikeway Cyclists			
Date			
2016-01-05	179.0	348.0	0.0
39.0			
2016-01-06	731.0	1758.0	0.0
143.0			
2016-01-07	589.0	1942.0	0.0
159.0			
2016-01-08	545.0	1799.0	0.0
140.0			
2016-01-09	561.0	1514.0	0.0
83.0			
2016-01-10	660.0	1582.0	670.0
105.0			
2016-01-11	689.0	2045.0	5.0
194.0			
2016-01-12	756.0	2292.0	0.0
168.0			
2016-01-13	688.0	2260.0	6.0
184.0			
23 Oct 2018			
Bicentennial Bikeway Pedestrians Bicentennial Bikeway Cyclists North Brisbane Bikeway Pedestrians North Brisbane Bikeway Cyclists			
Date			
2018-10-20	794.0	1491.0	60.0
126.0			
2018-10-21	652.0	1060.0	53.0
126.0			
2018-10-22	852.0	2147.0	152.0
304.0			
2018-10-23	918.0	2619.0	167.0
365.0			
2018-10-24	886.0	2420.0	179.0
285.0			
2018-10-25	720.0	2268.0	155.0
330.0			
2018-10-26	603.0	1973.0	103.0
228.0			

S1.T2 Python notebook evidence

Box 1 — Markdown cell

S1.T2: Weather Data Inspection

Bureau of Meteorology weather data for Brisbane, 1 January 2014 to 31 December 2018: daily rainfall, daily solar exposure and daily maximum temperature.

Box 2 — Markdown cell

Loading and checking the data

Box 3 — Code cell

```
# Box 3 - Import the libraries
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
```

Box 4 — Code cell

```
# Box 4 - Load the raw weather data and check its structure
weather = pd.read_csv('brisbane_weather (1).csv', header=0)
weather.info()

weather['Date'] = pd.to_datetime(weather['Date'], format='%Y-%m-%d')

print("Shape : ", weather.shape)
print("Rows : ", weather.shape[0])
print("Columns : ", weather.shape[1])
```

Output

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1826 entries, 0 to 1825
Data columns (total 4 columns):
#   Column                Non-Null Count  Dtype
---  -
0   Date                  1826 non-null   str
1   daily_rainfall        1787 non-null   float64
2   solar_exposure        1825 non-null   float64
3   maximum_temperature   1802 non-null   float64
dtypes: float64(3), str(1)
memory usage: 57.2+ KB
Shape : (1826, 4)
Rows : 1826
Columns : 4
```

Box 5 — Code cell

```
# Box 5 - First five rows
weather.head()
```

Output

	Date	daily_rainfall	solar_exposure	maximum_temperature
0	2014-01-01	0.0	31.2	30.6
1	2014-01-02	0.0	23.4	31.8
2	2014-01-03	1.0	29.6	34.5
3	2014-01-04	0.0	30.5	38.7
4	2014-01-05	0.0	15.7	33.6

Box 6 — Code cell

```
# Box 6 - Last five rows
weather.tail()
```

Output

	Date	daily_rainfall	solar_exposure	maximum_temperature
1821	2018-12-27	0.0	29.7	29.5
1822	2018-12-28	0.0	28.6	29.7
1823	2018-12-29	0.0	29.8	30.5
1824	2018-12-30	0.0	27.9	30.6
1825	2018-12-31	0.0	28.4	31.4

Box 7 — Code cell

```
# Box 7 - Reload with Date as the index and get summary statistics (quartiles for the IQR rule)
weather = pd.read_csv('brisbane_weather (1).csv', header=0, index_col='Date')
weather.index = pd.to_datetime(weather.index)

weather.describe()
```

Output

	daily_rainfall	solar_exposure	maximum_temperature
count	1787.000000	1825.000000	1802.000000
mean	2.698601	18.242575	26.983407
std	10.875995	6.427525	3.812884
min	0.000000	0.500000	16.100000
25%	0.000000	13.800000	24.000000
50%	0.000000	17.300000	27.100000

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75%	0.600000	23.300000	29.800000
max	182.600000	31.500000	38.900000

Box 8 — Markdown cell

(a) Box plots of each weather factor

Box 9 — Code cell

```
# Box 9 - Part (a): Separate box plots for rainfall, maximum temperature and solar exposure
fig, axs = plt.subplots(nrows=1, ncols=3, figsize=(18, 7))

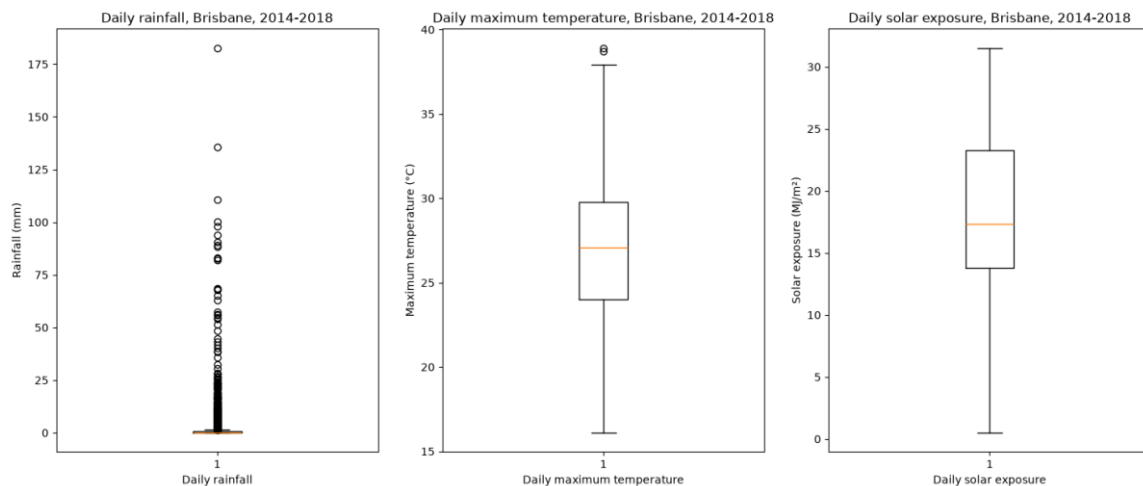
axs[0].boxplot([weather['daily_rainfall'][np.isfinite(weather['daily_rainfall'])]])
axs[0].set_title('Daily rainfall, Brisbane, 2014-2018')
axs[0].set_xlabel('Daily rainfall')
axs[0].set_ylabel('Rainfall (mm)')

axs[1].boxplot([weather['maximum_temperature'][np.isfinite(weather['maximum_temperature'])]])
axs[1].set_title('Daily maximum temperature, Brisbane, 2014-2018')
axs[1].set_xlabel('Daily maximum temperature')
axs[1].set_ylabel('Maximum temperature (°C)')

axs[2].boxplot([weather['solar_exposure'][np.isfinite(weather['solar_exposure'])]])
axs[2].set_title('Daily solar exposure, Brisbane, 2014-2018')
axs[2].set_xlabel('Daily solar exposure')
axs[2].set_ylabel('Solar exposure (MJ/m²)')

plt.show()
```

Output



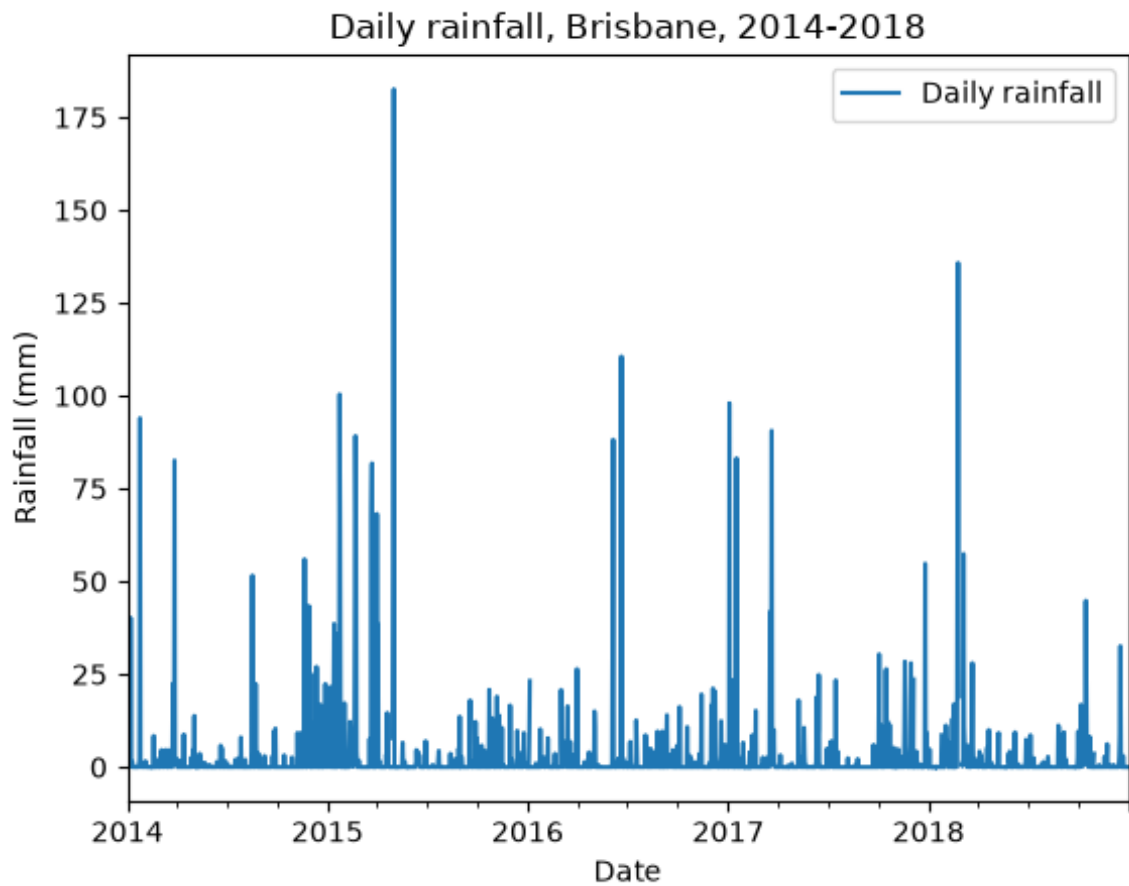
Box 10 — Markdown cell

(b) Line graphs of each weather factor

Box 11 — Code cell

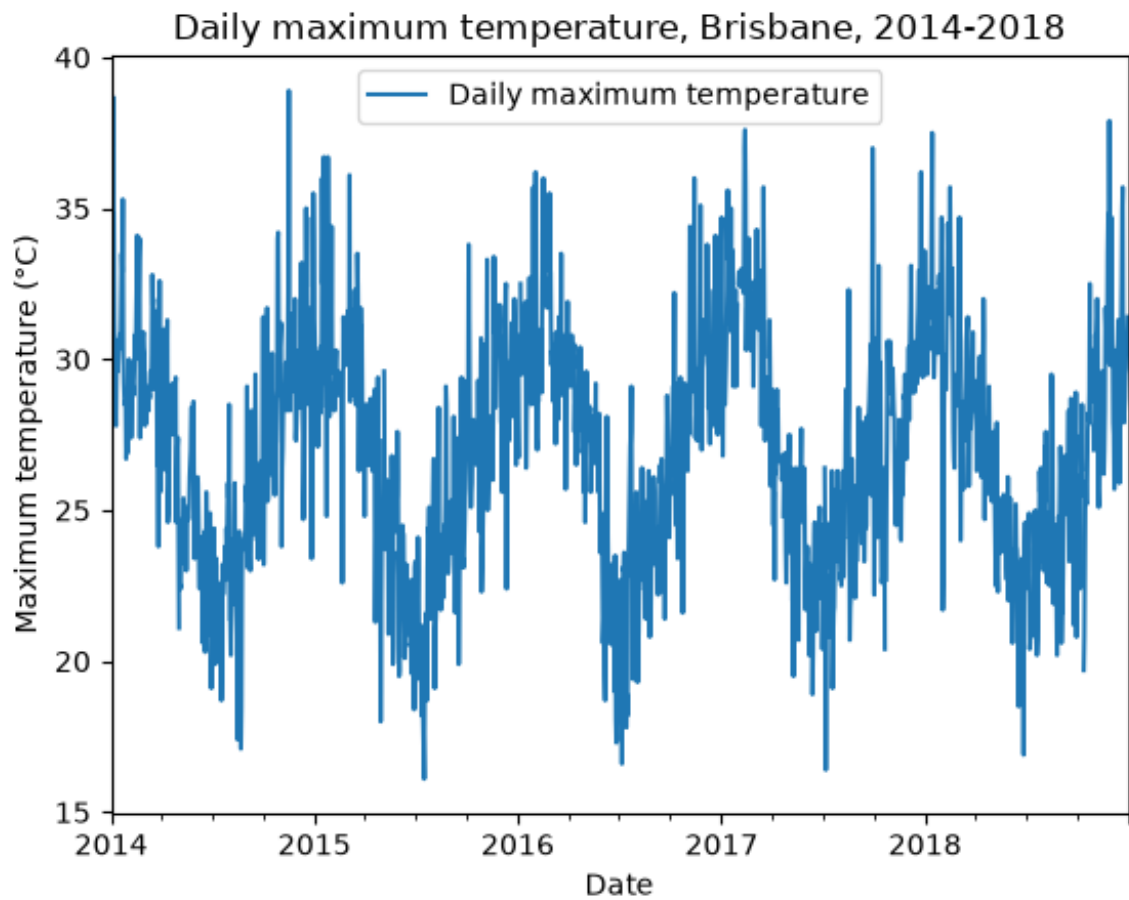
```
# Box 11 - Part (b): Line graph of daily rainfall
weather['daily_rainfall'].plot(kind='line', title='Daily rainfall, Brisbane, 2014-2018')
plt.xlabel('Date')
plt.ylabel('Rainfall (mm)')
plt.legend(['Daily rainfall'])
plt.show()
```

Output

**Box 12 — Code cell**

```
# Box 12 - Part (b): Line graph of daily maximum temperature
weather['maximum temperature'].plot(kind='line', title='Daily maximum temperature, Brisbane, 2014-2018')
plt.xlabel('Date')
plt.ylabel('Maximum temperature (°C)')
plt.legend(['Daily maximum temperature'])
plt.show()
```

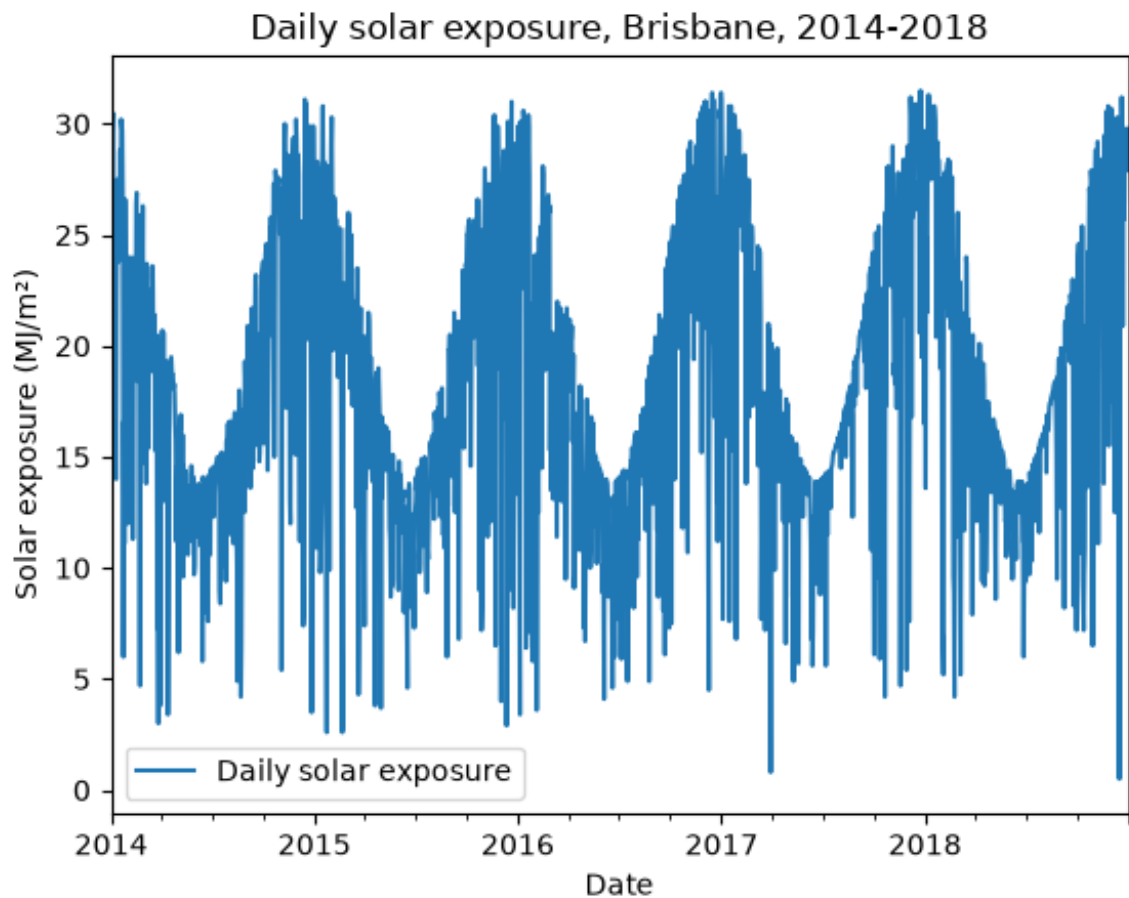
Output



Box 13 — Code cell

```
# Box 13 - Part (b): Line graph of daily solar exposure
weather['solar_exposure'].plot(kind='line', title='Daily solar exposure, Brisbane, 2014-2018')
plt.xlabel('Date')
plt.ylabel('Solar exposure (MJ/m²)')
plt.legend(['Daily solar exposure'])
plt.show()
```

Output

**Box 14 — Markdown cell**

(d) Checking for problematic data

Missing values

Box 15 — Code cell

```
# Box 15 - Part (d): Missing values in each column
print('Daily rainfall - missing days')
print(weather[np.isnan(weather['daily_rainfall'])])
print(weather[np.isnan(weather['daily_rainfall'])].shape)

print('Daily maximum temperature - missing days')
print(weather[np.isnan(weather['maximum_temperature'])])
print(weather[np.isnan(weather['maximum_temperature'])].shape)

print('Daily solar exposure - missing days')
print(weather[np.isnan(weather['solar_exposure'])])
print(weather[np.isnan(weather['solar_exposure'])].shape)
```

Output

```
Daily rainfall - missing days
   daily_rainfall  solar_exposure  maximum_temperature
Date
2014-02-14      NaN             18.7                 30.0
2014-05-07      NaN             12.9                 23.7
2014-08-06      NaN             15.1                 22.3
2015-01-15      NaN             28.6                 36.0
2015-02-04      NaN             23.5                 28.6
2015-04-29      NaN             13.3                 23.6
2015-05-06      NaN             16.4                 29.6
2015-05-08      NaN             16.4                 24.4
2015-05-27      NaN             12.5                 24.3
2015-09-02      NaN             18.2                 25.1
2016-01-06      NaN             30.1                 32.5
2016-02-04      NaN              3.6                 27.4
2016-02-05      NaN              9.8                 27.0
2017-01-31      NaN             29.6                 32.5
2017-02-07      NaN             27.0                 32.8
2017-03-30      NaN              0.8                 25.8
2017-03-31      NaN             17.4                 28.3
2017-04-22      NaN             12.1                 24.8
2017-11-02      NaN             26.5                 28.2
2017-12-06      NaN             31.2                 31.8
```

EGH404 – Assessment 2 Template

```

2017-12-23      NaN      29.6      31.7
2017-12-30      NaN      24.7      32.2
2018-01-07      NaN      31.3      31.5
2018-01-08      NaN      31.3      32.4
2018-01-12      NaN      27.5      32.6
2018-01-16      NaN      30.1      31.1
2018-01-26      NaN      27.4      32.5
2018-02-06      NaN      23.9      NaN
2018-02-07      NaN      27.1      29.3
2018-02-19      NaN      27.6      33.0
2018-02-20      NaN      20.2      30.9
2018-02-21      NaN      9.6      27.6
2018-02-26      NaN      23.8      NaN
2018-02-27      NaN      12.3      28.5
2018-03-01      NaN      26.0      NaN
2018-03-02      NaN      11.4      29.1
2018-05-22      NaN      15.0      25.5
2018-09-26      NaN      18.9      23.7
2018-10-31      NaN      25.8      27.0
(39, 3)
Daily maximum temperature - missing days
      daily_rainfall  solar_exposure  maximum_temperature
Date
2014-05-04      0.2      16.3      NaN
2015-02-03      2.2      19.8      NaN
2015-04-06      0.0      20.5      NaN
2015-05-05      0.0      15.9      NaN
2015-05-26      0.0      12.1      NaN
2016-01-05      23.2      3.4      NaN
2017-01-30      0.0      26.9      NaN
2017-03-26      0.2      21.0      NaN
2017-04-21      0.4      14.0      NaN
2017-07-04      0.0      13.8      NaN
2017-10-22      11.4      26.0      NaN
2017-11-01      0.0      22.7      NaN
2017-12-22      0.4      20.6      NaN
2018-01-25      8.6      28.4      NaN
2018-02-06      NaN      23.9      NaN
2018-02-18      0.0      24.0      NaN
2018-02-25      19.0      21.9      NaN
2018-02-26      NaN      23.8      NaN
2018-03-01      NaN      26.0      NaN
2018-05-21      0.0      15.1      NaN
2018-10-12      8.4      8.3      NaN
2018-10-26      0.6      27.9      NaN
2018-10-29      0.8      6.5      NaN
2018-10-30      3.6      22.3      NaN
(24, 3)
Daily solar exposure - missing days
      daily_rainfall  solar_exposure  maximum_temperature
Date
2017-11-26      0.0      NaN      29.5
(1, 3)

```

Box 16 — Code cell

```

# Box 16 - Part (d): Values around the missing days in the table
print('Rainfall missing 30-31 Mar 2017 (ex-Tropical Cyclone Debbie)')
print(weather.iloc[1181:1189, :])

print('Rainfall missing 6 Jan 2016; maximum temperature missing 5 Jan 2016')
print(weather.iloc[732:739, :])

print('Feb-Mar 2018: rainfall missing 19-21 Feb, 26-27 Feb and 1-2 Mar; maximum temperature missing 18 Feb, 25-26 Feb and 1 Mar')
print(weather.iloc[1507:1522, :])

print('Maximum temperature missing 4 May 2014')
print(weather.iloc[120:127, :])

print('Maximum temperature missing 29-30 Oct 2018')
print(weather.iloc[1759:1766, :])

print('Solar exposure missing 26 Nov 2017 (the only missing solar value)')
print(weather.iloc[1422:1429, :])

```

Output

```

Rainfall missing 30-31 Mar 2017 (ex-Tropical Cyclone Debbie)
      daily_rainfall  solar_exposure  maximum_temperature
Date
2017-03-27      0.0      18.4      30.1
2017-03-28      0.0      20.7      31.3
2017-03-29      0.0      11.8      29.5
2017-03-30      NaN      0.8      25.8
2017-03-31      NaN      17.4      28.3
2017-04-01      0.2      20.1      29.0
2017-04-02      0.0      15.0      27.7
2017-04-03      0.0      13.5      25.6
Rainfall missing 6 Jan 2016; maximum temperature missing 5 Jan 2016
      daily_rainfall  solar_exposure  maximum_temperature
Date
2016-01-03      0.0      14.9      28.0
2016-01-04      8.8      10.6      26.8
2016-01-05      23.2      3.4      NaN
2016-01-06      NaN      30.1      32.5

```

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2016-01-07	0.0	26.9	30.0
2016-01-08	0.0	22.6	29.5
2016-01-09	0.0	27.4	30.3
Feb-Mar 2018: rainfall missing 19-21 Feb, 26-27 Feb and 1-2 Mar; maximum temperature missing 18 Feb, 25-26 Feb and 1 Mar			
	daily_rainfall	solar_exposure	maximum_temperature
Date			
2018-02-16	0.0	20.8	33.0
2018-02-17	17.0	22.0	31.5
2018-02-18	0.0	24.0	NaN
2018-02-19	NaN	27.6	33.0
2018-02-20	NaN	20.2	30.9
2018-02-21	NaN	9.6	27.6
2018-02-22	0.2	8.9	26.7
2018-02-23	62.8	4.2	26.4
2018-02-24	135.8	15.5	29.1
2018-02-25	19.0	21.9	NaN
2018-02-26	NaN	23.8	NaN
2018-02-27	NaN	12.3	28.5
2018-02-28	0.2	22.5	29.5
2018-03-01	NaN	26.0	NaN
2018-03-02	NaN	11.4	29.1
Maximum temperature missing 4 May 2014			
	daily_rainfall	solar_exposure	maximum_temperature
Date			
2014-05-01	4.6	14.3	27.4
2014-05-02	1.2	6.2	23.5
2014-05-03	13.8	15.4	21.1
2014-05-04	0.2	16.3	NaN
2014-05-05	0.0	16.2	22.5
2014-05-06	0.0	16.9	22.4
2014-05-07	NaN	12.9	23.7
Maximum temperature missing 29-30 Oct 2018			
	daily_rainfall	solar_exposure	maximum_temperature
Date			
2018-10-26	0.6	27.9	NaN
2018-10-27	0.0	27.5	28.6
2018-10-28	0.0	21.8	30.3
2018-10-29	0.8	6.5	NaN
2018-10-30	3.6	22.3	NaN
2018-10-31	NaN	25.8	27.0
2018-11-01	0.0	27.9	27.8
Solar exposure missing 26 Nov 2017 (the only missing solar value)			
	daily_rainfall	solar_exposure	maximum_temperature
Date			
2017-11-23	0.0	23.9	26.5
2017-11-24	1.2	28.4	28.8
2017-11-25	0.0	24.2	28.7
2017-11-26	0.0	NaN	29.5
2017-11-27	0.0	26.4	29.0
2017-11-28	0.0	24.0	29.2
2017-11-29	18.2	5.4	26.7

Box 17 — Markdown cell

Outliers (1.5 x IQR rule)

Box 18 — Code cell

```
# Box 18 - Part (d): Outliers using the 1.5 x IQR rule
# Bounds from the quartiles in Box 7:
# Rainfall:    upper = 0.6 + 1.5*(0.6-0) = 1.5 mm, lower = 0 - 1.5*(0.6-0) = -0.9 mm
# Max temp:    upper = 29.8 + 1.5*(29.8-24.0) = 38.5 °C, lower = 24.0 - 1.5*(29.8-24.0) = 15.3 °C
# Solar:       upper = 23.3 + 1.5*(23.3-13.8) = 37.55 MJ/m², lower = 13.8 - 1.5*(23.3-13.8) = -0.45 MJ/m²

print('Rainfall above 1.5 mm')
print(weather[weather['daily_rainfall'] > 1.5].shape)
print('Days with no rain')
print(weather[weather['daily_rainfall'] == 0].shape)
print('Rainfall above 100 mm')
print(weather[weather['daily_rainfall'] > 100])

print('Maximum temperature above 38.5 °C')
print(weather[weather['maximum_temperature'] > 38.5])
print('Maximum temperature below 15.3 °C')
print(weather[weather['maximum_temperature'] < 15.3])

print('Solar exposure above 37.55 MJ/m²')
print(weather[weather['solar_exposure'] > 37.55])
print('Solar exposure below 1 MJ/m² (lowest values on the line graph)')
print(weather[weather['solar_exposure'] < 1])
```

Output

```
Rainfall above 1.5 mm
(338, 3)
Days with no rain
(1161, 3)
Rainfall above 100 mm
daily_rainfall  solar_exposure  maximum_temperature
Date
2015-01-23      100.4           2.6                24.8
2015-05-02      182.6          16.8                24.9
2016-06-20      110.6          13.2                20.5
2018-02-24      135.8          15.5                29.1
Maximum temperature above 38.5 °C
```

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```
daily_rainfall solar_exposure maximum_temperature
Date
2014-01-04      0.0          30.5          38.7
2014-11-16      0.0          24.3          38.9
Maximum temperature below 15.3 °C
Empty DataFrame
Columns: [daily_rainfall, solar_exposure, maximum_temperature]
Index: []
Solar exposure above 37.55 MJ/m²
Empty DataFrame
Columns: [daily_rainfall, solar_exposure, maximum_temperature]
Index: []
Solar exposure below 1 MJ/m² (lowest values on the line graph)
daily_rainfall solar_exposure maximum_temperature
Date
2017-03-30      NaN          0.8          25.8
2018-12-16      9.0          0.5          25.9
```

Box 19 — Code cell

```
# Box 19 - Part (d): Values around selected extreme examples
print('Rainfall 182.6 mm on 2 May 2015')
print(weather.iloc[482:489, :])

print('Rainfall 110.6 mm on 20 Jun 2016')
print(weather.iloc[897:905, :])

print('Maximum temperature 38.7 °C on 4 Jan 2014')
print(weather.iloc[0:7, :])

print('Maximum temperature 38.9 °C on 16 Nov 2014')
print(weather.iloc[316:322, :])

print('Solar exposure 0.5 MJ/m² on 16 Dec 2018')
print(weather.iloc[1806:1815, :])
```

Output

```
Rainfall 182.6 mm on 2 May 2015
daily_rainfall solar_exposure maximum_temperature
Date
2015-04-28      0.0          17.0          25.2
2015-04-29      NaN          13.3          23.6
2015-04-30      7.8          3.7          18.0
2015-05-01     56.2          3.7          21.0
2015-05-02     182.6          16.8          24.9
2015-05-03      0.0          13.2          24.2
2015-05-04      0.0          16.1          27.3
Rainfall 110.6 mm on 20 Jun 2016
daily_rainfall solar_exposure maximum_temperature
Date
2016-06-16      2.4          11.0          23.2
2016-06-17      0.0          11.8          22.9
2016-06-18      0.0           8.3          22.5
2016-06-19      3.8           4.6          22.5
2016-06-20     110.6          13.2          20.5
2016-06-21      0.2          12.7          20.6
2016-06-22      0.0           9.2          21.5
2016-06-23      2.4           6.0          19.0
Maximum temperature 38.7 °C on 4 Jan 2014
daily_rainfall solar_exposure maximum_temperature
Date
2014-01-01      0.0          31.2          30.6
2014-01-02      0.0          23.4          31.8
2014-01-03      1.0          29.6          34.5
2014-01-04      0.0          30.5          38.7
2014-01-05      0.0          15.7          33.6
2014-01-06      0.0          18.7          34.8
2014-01-07     40.2          27.2          31.2
Maximum temperature 38.9 °C on 16 Nov 2014
daily_rainfall solar_exposure maximum_temperature
Date
2014-11-13      9.2          21.5          28.6
2014-11-14      0.2          28.6          30.0
2014-11-15      0.0          28.5          32.2
2014-11-16      0.0          24.3          38.9
2014-11-17      0.0          26.9          30.3
2014-11-18      2.6          13.1          28.3
Solar exposure 0.5 MJ/m² on 16 Dec 2018
daily_rainfall solar_exposure maximum_temperature
Date
2018-12-12      0.0          30.3          29.4
2018-12-13      0.0          25.6          30.2
2018-12-14      0.0          21.7          31.3
2018-12-15      3.0          23.9          31.0
2018-12-16      9.0           0.5          25.9
2018-12-17     32.6          18.8          28.8
2018-12-18      2.8          22.1          29.8
2018-12-19      0.0          27.0          30.9
2018-12-20      0.0          31.2          32.5
```

Box 20 — Markdown cell

Errors

Box 21 — Code cell

```
# Box 21 - Part (d): Checks for impossible negative values
print('Negative values')
print(weather[weather['daily_rainfall'] < 0])
print(weather[weather['maximum_temperature'] < 0])
print(weather[weather['solar_exposure'] < 0])
```

Output

```
Negative values
Empty DataFrame
Columns: [daily rainfall, solar exposure, maximum temperature]
Index: []
Empty DataFrame
Columns: [daily_rainfall, solar_exposure, maximum_temperature]
Index: []
Empty DataFrame
Columns: [daily_rainfall, solar_exposure, maximum_temperature]
Index: []
```

Section 2 Python code and outputs

The code and its corresponding outputs are included below in S2 Box order so the calculations, model summaries, printed evidence and figures can be checked directly.

S2 Box 1

```
# Box 1 - Import the libraries
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
```

S2 Box 2

```
# Box 2 - Load the supplied cleaned data and inspect its structure
df = pd.read_csv('brisbane_bikeway_counters_cleaned (1).csv', header=0)
df['Date'] = pd.to_datetime(df['Date'])
df.info()
print(df.shape)
```

Output

```
<class 'pandas.DataFrame'>
RangeIndex: 1461 entries, 0 to 1460
Data columns (total 5 columns):
#   Column                                Non-Null Count  Dtype
---  -
0   Date                                1461 non-null   datetime64[us]
1   Bicentennial Bikeway Pedestrians    1461 non-null   float64
2   Bicentennial Bikeway Cyclists       1461 non-null   float64
3   North Brisbane Bikeway Pedestrians  1461 non-null   float64
4   North Brisbane Bikeway Cyclists     1461 non-null   float64
dtypes: datetime64[us](1), float64(4)
memory usage: 57.2 KB
(1461, 5)
```

S2 Box 3

```
# Box 3 - First five rows
df.head()
```

Output

```
Date ... North Brisbane Bikeway Cyclists
0 2014-06-19 ... 0.0
1 2014-06-20 ... 0.0
2 2014-06-21 ... 0.0
3 2014-06-22 ... 0.0
4 2014-06-23 ... 0.0

[5 rows x 5 columns]
```

S2 Box 4

```
# Box 4 - Last five rows
df.tail()
```

Output

```
Date ... North Brisbane Bikeway Cyclists
1456 2018-12-27 ... 23.0
1457 2018-12-28 ... 23.0
1458 2018-12-29 ... 20.0
1459 2018-12-30 ... 32.0
1460 2018-12-31 ... 17.0

[5 rows x 5 columns]
```

S2 Box 5

```
# Box 5 - S2.T1: Pearson correlation coefficients
bc = df['Bicentennial Bikeway Cyclists']
bp = df['Bicentennial Bikeway Pedestrians']
nc = df['North Brisbane Bikeway Cyclists']
nped = df['North Brisbane Bikeway Pedestrians']

corrcoef_bc_bp = np.corrcoef(bc, bp)[0, 1]
print('(a) Bicentennial cyclists and pedestrians =', corrcoef_bc_bp)
corrcoef_nc_nped = np.corrcoef(nc, nped)[0, 1]
print('(b) North Brisbane cyclists and pedestrians =', corrcoef_nc_nped)
corrcoef_bc_nc = np.corrcoef(bc, nc)[0, 1]
print('(c) Bicentennial cyclists and North Brisbane cyclists =', corrcoef_bc_nc)
corrcoef_bp_nped = np.corrcoef(bp, nped)[0, 1]
print('(d) Bicentennial pedestrians and North Brisbane cyclists =', corrcoef_bp_nped)
```

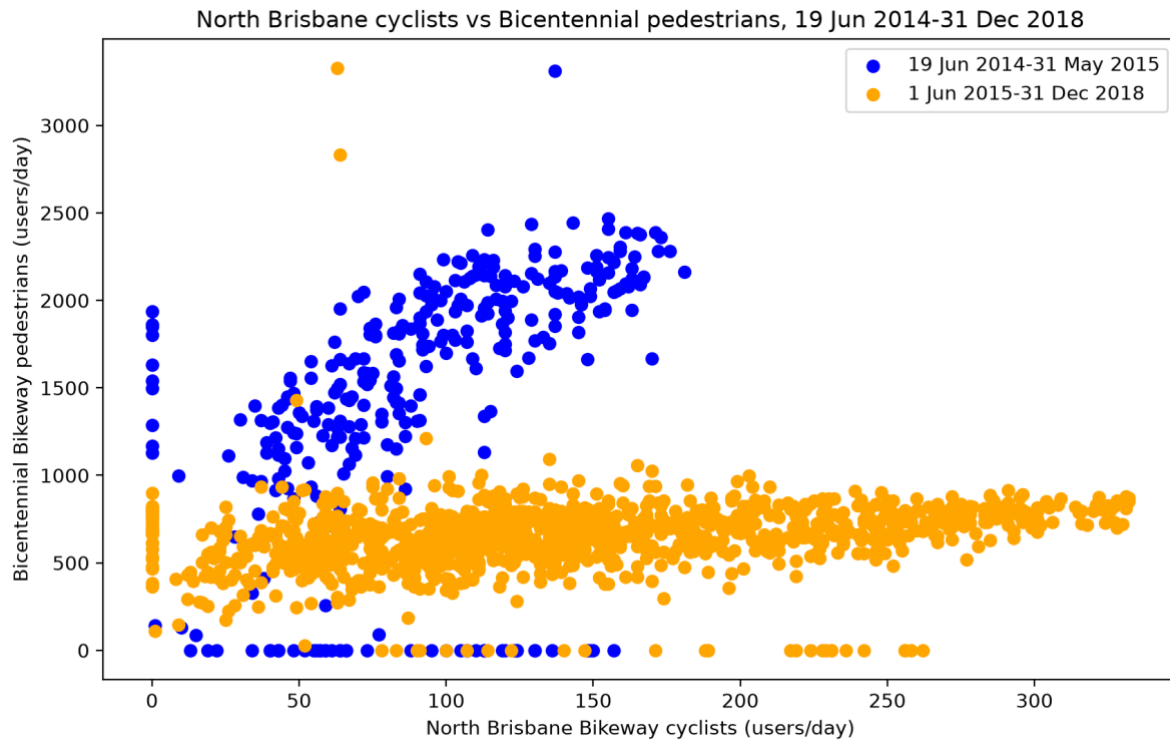
Output

```
(a) Bicentennial cyclists and pedestrians = -0.35538759226047
(b) North Brisbane cyclists and pedestrians = 0.27612018239033936
(c) Bicentennial cyclists and North Brisbane cyclists = 0.48183668055797946
(d) Bicentennial pedestrians and North Brisbane cyclists = -0.02774595425430586
```

S2 Box 6

```
# Box 6 - S2.T2: Scatter plot, coloured by period
before = df.iloc[0:320, :]
after = df.iloc[320:1461, :]

plt.figure(figsize=(10, 6))
plt.scatter(before['North Brisbane Bikeway Cyclists'], before['Bicentennial Bikeway Pedestrians'], color='blue',
            marker='o')
plt.scatter(after['North Brisbane Bikeway Cyclists'], after['Bicentennial Bikeway Pedestrians'], color='orange',
            marker='o')
plt.xlabel('North Brisbane Bikeway cyclists (users/day)')
plt.ylabel('Bicentennial Bikeway pedestrians (users/day)')
plt.title('North Brisbane cyclists vs Bicentennial pedestrians, 19 Jun 2014-31 Dec 2018')
plt.legend(['19 Jun 2014-31 May 2015', '1 Jun 2015-31 Dec 2018'])
plt.show()
```

Output

```

Date      ... North Brisbane Bikeway Cyclists
318 2015-05-30 ... 90.0
319 2015-05-31 ... 64.0
320 2015-06-01 ... 148.0
321 2015-06-02 ... 156.0

[4 rows x 5 columns]
19 Jun 2014-31 May 2015
      Bicentennial Bikeway Pedestrians North Brisbane Bikeway Cyclists
count      320.000000      320.000000
mean      1439.646875      91.396875
std       740.050035      43.177936
min        0.000000        0.000000
25%      1130.750000      59.750000
50%      1628.000000      91.000000
75%      2022.250000      123.000000
max      3312.000000      181.000000
r = 0.4689986197678042
1 Jun 2015-31 Dec 2018
      Bicentennial Bikeway Pedestrians North Brisbane Bikeway Cyclists
count      1141.000000      1141.000000
mean      648.325153      143.794040
std       204.829245      79.539677
min        0.000000        0.000000
25%      561.000000      86.000000
50%      665.000000      132.000000
75%      761.000000      202.000000
max      3328.000000      332.000000
r = 0.21575472608009483

```

S2 Box 8

```
# Box 8 - S2.T3: Correlation within each calendar year
print(df.iloc[194:196, :])
print(df.iloc[531:533, :])
print(df.iloc[869:871, :])
print(df.iloc[1144:1146, :])

year_2014 = df.iloc[0:195, :]
year_2015 = df.iloc[195:532, :]
year_2016 = df.iloc[532:870, :]
year_2017 = df.iloc[870:1145, :]
year_2018 = df.iloc[1145:1461, :]

print('2014 r =', np.corrcoef(year_2014['North Brisbane Bikeway Cyclists'], year_2014['Bicentennial Bikeway Pedestrians'])[0, 1])
print('2015 r =', np.corrcoef(year_2015['North Brisbane Bikeway Cyclists'], year_2015['Bicentennial Bikeway Pedestrians'])[0, 1])
print('2016 r =', np.corrcoef(year_2016['North Brisbane Bikeway Cyclists'], year_2016['Bicentennial Bikeway Pedestrians'])[0, 1])
print('2017 r =', np.corrcoef(year_2017['North Brisbane Bikeway Cyclists'], year_2017['Bicentennial Bikeway Pedestrians'])[0, 1])
print('2018 r =', np.corrcoef(year_2018['North Brisbane Bikeway Cyclists'], year_2018['Bicentennial Bikeway Pedestrians'])[0, 1])
```

Output

```
Date ... North Brisbane Bikeway Cyclists
194 2014-12-31 ... 57.0
195 2015-01-01 ... 43.0

[2 rows x 5 columns]
Date ... North Brisbane Bikeway Cyclists
531 2015-12-31 ... 63.0
532 2016-01-01 ... 43.0

[2 rows x 5 columns]
Date ... North Brisbane Bikeway Cyclists
869 2016-12-31 ... 55.0
870 2017-01-01 ... 68.0

[2 rows x 5 columns]
Date ... North Brisbane Bikeway Cyclists
1144 2017-12-31 ... 61.0
1145 2018-01-01 ... 62.0

[2 rows x 5 columns]
2014 r = 0.36565978655074965
2015 r = 0.38174936451926184
2016 r = 0.1765332895003914
2017 r = 0.10897612786288775
2018 r = 0.3813049830851158
```

S2 Box 9

```
# Box 9 - S2.T3: Check zero counts retained in the supplied file
print('Bicentennial pedestrian zero days:')
print(df[df['Bicentennial Bikeway Pedestrians'] == 0].shape)
print('North Brisbane cyclist zero days:')
print(df[df['North Brisbane Bikeway Cyclists'] == 0].shape)
```

Output

```
Bicentennial pedestrian zero days:
(72, 5)
North Brisbane cyclist zero days:
(40, 5)
```

```

Training shape: (1145, 7)
Testing shape: (316, 7)
First training row
Date      Bicentennial Bikeway Pedestrians  ...  maximum_temperature
2014-06-19      1630.0  ...  20.3

[1 rows x 7 columns]
Last training row
Date      Bicentennial Bikeway Pedestrians  ...  maximum_temperature
2017-12-31      520.0  ...  33.6

[1 rows x 7 columns]
First testing row
Date      Bicentennial Bikeway Pedestrians  ...  maximum_temperature
2018-01-01      553.0  ...  32.1

```

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```
[1 rows x 7 columns]
Last testing row
      Bicentennial Bikeway Pedestrians  ...  maximum_temperature
Date
2018-12-31                        663.0  ...                31.4

[1 rows x 7 columns]
```

S3 Box 4

```
# Box 4 - Pedestrian model with all three weather factors
X_p_full = sm.add_constant(training_data[
    ['daily_rainfall', 'solar_exposure', 'maximum_temperature']
])
y_p = training_data['Bicentennial Bikeway Pedestrians']
ped_full = sm.OLS(y_p, X_p_full).fit()
print(ped_full.summary())
```

Output

```
OLS Regression Results
=====
Dep. Variable:      Bicentennial Bikeway Pedestrians      R-squared:      0.026
Model:              OLS                                  Adj. R-squared: 0.023
Method:             Least Squares                       F-statistic:    10.15
Date:               Tue, 29 Sep 2026                     Prob (F-statistic): 1.33e-06
Time:               10:40:27                             Log-Likelihood: -8864.5
No. Observations:   1145                                 AIC:           1.774e+04
Df Residuals:       1141                                 BIC:           1.776e+04
Df Model:           3
Covariance Type:    nonrobust
=====
               coef      std err          t      P>|t|      [0.025      0.975]
-----
const          1278.5812    119.033     10.741     0.000    1045.033    1512.129
daily_rainfall    -2.6686     1.494     -1.786     0.074     -5.600     0.262
solar_exposure    14.7729     3.379      4.372     0.000      8.143    21.403
maximum_temperature -25.5964     5.517     -4.640     0.000    -36.420    -14.772
=====
Omnibus:          224.625    Durbin-Watson:      0.319
Prob(Omnibus):    0.000    Jarque-Bera (JB):    375.716
Skew:             1.268    Prob(JB):            2.60e-82
Kurtosis:         4.203    Cond. No.            241.
=====
```

```
Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.
```

S3 Box 5

```
# Box 5 - Evaluate the full pedestrian model
ped_full_train = ped_full.predict(X_p_full)
ped_full_test = ped_full.predict(sm.add_constant(testing_data[
    ['daily_rainfall', 'solar_exposure', 'maximum_temperature']
]))

ped_full_train_mse = mean_squared_error(y_p, ped_full_train)
ped_full_test_mse = mean_squared_error(
    testing_data['Bicentennial Bikeway Pedestrians'],
    ped_full_test
)

# RMSE = square root of MSE (Week 8 lecture)
print('Full pedestrian training RMSE:', np.sqrt(ped_full_train_mse))
print('Full pedestrian testing RMSE:', np.sqrt(ped_full_test_mse))
```

Output

```
Full pedestrian training RMSE: 557.2489499473432
Full pedestrian testing RMSE: 215.68989736550222
```

S3 Box 6

```
# Box 6 - Remove rainfall because its training p-value exceeds 0.05
X_p = sm.add_constant(training_data[
    ['solar_exposure', 'maximum_temperature']
])
ped_final = sm.OLS(y_p, X_p).fit()
print(ped_final.summary())
```

Output

```
OLS Regression Results
=====
Dep. Variable:    Bicentennial Bikeway Pedestrians    R-squared:        0.023
Model:            OLS                                Adj. R-squared:    0.022
Method:            Least Squares                     F-statistic:       13.61
Date:              Tue, 29 Sep 2026                   Prob (F-statistic): 1.44e-06
Time:              10:40:27                           Log-Likelihood:    -8866.1
No. Observations: 1145                                AIC:               1.774e+04
Df Residuals:      1142                                BIC:               1.775e+04
Df Model:           2
Covariance Type:   nonrobust
=====
               coef      std err          t      P>|t|      [0.025      0.975]
-----
const                1273.4554      119.113      10.691      0.000      1039.751      1507.159
solar_exposure         15.6765         3.344         4.687      0.000         9.115         22.238
maximum_temperature    -26.2994         5.508        -4.775      0.000        -37.106        -15.493
=====
Omnibus:                 226.505    Durbin-Watson:           0.320
Prob(Omnibus):            0.000    Jarque-Bera (JB):        380.774
Skew:                     1.274    Prob(JB):                 2.07e-83
Kurtosis:                 4.221    Cond. No.                  240.
=====
```

Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

S3 Box 7

```
# Box 7 - Evaluate the final pedestrian model
ped_train = ped_final.predict(X_p)
ped_test = ped_final.predict(sm.add_constant(testing_data[
    ['solar_exposure', 'maximum_temperature']
]))

ped_train_mse = mean_squared_error(y_p, ped_train)
ped_test_mse = mean_squared_error(
    testing_data['Bicentennial Bikeway Pedestrians'],
    ped_test
)

# RMSE = square root of MSE (Week 8 lecture)
print('Final pedestrian training RMSE:', np.sqrt(ped_train_mse))
print('Final pedestrian testing RMSE:', np.sqrt(ped_test_mse))
```

Output

```
Final pedestrian training RMSE: 558.0277512674206
Final pedestrian testing RMSE: 218.26934431394153
```

S3 Box 8

```

# Box 8 - Final pedestrian model: fitted lines for retained predictors
# Each line comes from the same multiple regression model.
# The other retained predictor is held at its training mean.
mean_temperature = training_data['maximum_temperature'].mean()
mean_solar = training_data['solar_exposure'].mean()

ped_solar_line = sm.add_constant(training_data[
    ['solar_exposure', 'maximum_temperature']
].sort_values('solar_exposure'))
ped_solar_line['maximum_temperature'] = mean_temperature

ped_temp_line = sm.add_constant(training_data[
    ['solar_exposure', 'maximum_temperature']
].sort_values('maximum_temperature'))
ped_temp_line['solar_exposure'] = mean_solar

fig, axs = plt.subplots(2, 1, figsize=(10, 11))

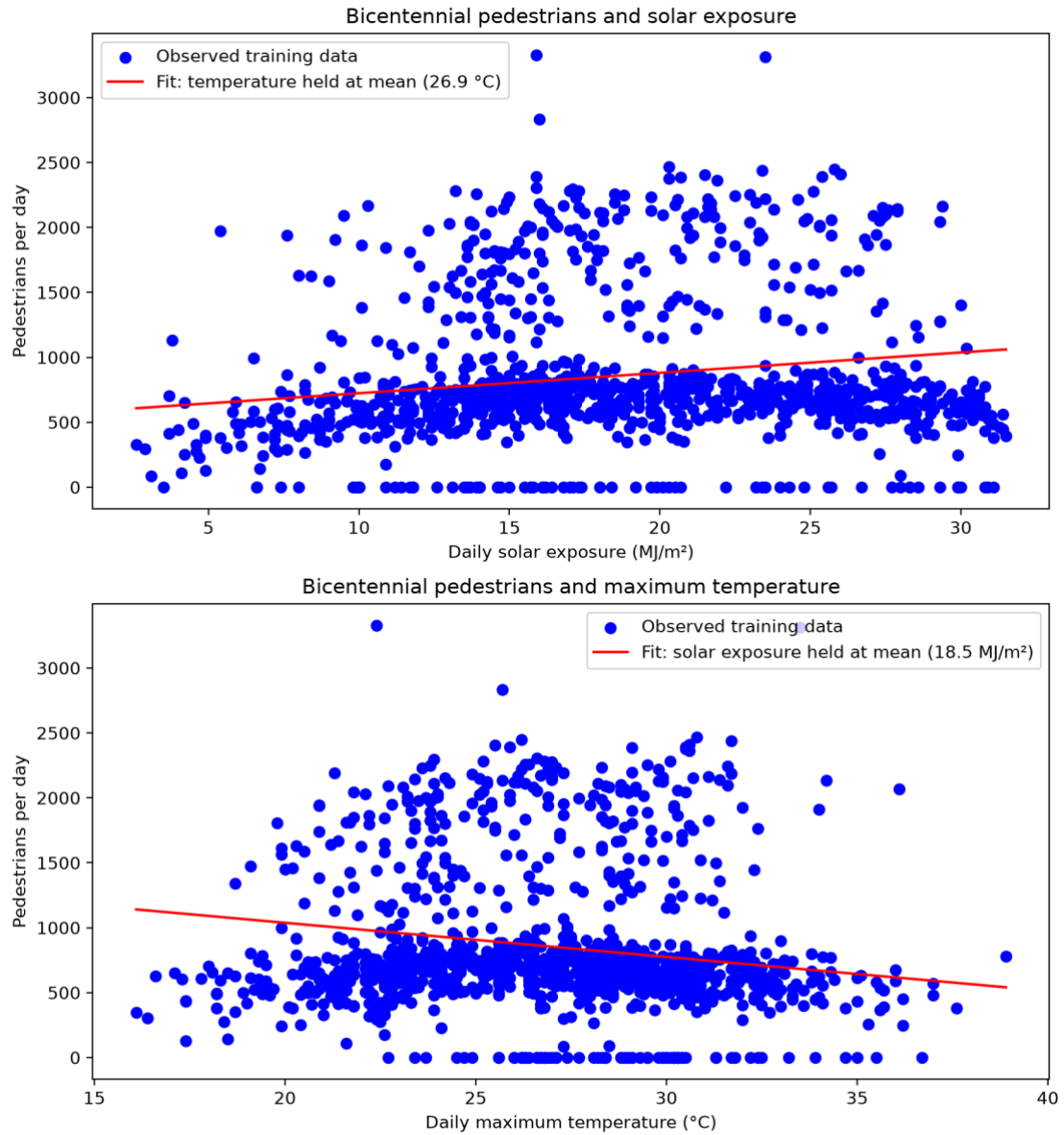
axs[0].scatter(
    training_data['solar_exposure'],
    Y_P,
    color='blue',
    label='Observed training data'
)
axs[0].plot(
    ped_solar_line['solar_exposure'],
    ped_final.predict(ped_solar_line),
    color='red',
    label=f'Fit: temperature held at mean ({mean_temperature:.1f} °C)'
)
axs[0].set_xlabel('Daily solar exposure (MJ/m²)')
axs[0].set_ylabel('Pedestrians per day')
axs[0].set_title('Bicentennial pedestrians and solar exposure')
axs[0].legend()

axs[1].scatter(
    training_data['maximum_temperature'],
    Y_P,
    color='blue',
    label='Observed training data'
)
axs[1].plot(
    ped_temp_line['maximum_temperature'],
    ped_final.predict(ped_temp_line),
    color='red',
    label=f'Fit: solar exposure held at mean ({mean_solar:.1f} MJ/m²)'
)
axs[1].set_xlabel('Daily maximum temperature (°C)')
axs[1].set_ylabel('Pedestrians per day')
axs[1].set_title('Bicentennial pedestrians and maximum temperature')
axs[1].legend()

plt.show()

```

Output



S3 Box 9

```
# Box 9 - Cyclist model with all three weather factors
X_c_full = sm.add_constant(training_data[
    ['daily_rainfall', 'solar_exposure', 'maximum_temperature']
])
y_c = training_data['Bicentennial Bikeway Cyclists']
cyc_full = sm.OLS(y_c, X_c_full).fit()
print(cyc_full.summary())
```

Output

```
OLS Regression Results
=====
Dep. Variable:    Bicentennial Bikeway Cyclists    R-squared:        0.043
Model:            OLS                            Adj. R-squared:    0.040
Method:            Least Squares                  F-statistic:       16.94
Date:              Tue, 29 Sep 2026                Prob (F-statistic): 9.10e-11
Time:              10:40:28                        Log-Likelihood:    -9219.9
No. Observations: 1145                             AIC:               1.845e+04
Df Residuals:      1141                            BIC:               1.847e+04
Df Model:           3
Covariance Type:   nonrobust
=====
                    coef    std err          t      P>|t|      [0.025      0.975]
-----
const                911.3816    162.349      5.614    0.000     592.846    1229.917
daily_rainfall        -7.8739      2.037     -3.865    0.000    -11.871     -3.876
solar_exposure        18.8146      4.609      4.082    0.000      9.772    27.857
maximum_temperature    0.4208      7.524      0.056    0.955    -14.342    15.183
=====
Omnibus:                 342.436   Durbin-Watson:           0.238
Prob(Omnibus):            0.000   Jarque-Bera (JB):         87.885
Skew:                    -0.435   Prob(JB):                 8.24e-20
Kurtosis:                 1.958   Cond. No.                  241.
=====
```

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

S3 Box 10

```
# Box 10 - Evaluate the full cyclist model
cyc_full_train = cyc_full.predict(X_c_full)
cyc_full_test = cyc_full.predict(sm.add_constant(testing_data[
    ['daily_rainfall', 'solar_exposure', 'maximum_temperature']
]))

cyc_full_train_mse = mean_squared_error(y_c, cyc_full_train)
cyc_full_test_mse = mean_squared_error(
    testing_data['Bicentennial Bikeway Cyclists'],
    cyc_full_test
)

# RMSE = square root of MSE (Week 8 lecture)
print('Full cyclist training RMSE:', np.sqrt(cyc_full_train_mse))
print('Full cyclist testing RMSE:', np.sqrt(cyc_full_test_mse))
```

Output

```
Full cyclist training RMSE: 760.029744364261
Full cyclist testing RMSE: 586.7320716293268
```

S3 Box 11

```
# Box 11 - Remove maximum temperature because its p-value exceeds 0.05
X_c = sm.add_constant(training_data[
    ['daily_rainfall', 'solar_exposure']
])
cyc_final = sm.OLS(y_c, X_c).fit()
print(cyc_final.summary())
```

Output

```
OLS Regression Results
=====
Dep. Variable:    Bicentennial Bikeway Cyclists    R-squared:        0.043
Model:            OLS                            Adj. R-squared:    0.041
Method:            Least Squares                  F-statistic:       25.43
Date:              Tue, 29 Sep 2026                Prob (F-statistic): 1.57e-11
Time:              10:40:28                        Log-Likelihood:    -9219.9
No. Observations: 1145                            AIC:               1.845e+04
Df Residuals:      1142                            BIC:               1.846e+04
Df Model:           2
Covariance Type:   nonrobust
=====
               coef      std err          t      P>|t|      [0.025      0.975]
-----
const          919.5806     69.734     13.187     0.000     782.759     1056.402
daily_rainfall    -7.8658      2.031     -3.872     0.000    -11.851     -3.880
solar_exposure    18.9815      3.512      5.405     0.000     12.092     25.871
=====
Omnibus:                 342.423   Durbin-Watson:           0.238
Prob(Omnibus):            0.000   Jarque-Bera (JB):         87.755
Skew:                     -0.434   Prob(JB):                 8.79e-20
Kurtosis:                  1.958   Cond. No.                  61.5
=====
```

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

S3 Box 12

```
# Box 12 - Evaluate the final cyclist model
cyc_train = cyc_final.predict(X_c)
cyc_test = cyc_final.predict(sm.add_constant(testing_data[
    ['daily_rainfall', 'solar_exposure']
]))

cyc_train_mse = mean_squared_error(y_c, cyc_train)
cyc_test_mse = mean_squared_error(
    testing_data['Bicentennial Bikeway Cyclists'],
    cyc_test
)

# RMSE = square root of MSE (Week 8 lecture)
print('Final cyclist training RMSE:', np.sqrt(cyc_train_mse))
print('Final cyclist testing RMSE:', np.sqrt(cyc_test_mse))
```

Output

```
Final cyclist training RMSE: 760.0307861899571
Final cyclist testing RMSE: 586.7363876381361
```

S3 Box 13

```

# Box 13 - Final cyclist model: fitted lines for retained predictors
# Each line comes from the same multiple regression model.
# The other retained predictor is held at its training mean.
mean_rainfall = training_data['daily_rainfall'].mean()

cyc_rain_line = sm.add_constant(training_data[
    ['daily_rainfall', 'solar_exposure']
].sort_values('daily_rainfall'))
cyc_rain_line['solar_exposure'] = mean_solar

cyc_solar_line = sm.add_constant(training_data[
    ['daily_rainfall', 'solar_exposure']
].sort_values('solar_exposure'))
cyc_solar_line['daily_rainfall'] = mean_rainfall

fig, axs = plt.subplots(2, 1, figsize=(10, 11))

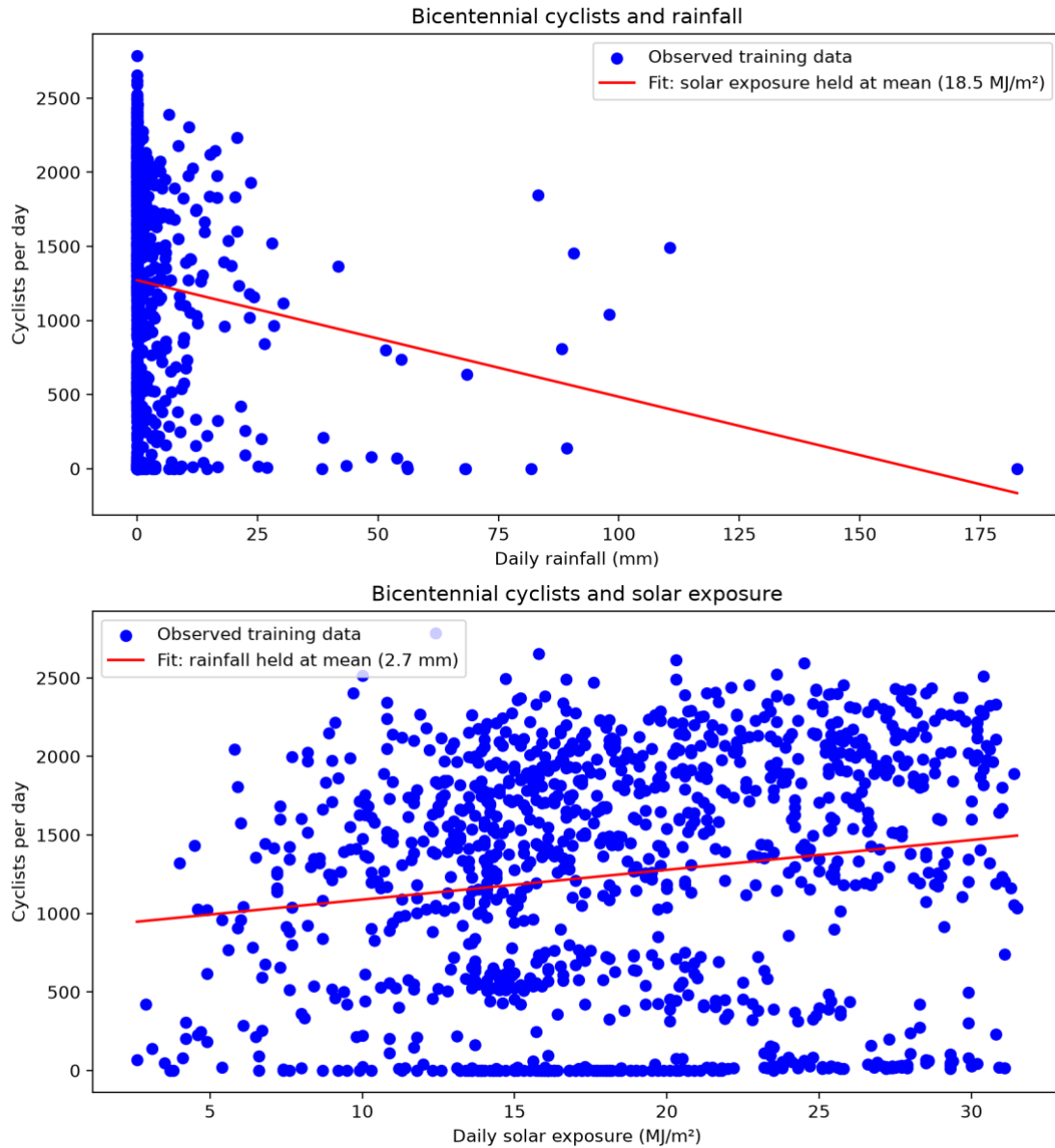
axs[0].scatter(
    training_data['daily_rainfall'],
    y_c,
    color='blue',
    label='Observed training data'
)
axs[0].plot(
    cyc_rain_line['daily_rainfall'],
    cyc_final.predict(cyc_rain_line),
    color='red',
    label=f'Fit: solar exposure held at mean ({mean_solar:.1f} MJ/m²)'
)
axs[0].set_xlabel('Daily rainfall (mm)')
axs[0].set_ylabel('Cyclists per day')
axs[0].set_title('Bicentennial cyclists and rainfall')
axs[0].legend()

axs[1].scatter(
    training_data['solar_exposure'],
    y_c,
    color='blue',
    label='Observed training data'
)
axs[1].plot(
    cyc_solar_line['solar_exposure'],
    cyc_final.predict(cyc_solar_line),
    color='red',
    label=f'Fit: rainfall held at mean ({mean_rainfall:.1f} mm)'
)
axs[1].set_xlabel('Daily solar exposure (MJ/m²)')
axs[1].set_ylabel('Cyclists per day')
axs[1].set_title('Bicentennial cyclists and solar exposure')
axs[1].legend()

plt.show()

```

Output



S3 Box 14

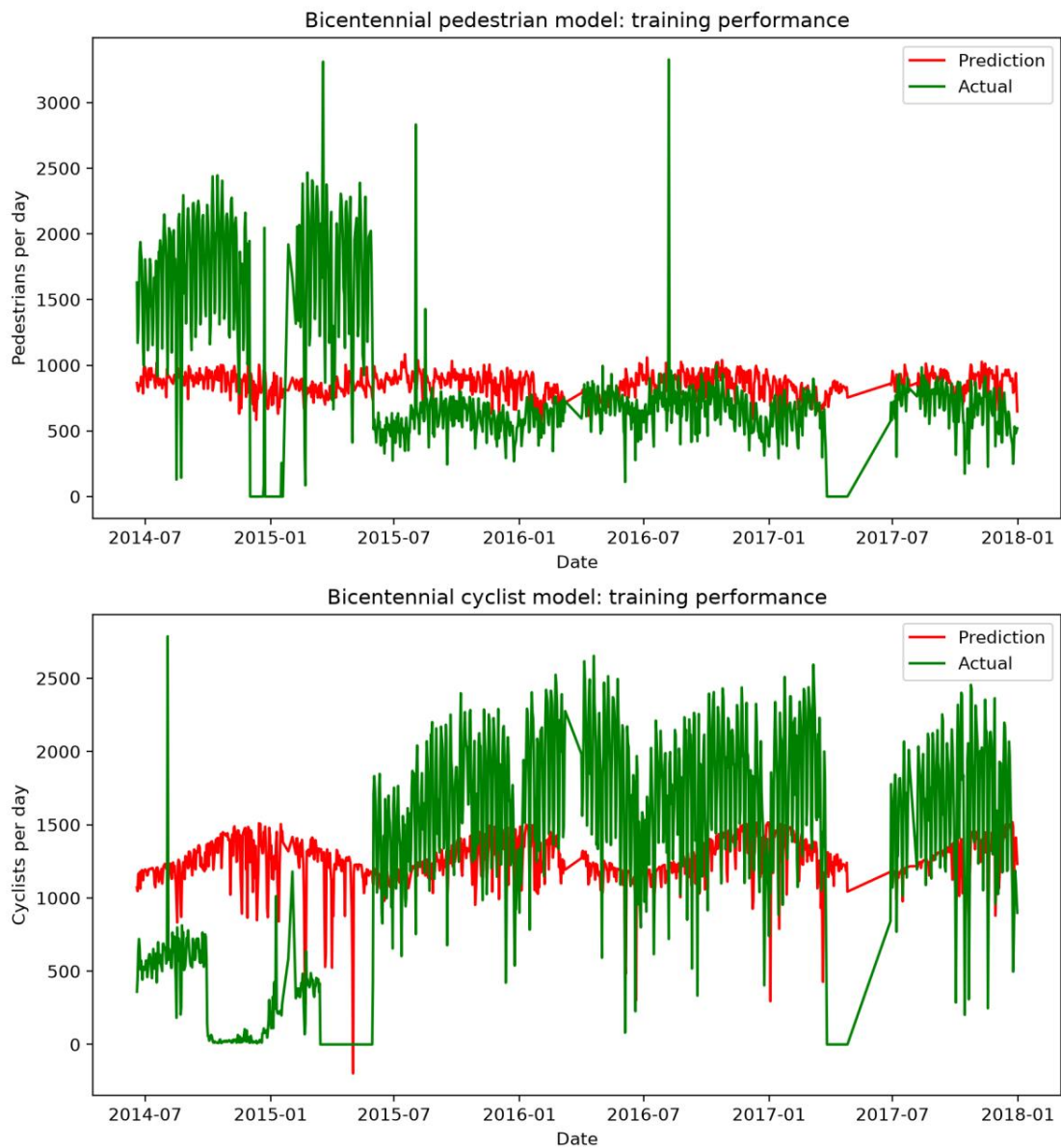
```
# Box 14 - Training performance of both final models over time
fig, axs = plt.subplots(2, 1, figsize=(10, 11))

axs[0].plot(training_data.index, ped_train,
            color='red', label='Prediction')
axs[0].plot(training_data.index, y_p,
            color='green', label='Actual')
axs[0].set_xlabel('Date')
axs[0].set_ylabel('Pedestrians per day')
axs[0].set_title('Bicentennial pedestrian model: training performance')
axs[0].legend()

axs[1].plot(training_data.index, cyc_train,
            color='red', label='Prediction')
axs[1].plot(training_data.index, y_c,
            color='green', label='Actual')
axs[1].set_xlabel('Date')
axs[1].set_ylabel('Cyclists per day')
axs[1].set_title('Bicentennial cyclist model: training performance')
axs[1].legend()

plt.show()
```

Output



S3 Box 15

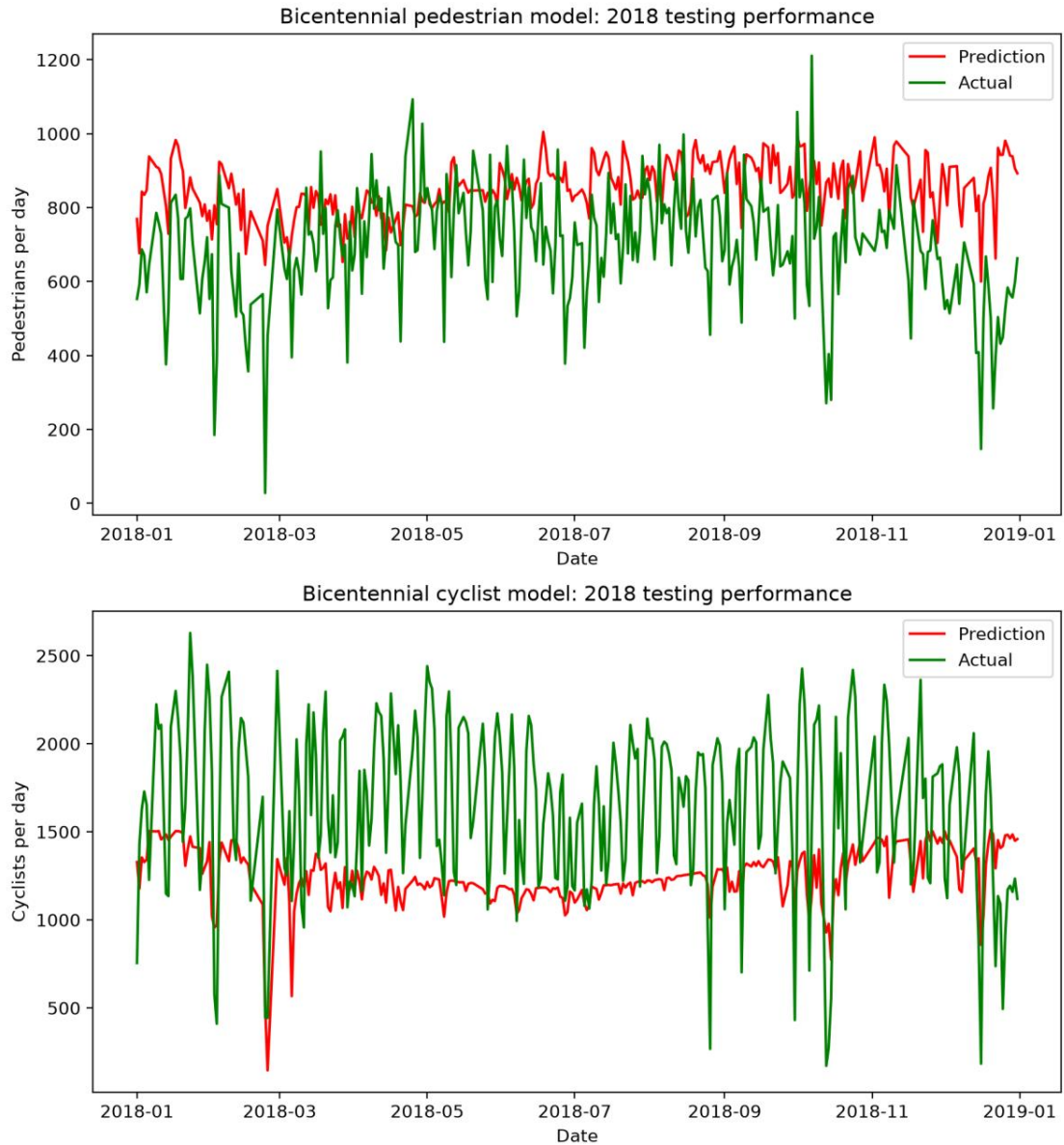
```
# Box 15 - Testing performance of both final models over time
fig, axs = plt.subplots(2, 1, figsize=(10, 11))

axs[0].plot(testing_data.index, ped_test,
            color='red', label='Prediction')
axs[0].plot(
    testing_data.index,
    testing_data['Bicentennial Bikeway Pedestrians'],
    color='green',
    label='Actual'
)
axs[0].set_xlabel('Date')
axs[0].set_ylabel('Pedestrians per day')
axs[0].set_title('Bicentennial pedestrian model: 2018 testing performance')
axs[0].legend()

axs[1].plot(testing_data.index, cyc_test,
            color='red', label='Prediction')
axs[1].plot(
    testing_data.index,
    testing_data['Bicentennial Bikeway Cyclists'],
    color='green',
    label='Actual'
)
axs[1].set_xlabel('Date')
axs[1].set_ylabel('Cyclists per day')
axs[1].set_title('Bicentennial cyclist model: 2018 testing performance')
axs[1].legend()

plt.show()
```

Output



S3 Box 16

```
# Box 16 - Predicted versus actual values on the testing set
ped_actual = testing_data[
    'Bicentennial Bikeway Pedestrians'
].sort_values()
cyc_actual = testing_data[
    'Bicentennial Bikeway Cyclists'
].sort_values()

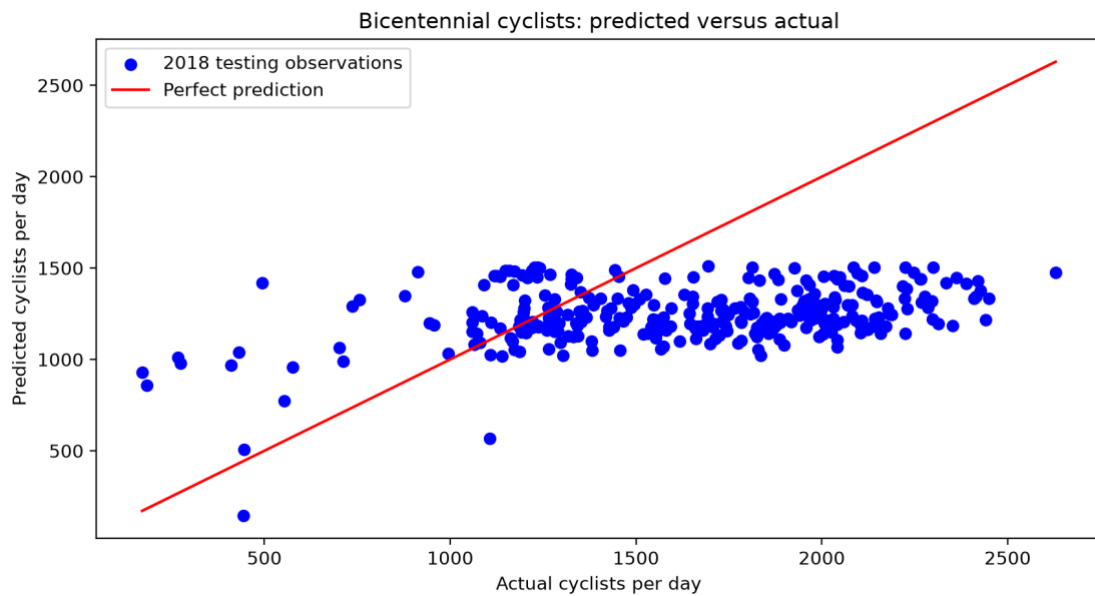
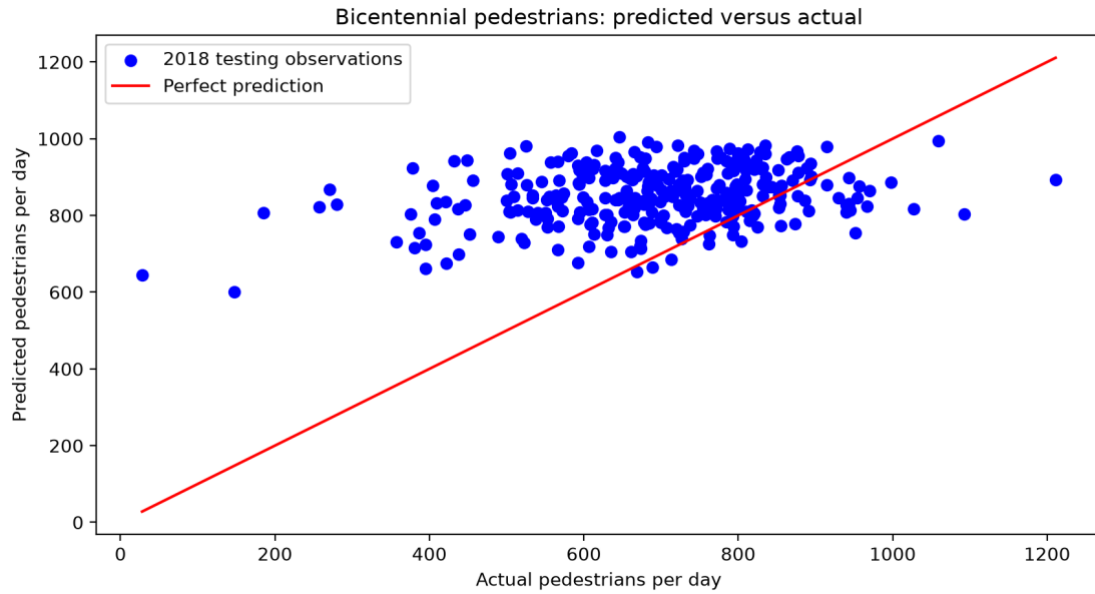
fig, axs = plt.subplots(2, 1, figsize=(10, 11))

axs[0].scatter(
    testing_data['Bicentennial Bikeway Pedestrians'],
    ped_test,
    color='blue',
    label='2018 testing observations'
)
axs[0].plot(ped_actual, ped_actual,
            color='red', label='Perfect prediction')
axs[0].set_xlabel('Actual pedestrians per day')
axs[0].set_ylabel('Predicted pedestrians per day')
axs[0].set_title('Bicentennial pedestrians: predicted versus actual')
axs[0].legend()

axs[1].scatter(
    testing_data['Bicentennial Bikeway Cyclists'],
    cyc_test,
    color='blue',
    label='2018 testing observations'
)
axs[1].plot(cyc_actual, cyc_actual,
            color='red', label='Perfect prediction')
axs[1].set_xlabel('Actual cyclists per day')
axs[1].set_ylabel('Predicted cyclists per day')
axs[1].set_title('Bicentennial cyclists: predicted versus actual')
axs[1].legend()

plt.show()
```

Output



S3 Box 17

```
# Box 17 - Evidence for interpreting training and testing performance
print('Training pedestrians and cyclists')
print(training_data[[
    'Bicentennial Bikeway Pedestrians',
    'Bicentennial Bikeway Cyclists'
]].describe())

print('Testing pedestrians and cyclists')
print(testing_data[[
    'Bicentennial Bikeway Pedestrians',
    'Bicentennial Bikeway Cyclists'
]].describe())

print(
    'Testing pedestrian actual mean:',
    testing_data['Bicentennial Bikeway Pedestrians'].mean()
)
print('Testing pedestrian predicted mean:', ped_test.mean())
print(
    'Testing cyclist actual mean:',
    testing_data['Bicentennial Bikeway Cyclists'].mean()
)
print('Testing cyclist predicted mean:', cyc_test.mean())

print('Training weather factors')
print(training_data[[
    'daily_rainfall',
    'solar_exposure',
    'maximum_temperature'
]].describe())
```

Output

```
Training pedestrians and cyclists
      Bicentennial Bikeway Pedestrians  Bicentennial Bikeway Cyclists
count                1145.000000                1145.000000
mean                 856.860262                1249.562445
std                  564.885038                777.108224
min                   0.000000                 0.000000
25%                  561.000000                547.000000
50%                  700.000000                1409.000000
75%                  883.000000                1912.000000
max                  3328.000000                2786.000000

Testing pedestrians and cyclists
      Bicentennial Bikeway Pedestrians  Bicentennial Bikeway Cyclists
count                316.000000                316.000000
mean                 694.053797                1642.458861
std                  153.472857                460.214672
min                   28.000000                171.000000
25%                  610.250000                1292.000000
50%                  706.000000                1711.500000
75%                  794.250000                2003.500000
max                  1211.000000                2629.000000

Testing pedestrian actual mean: 694.0537974683544
Testing pedestrian predicted mean: 852.6008134720012
Testing cyclist actual mean: 1642.4588607594937
Testing cyclist predicted mean: 1243.1934295729304

Training weather factors
      daily_rainfall  solar_exposure  maximum_temperature
count                1145.000000                1145.000000
mean                 2.732402                18.516681
std                  11.180809                 6.467658
min                   0.000000                 2.600000
25%                  0.000000                14.000000
50%                  0.000000                17.800000
75%                  0.400000                23.800000
max                  182.600000                31.500000
```

Section 5 Python code and outputs

The code and corresponding outputs used for the preliminary-data correlation analysis are included below in S5 Box order. The public dataset is available from Mendeley Data at <https://data.mendeley.com/datasets/3xvfyycp8d/1> (DOI: 10.17632/3xvfyycp8d.1). The Excel workbook was saved as compressive_strength_database.csv in CSV UTF-8 format before the analysis was run.

S5 Box 1

```
# Box 1 - Import the libraries
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
```

S5 Box 2

```
# Box 2 - Load the data, calculate water-to-binder ratio and retain the authors' modelling subset
df = pd.read_csv('compressive_strength_database.csv', header=0)
df.info()

print('Original data shape:', df.shape)
print('Rows flagged as outliers:', df.loc[df['outlier_flag'] == 1, :].shape)

df['Water-to-binder ratio'] = df['Water (kg/m3)'] / (
    df['Cement (kg/m3)'] + df['FA (kg/m3)'] + df['GGBS (kg/m3)']
)

data = df.loc[df['outlier_flag'] == 0, :]
rca_data = data.loc[data['RAR (%)'] > 0, :]

print('Unflagged data shape:', data.shape)
print('RCA-containing subset shape:', rca_data.shape)
```

Output

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1736 entries, 0 to 1735
Data columns (total 15 columns):
#   Column                Non-Null Count  Dtype
---  -
0   record_id              1736 non-null   object
1   source_reference       116 non-null    object
2   p-RCA (kg/m3)          1736 non-null   float64
3   CV-RCA (%)             1736 non-null   float64
4   W-RCA (%)              1736 non-null   float64
5   Cement (kg/m3)         1736 non-null   float64
6   FA (kg/m3)             1736 non-null   float64
7   GGBS (kg/m3)           1736 non-null   float64
8   Water (kg/m3)          1736 non-null   float64
9   Sand (kg/m3)           1736 non-null   float64
10  CA (kg/m3)             1736 non-null   float64
11  RAR (%)                1736 non-null   float64
12  SP (kg/m3)             1736 non-null   float64
13  CS (MPa)               1736 non-null   float64
14  outlier_flag           1736 non-null   int64
dtypes: float64(12), int64(1), object(2)
memory usage: 203.6+ KB
Original data shape: (1736, 15)
Rows flagged as outliers: (87, 15)
Unflagged data shape: (1649, 16)
RCA-containing subset shape: (1422, 16)
```

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S5 Box 3

```
# Box 3 - Select the RCA properties, water-to-binder ratio and target
rca_analysis = rca_data[['p-RCA (kg/m3)', 'CV-RCA (%)', 'W-RCA (%)',
                        'RAR (%)', 'Water-to-binder ratio', 'CS (MPa)']]
rca_analysis.describe()
```

Output

	p-RCA (kg/m3)	CV-RCA (%)	W-RCA (%)	RAR (%)	Water-to-binder ratio	CS (MPa)
count	1422.000000	1422.000000	1422.000000	1422.000000	1422.000000	1422.000000
mean	2508.996343	14.813570	4.356944	67.905977	0.472944	38.764571
std	109.203177	3.809194	2.256800	28.808747	0.098106	10.959427
min	2100.000000	6.000000	0.400000	5.000000	0.250000	14.200000
25%	2461.000000	12.740000	2.700000	46.250000	0.400000	30.900000
50%	2510.000000	15.000000	4.400000	70.000000	0.467978	37.800000
75%	2594.000000	16.300000	6.000000	100.000000	0.539773	45.700000
max	2880.000000	32.000000	10.100000	100.000000	1.000000	73.100000

S5 Box 4

```
# Box 4 - Pearson correlation matrix for the RCA-containing subset
rca_correlation = rca_analysis.corr()
rca_correlation
```

Output

	p-RCA (kg/m3)	CV-RCA (%)	W-RCA (%)	RAR (%)	Water-to-binder ratio	CS (MPa)
p-RCA (kg/m3)	1.000000	-0.234676	-0.201007	0.105184	0.033629	0.107385
CV-RCA (%)	-0.234676	1.000000	0.298420	0.019120	0.099509	-0.237731
W-RCA (%)	-0.201007	0.298420	1.000000	0.026986	0.141470	-0.350436
RAR (%)	0.105184	0.019120	0.026986	1.000000	0.066463	-0.082339
Water-to-binder ratio	0.033629	0.099509	0.141470	0.066463	1.000000	-0.569357
CS (MPa)	0.107385	-0.237731	-0.350436	-0.082339	-0.569357	1.000000

S5 Box 5

```
# Box 5 - Correlations between the selected RCA properties and compressive strength
r_w = np.corrcoef(rca_data['W-RCA (%)'], rca_data['CS (MPa)'])[0, 1]
r_cv = np.corrcoef(rca_data['CV-RCA (%)'], rca_data['CS (MPa)'])[0, 1]
r_density = np.corrcoef(rca_data['p-RCA (kg/m3)'], rca_data['CS (MPa)'])[0, 1]
r_rar = np.corrcoef(rca_data['RAR (%)'], rca_data['CS (MPa)'])[0, 1]

print('Water absorption and strength:', r_w)
print('Crushing value and strength:', r_cv)
print('Apparent density and strength:', r_density)
print('Replacement ratio and strength:', r_rar)
```

Output

```
Water absorption and strength: -0.3504361980736066
Crushing value and strength: -0.23773147123182695
Apparent density and strength: 0.10738460505256883
Replacement ratio and strength: -0.08233864817981153
```

S5 Box 6

```
# Box 6 - Key mix-design relationships across all unflagged mixtures
mix_analysis = data[['Water-to-binder ratio', 'SP (kg/m3)', 'Cement (kg/m3)',
                    'Water (kg/m3)', 'CS (MPa)']]
mix_correlation = mix_analysis.corr()
mix_correlation
```

Output

	Water-to-binder ratio	SP (kg/m3)	Cement (kg/m3)	Water (kg/m3)	CS (MPa)
Water-to-binder ratio	1.000000	-0.305171	-0.529016	0.456833	-0.582346
SP (kg/m3)	-0.305171	1.000000	-0.062529	-0.432768	0.455821
Cement (kg/m3)	-0.529016	-0.062529	1.000000	0.254750	0.357533
Water (kg/m3)	0.456833	-0.432768	0.254750	1.000000	-0.330792
CS (MPa)	-0.582346	0.455821	0.357533	-0.330792	1.000000

S5 Box 7

```
# Box 7 - Scatter plots of RCA properties against compressive strength
fig, axs = plt.subplots(2, 2, figsize=(12, 10))

axs[0, 0].scatter(rca_data['W-RCA (%)'], rca_data['CS (MPa)'],
                  color='blue', marker='o', label=f'r = {r_w:.3f}')
axs[0, 0].set_xlabel('RCA water absorption (%)')
axs[0, 0].set_ylabel('28-day compressive strength (MPa)')
axs[0, 0].legend(loc='upper right')

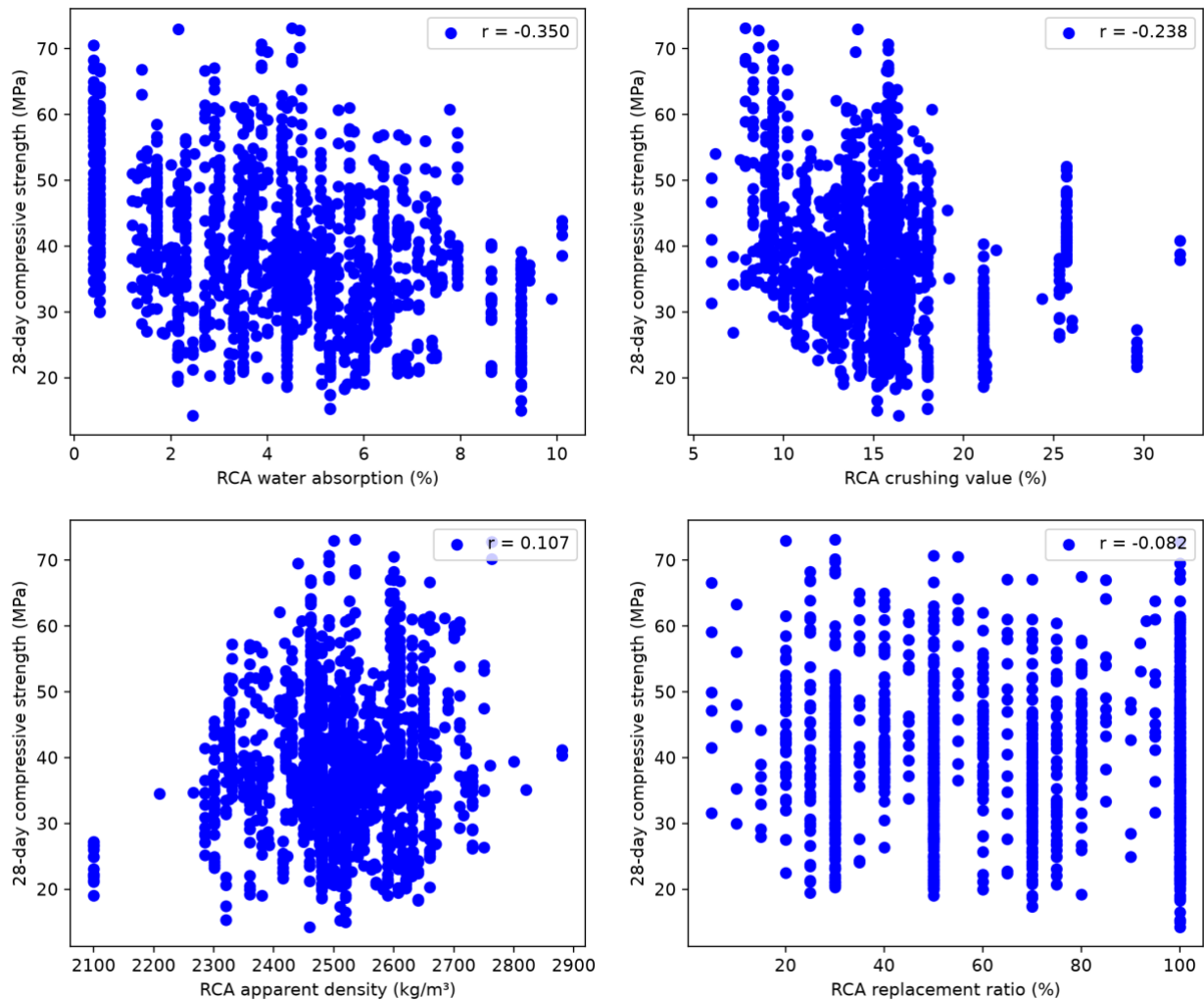
axs[0, 1].scatter(rca_data['CV-RCA (%)'], rca_data['CS (MPa)'],
                  color='blue', marker='o', label=f'r = {r_cv:.3f}')
axs[0, 1].set_xlabel('RCA crushing value (%)')
axs[0, 1].set_ylabel('28-day compressive strength (MPa)')
axs[0, 1].legend(loc='upper right')

axs[1, 0].scatter(rca_data['p-RCA (kg/m3)'], rca_data['CS (MPa)'],
                  color='blue', marker='o', label=f'r = {r_density:.3f}')
axs[1, 0].set_xlabel('RCA apparent density (kg/m³)')
axs[1, 0].set_ylabel('28-day compressive strength (MPa)')
axs[1, 0].legend(loc='upper right')

axs[1, 1].scatter(rca_data['RAR (%)'], rca_data['CS (MPa)'],
                  color='blue', marker='o', label=f'r = {r_rar:.3f}')
axs[1, 1].set_xlabel('RCA replacement ratio (%)')
axs[1, 1].set_ylabel('28-day compressive strength (MPa)')
axs[1, 1].legend(loc='upper right')

plt.show()
```

Output



S5 Box 8

```
# Box 8 - Water-to-binder ratio against compressive strength
r_wb = np.corrcoef(data['Water-to-binder ratio'], data['CS (MPa)'])[0, 1]

plt.figure(figsize=(9, 6))
plt.scatter(data['Water-to-binder ratio'], data['CS (MPa)'],
            color='blue', marker='o', label=f'r = {r_wb:.3f}')
plt.xlabel('Water-to-binder ratio')
plt.ylabel('28-day compressive strength (MPa)')
plt.legend(loc='upper right')
plt.show()
```

Output